Mathematical Programming and Operations Research

Modeling, Algorithms, and Complexity

Examples in Excel and Python

(Work in progress)

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Version Compilation date: January 21, 2025

Preface

This entire book is a working manuscript. The first draft of the book is yet to be completed.

MAJOR ACKNOWLEDGEMENTS

I would like to acknowledge that substantial parts of this book were borrowed under a CC-BY-SA license. These substantial pieces include:

- "A First Course in Linear Algebra" by Lyryx Learning (based on original text by Ken Kuttler). A majority of their formatting was used along with selected sections that make up the appendix sections on linear algebra. We are extremely grateful to Lyryx for sharing their files with us. They do an amazing job compiling their books and the templates and formatting that we have borrowed here clearly took a lot of work to set up. Thank you for sharing all of this material to make structuring and formating this book much easier! See subsequent page for list of contributors.
- "Foundations of Applied Mathmatics" with many contributors. See `https://github.com/Foundations-of-Applied-Mathematics`. Several sections from these notes were used along with some formatting. Some of this content has been edited or rearranged to suit the needs of this book. This content comes with some great references to code and nice formatting to present code within the book. See subsequent page with list of contributors.
- "Linear Inequalities and Linear Programming" by Kevin Cheung. See `https://github.com/dataopt/lineqlpbook`. These notes are posted on GitHub in a ".Rmd" format for nice reading online. This content was converted to LaTeXusing Pandoc. These notes make up a substantial section of the Linear Programming part of this book.
- Linear Programming notes by Douglas Bish. These notes also make up a substantial section of the Linear Programming part of this book.

I would also like to acknowledge Laurent Porrier and Diego Moran for contributing various notes on linear and integer programming.

I would also like to thank Jamie Fravel for helping to edit this book and for contributing chapters, examples, and code.

Contents

Introduction

Letter to instructors

This is an introductory book for students to learn optimization theory, tools, and applications. The two main goals are (1) students are able to apply the of optimization in their future work or research (2) students understand the concepts underlying optimization solver and use this knowledge to use solvers effectively.

This book was based on a sequence of course at Virginia Tech in the Industrial and Systems Engineering Department. The courses are *Deterministic Operations Research I* and *Deterministic Operations Reserach II*. The first course focuses on linear programming, while the second course covers integer programming an nonlinear programming. As such, the content in this book is meant to cover 2 or more full courses in optimization.

The book is designed to be read in a linear fashion. That said, many of the chapters are mostly independent and could be rearrranged, removed, or shortened as desited. This is an open source textbook, so use it as you like. You are encouraged to share adaptations and improvements so that this work can evolve over time.

Letter to students

This book is designed to be a resource for students interested in learning what optimizaiton is, how it works, and how you can apply it in your future career. The main focus is being able to apply the techniques of optimization to problems using computer technology while understanding (at least at a high level) what the computer is doing and what you can claim about the output from the computer.

Quite importantly, keep in mind that when someone claims to have *optimized* a problem, we want to know what kind of gaurantees they have about how good their solution is. Far too often, the solution that is provided is suboptimal by 10%, 20%, or even more. This can mean spending excess amounts of money, time, or energy that could have been saved. And when problems are at a large scale, this can easily result in millions of dollars in savings.

For this reason, we will learn the perspective of *mathematical programming* (a.k.a. mathematical optimization). The key to this study is that we provide gaurantees on how good a solution is to a given problem. We will also study how difficult a problem is to solve. This will help us know (a) how long it might take to solve it and (b) how good a of a solution we might expect to be able to find in reasonable amount of time.

We will later study *heuristic* methods - these methods typically do not come with gaurantees, but tend to help find quality solutions.

Note: Although there is some computer programming required in this work, this is not a course on programming. Thanks to the fantastic modelling packages available these days, we are able to solve complicated problems with little programming effort. The key skill we will need to learn is *mathematical modeling*: converting words and ideas into numbers and variables in order to communicate problems to a compter so that it can solve a problem for you.

As a main element of this book, we would like to make the process of using code and software as easy as possible. Attached to most examples in the book, there will be links to code that implements and solves the problem using several different tools from Excel and Python. These examples can should make it easy to solve a similar problem with different data, or more generally, can serve as a basis for solving related problems with similar structure.

How to use this book

Skim ahead. We recommend that before you come across a topic in lecture, that you skim the relevant sections ahead of time to get a broad overview of what is to come. This may take only a fraction of the time that it may take for you to read it.

Read the expected outcomes. At the beginning of each section, there will be a list of expected outcomes from that section. Review these outcomes before reading the section to help guide you through what is most relevant for you to take away from the material. This will also provide a brief look into what is to follow in that section.

Read the text. Read carefully the text to understand the problems and techniques. We will try to provide a plethora of examples, therefore, depending on your understanding of a topic, you many need to go carefully over all of the examples.

Explore the resources. Lastly, we recognize that there are many alternative methods of learning given the massive amounts of information and resources online. Thus, at the end of each section, there will be a number of superb resources that available on the internet in different formats. There are other free textbooks, informational websites, and also number of fantastic videos posted to youtube. We encourage to explore the resources to get another perspective on the material or to hear/read it taught from a differnt point of view or in presentation style.

Outline of this book

This book is divided in to 4 Parts:

Part I Linear Programming,

Part II Integer Programming,

Part III Discrete Algorithms,

Part IV Nonlinear Programming.

There are also a number of chapters of background material in the Appendix.

The content of this book is designed to encompass 2-3 full semester courses in an industrial engineering department.

Work in progress

This book is still a work in progress, so please feel free to send feedback, questions, comments, and edits to Robert Hildebrand at `open.optimization@gmail.com`.

1. Resources and Notation

Here are a list of resources that may be useful as alternative references or additional references.

Free notes and textbooks

- Mathematical Programming with Julia by Richard Lusby & Thomas Stidsen
- Linear Programming by K.J. Mtetwa, David
- A first course in optimization by Jon Lee
- Introduction to Optimizaiton Notes by Komei Fukuda
- Convex Optimization by Bord and Vandenberghe
- LP notes of Michel Goemans from MIT
- Understanding and Using Linear Programming - Matousek and Gärtner [Downloadable from Springer with University account]
- Operations Research Problems Statements and Solutions - Raúl PolerJosefa Mula Manuel Díaz-Madroñero [Downloadable from Springer with University account]

Notes, books, and videos by various solver groups

- AIMMS Optimization Modeling
- Optimization Modeling with LINGO by Linus Schrage
- The AMPL Book
- Microsoft Excel 2019 Data Analysis and Business Modeling, Sixth Edition, by Wayne Winston - Available to read for free as an e-book through Virginia Tech library at Orielly.com.
- Lesson files for the Winston Book
- Video instructions for solver and an example workbook
- youtube-OR-course

Gurobi Links

- Go to `https://github.com/Gurobi` and download the example files.
- Essential ingredients
- Gurobi Linear Programming tutorial
- Gurobi tutorial MILP
- GUROBI - Python 1 - Modeling with GUROBI in Python
- GUROBI - Python II: Advanced Algebraic Modeling with Python and Gurobi
- GUROBI - Python III: Optimization and Heuristics
- Webinar Materials
- GUROBI Tutorials

How to prove things

- Hammack - Book of Proof

Statistics

- Open Stax - Introductory Statistics

Linear Algebra

- Beezer - A first course in linear algebra
- Selinger - Linear Algebra
- Cherney, Denton, Thomas, Waldron - Linear Algebra

Real Analysis

- Mathematical Analysis I by Elias Zakon

Discrete Mathematics, Graphs, Algorithms, and Combinatorics

- Levin - Discrete Mathematics - An Open Introduction, 3rd edition
- Github - Discrete Mathematics: an Open Introduction CC BY SA
- Keller, Trotter - Applied Combinatorics (CC-BY-SA 4.0)
- Keller - Github - Applied Combinatorics

Mixed Integer Nonlinear Programming

- Mixed-Integer Nonlinear Optimization Pietro Belotti, Christian Kirches, Sven Leyffer, Jeff Linderoth, Jim Luedtke, and Ashutosh Mahajan

Programming with Python

- A Byte of Python
- Github - Byte of Python (CC-BY-SA)

Also, go to `https://github.com/open-optimization/open-optimization-or-examples` to look at more examples.

Notation

- $\mathbb{1}$ - a vector of all ones (the size of the vector depends on context)
- $\forall$ - for all
- $\exists$ - there exists
- $\in$ - in
- $\therefore$ - therefore
- $\Rightarrow$ - implies
- s.t. - such that (or sometimes "subject to".... from context?)
- $\{0,1\}$ - the set of numbers 0 and 1
- $\mathbb{Z}$ - the set of integers (e.g. $1,2,3,-1,-2,-3,...$)
- $\mathbb{Q}$ - the set of rational numbers (numbers that can be written as p/q for $p,q \in \mathbb{Z}$ (e.g. $1,1/6,27/2$)
- $\mathbb{R}$ - the set of all real numbers (e.g. 1, 1.5, π, e, $-11/5$)
- $\setminus$ - setminus, (e.g. $\{0,1,2,3\} \setminus \{0,3\} = \{1,2\}$)
- $\cup$ - union (e.g. $\{1,2\} \cup \{3,5\} = \{1,2,3,5\}$

- $\cap$ - intersection (e.g. $\{1,2,3,4\} \cap \{3,4,5,6\} = \{3,4\}$)
- $\{0,1\}^4$ - the set of 4 dimensional vectors taking values 0 or 1, (e.g. $[0,0,1,0]$ or $[1,1,1,1]$)
- $\mathbb{Z}^4$ - the set of 4 dimensional vectors taking integer values (e.g., $[1,-5,17,3]$ or $[6,2,-3,-11]$)
- $\mathbb{Q}^4$ - the set of 4 dimensional vectors taking rational values (e.g. $[1.5,3.4,-2.4,2]$)
- $\mathbb{R}^4$ - the set of 4 dimensional vectors taking real values (e.g. $[3,\pi,-e,\sqrt{2}]$)
- $\sum_{i=1}^4 i = 1+2+3+4$
- $\sum_{i=1}^4 i^2 = 1^2+2^2+3^2+4^4$
- $\sum_{i=1}^4 x_i = x_1+x_2+x_3+x_4$
- $\square$ - this is a typical Q.E.D. symbol that you put at the end of a proof meaning "I proved it."
- For $x, y \in \mathbb{R}^3$, the following are equivalent (note, in other contexts, these notations can mean different things)
  - $x^\top y$ *matrix multiplication*
  - $x \cdot y$ *dot product*
  - $\langle x, y\rangle$ *inner product*

  and evaluate to $\sum_{i=1}^3 x_i y_i = x_1y_1 + x_2y_2 + x_3y_3$.

A sample sentence:

$$\forall x \in \mathbb{Q}^n \; \exists y \in \mathbb{Z}^n \setminus \{0\}^n s.t. x^\top y \in \{0,1\}$$

"For all non-zero rational vectors x in n-dimensions, there exists a non-zero n-dimensional integer vector y such that the dot product of x with y evaluates to either 0 or 1."

2. Mathematical Programming

Outcomes

- *Consider an introductory problem*
- *Identify reasons for studying operations research*
- *Define "Mathematical Programming"*
- *Learn about different applications of the tools in this book*
- *Explore the different types of optimization models and what types we will see in this book*

2.1 Warm-Up Scenario

Imagine you're starting a new clothing company and decide to sell shirts. You've opened a store in Roanoke and, based on market research, you estimate the demand for the upcoming weekend:

- **Friday**: 5 shirts
- **Saturday**: 10 shirts
- **Sunday**: 7 shirts

Your supplier can deliver shirts in one day, meaning an order placed on Thursday will arrive on Friday, and so on. As a business owner, you need to decide how many shirts to order and when to place each order to ensure you meet the daily demand without overstocking.

Defining the Problem

To tackle this challenge, you'll need to think strategically about two main factors:

- **Ordering Quantities**: How many shirts to order on each day.
- **Inventory Levels**: How many shirts to keep in stock at the end of each day.

You can define these decisions using variables:

- Let the number of shirts you order on a specific day be a decision variable. For example, you could track the number of shirts ordered on Thursday, Friday, and Saturday.
- Define variables to represent the number of shirts you have in stock at the end of each day.

Your goal is to meet the projected demand while managing costs. Costs could include:

- The expense of ordering shirts.
- The cost of holding inventory.
- Penalties for running out of stock.

Refining the Scenario

As your business grows, additional complexities may arise. For instance:

- The cost of ordering shirts might vary by day due to supplier pricing.
- Holding inventory could incur storage costs.
- A budget might limit how much you can spend on orders.
- You might expand your product line to include pants and socks, each with its own demand and costs.

By clearly defining variables and constraints for these decisions, you can systematically approach the problem and develop an optimal strategy. This step-by-step process of defining the problem, incorporating real-world details, and refining the solution is a fundamental aspect of optimization.

Model Development Cycle

This multi-stage example illustrates the iterative cycle of modeling:

1. **Initial Problem**: Determine ordering quantities to meet demand.
2. **Modeling**: Formulate an optimization model based on available information.
3. **Solution**: Solve the model to find the optimal ordering plan.
4. **New Information**: Incorporate additional real-world complexities (varying costs, inventory costs, budget constraints, additional products, fixed ordering costs).
5. **Refinement**: Update the model to reflect new information and solve again.

Further Extensions

Looking ahead, there are several ways to expand this scenario:

- **Expanding to New Stores**: Opening additional stores introduces location-based demand, transportation costs, and coordination of inventory across multiple sites.
- **Hiring New Staff**: Incorporate labor costs, scheduling constraints, and productivity rates into the model.

- **Producing Clothing In-House**: Transitioning to manufacturing involves modeling production processes, including purchasing raw materials, machine capacities, and processing times.
- **Uncertain Demand**: In reality, demand may be uncertain. Advanced models can incorporate stochastic elements or use probabilistic methods to handle randomness.

In this course, we will focus on deterministic models where all data is known and fixed. Building a strong foundation with these models will prepare us to tackle more complex problems involving uncertainty and randomness in future studies.

2.2 Why Study Operations Research?

Operations Research (OR) is a discipline that applies advanced analytical methods to help make better decisions. In a world where resources are limited and organizations face complex challenges, OR provides the tools and methodologies to model, analyze, and solve problems involving the optimal allocation of scarce resources. Studying OR equips individuals with a systematic and quantitative approach to decision-making, which is invaluable in industries such as logistics, finance, healthcare, and manufacturing.

By understanding OR, you gain the ability to construct mathematical models of real-world systems, allowing for the simulation and optimization of different scenarios. This leads to improved efficiency, reduced costs, and enhanced performance. Moreover, the skills developed through studying OR—such as critical thinking, problem-solving, and data analysis—are highly transferable and sought after in today's data-driven economy.

Operations Research also bridges the gap between theory and practice. It transforms abstract mathematical concepts into actionable strategies that can have a tangible impact on organizational success. Whether it's optimizing supply chains, scheduling flights, or managing portfolios, the applications of OR are vast and impactful.

2.3 What is Mathematical Programming?

Mathematical Programming is a core area within Operations Research that focuses on the formulation and solution of optimization problems. It involves creating mathematical models that represent the decision-making process of a system, where the goal is to find the best possible outcome under given constraints. The term "programming" in this context refers to planning or scheduling, not computer programming.

At the heart of Mathematical Programming is the objective function—a mathematical expression that defines the criterion to be optimized, such as minimizing costs or maximizing profits. The decision variables represent the choices available, and the constraints reflect the limitations or requirements of the problem, such as resource capacities or regulatory conditions.

There are various types of mathematical programming, each suited to different kinds of problems:

- **Linear Programming (LP)**: Deals with problems where both the objective function and constraints are linear.
- **Integer Programming (IP)**: Involves decision variables that must take on integer values.
- **Nonlinear Programming (NLP)**: Handles problems with nonlinear objective functions or constraints.
- **Dynamic Programming (DP)**: Breaks down problems into simpler subproblems and is particularly useful for multistage decision processes.

Mathematical Programming provides a powerful framework for optimizing complex systems and making informed decisions. It combines elements of mathematics, economics, and computer science to develop solutions that are both efficient and effective.

2.4 Applications

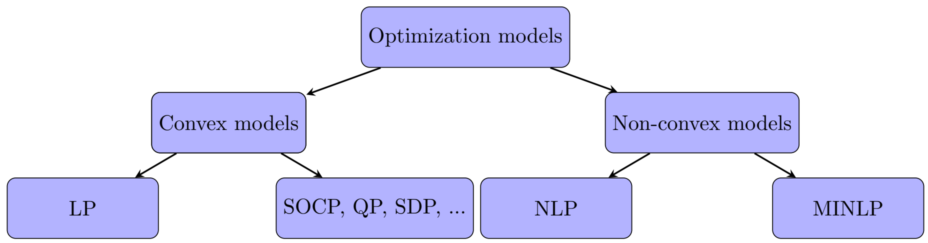
Operations Research and Mathematical Programming have a wide range of applications across various industries and sectors:

- **Transportation and Logistics**: Optimizing routing and scheduling for delivery vehicles, reducing transportation costs, and improving supply chain efficiency.
- **Manufacturing**: Streamlining production processes, managing inventory levels, and scheduling machinery to maximize productivity and minimize waste.
- **Finance**: Portfolio optimization, risk assessment, and capital budgeting to enhance financial performance and manage uncertainties.
- **Healthcare**: Allocating resources such as staff and equipment, scheduling surgeries, and managing patient flow to improve service quality and reduce wait times.
- **Energy Sector**: Optimizing power generation and distribution, planning for renewable energy integration, and managing energy consumption.
- **Telecommunications**: Network design and optimization, bandwidth allocation, and improving the reliability and efficiency of communication systems.
- **Agriculture**: Planning crop rotations, optimizing the use of fertilizers and water, and managing supply chains for agricultural products.
- **Public Sector**: Urban planning, emergency response logistics, and policy analysis to enhance public services and resource utilization.

These applications demonstrate the versatility and impact of OR and Mathematical Programming. By leveraging these tools, organizations can make data-driven decisions that lead to significant improvements in performance, cost savings, and competitive advantage. As global challenges become more complex, the importance of skilled professionals in Operations Research continues to grow.

2.5 Types of Optimization problems

We will state main general problem classes to be associated with in these notes. These are Linear Programming (LP), Mixed-Integer Linear Programming (MILP), Non-Linear Programming (NLP), and Mixed-Integer Non-Linear Programming (MINLP).



Along with each problem class, we will associate a complexity class for the general version of the problem. Complexity classes are discussed in a later portion of the book. Although we will often state that input data for a problem comes from $\mathbb{R}$ (as a real number), when we discuss complexity of such a problem, we actually mean that the data is rational, i.e., from $\mathbb{Q}$ (rational numbers that are encoded in the computer typically), and is given in binary encoding.

CAUTION!

In the following subsections, we will use advanced notation that may be unfamiliar at this point. These mathematical notations will be developed throughout the book.

Furthermore, don't be too concerned about the specific definitions of the complexity classes at this point. For now, you can read that *Polynomial time (P)* means it should be relatively easy to solve, while the other classes might be much more difficult to solve.

2.6 Linear Programming (LP)

Some linear programming background, theory, and examples will be provided in **??**.

Linear Programming (LP):

Polynomial time (P)

Given a matrix $A \in \mathbb{R}^{m\times n}$, vector $b \in \mathbb{R}^m$ and vector $c \in \mathbb{R}^n$, the *linear programming* problem is

$$\begin{aligned} \max \quad & c^\top x \\ \text{s.t.} \quad & Ax \le b \\ & x \ge 0 \end{aligned} \tag{2.1}$$

Linear programming can come in several forms, whether we are maximizing or minimizing, or if the constraints are $\leq, =$ or $\geq$. One form commonly used is *Standard Form* given as

Linear Programming (LP) Standard Form:

Polynomial time (P)

Given a matrix $A \in \mathbb{R}^{m\times n}$, vector $b \in \mathbb{R}^m$ and vector $c \in \mathbb{R}^n$, the *linear programming* problem in *standard form* is

$$\begin{aligned} \max \quad & c^\top x \\ \text{s.t.} \quad & Ax = b \\ & x \geq 0 \end{aligned} \tag{2.2}$$

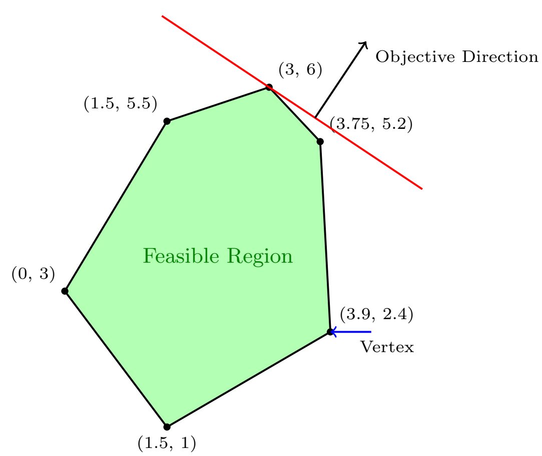


Figure 2.1: Linear programming constraints and objective.

2.7 Mixed-Integer Linear Programming (MILP)

Mixed-integer linear programming will be the focus of Sections **??**,**??**, **??**, and **??**. Recall that the notation $\mathbb{Z}$ means the set of integers and the set $\mathbb{R}$ means the set of real numbers. The first problem of interest here is a *binary integer program* (BIP) where all n variables are binary (either 0 or 1).

Binary Integer programming (BIP):

NP-Complete

Given a matrix $A \in \mathbb{R}^{m \times n}$, vector $b \in \mathbb{R}^m$ and vector $c \in \mathbb{R}^n$, the *binary integer programming* problem is

$$\begin{aligned} \max \quad & c^\top x \\ \text{s.t.} \quad & Ax \leq b \\ & x \in \{0,1\}^n \end{aligned} \tag{2.1}$$

A slightly more general class is the class of *Integer Linear Programs* (ILP). Often this is referred to as *Integer Program* (IP), although this term could leave open the possibility of non-linear parts.

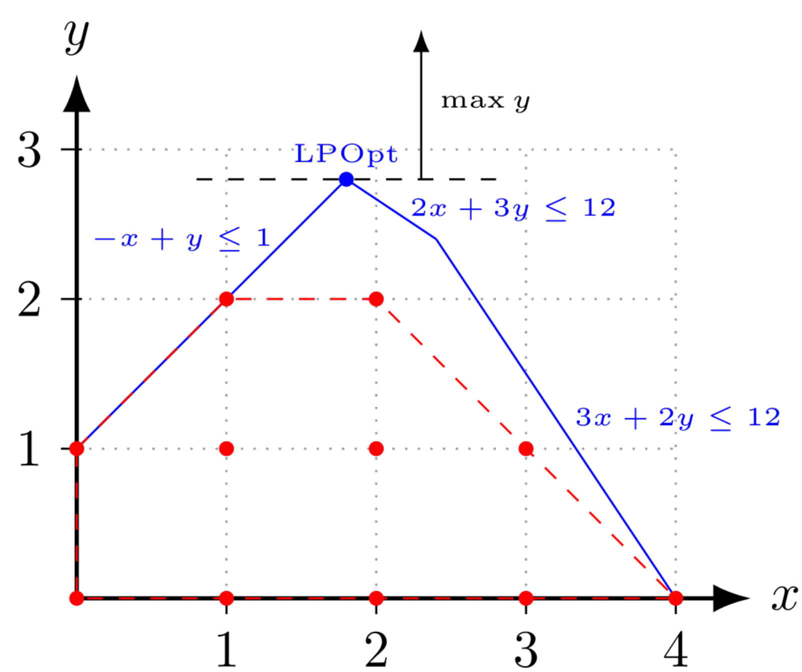


Figure 2.2: Comparing the LP relaxation to the IP solutions.

Figure 2.2

Integer Linear Programming (ILP):

NP-Complete

[1] *IP polytope with LP relaxation*, from `https://en.wikipedia.org/wiki/Integer_programming#/media/File:IP_polytope_with_LP_relaxation.svg`. Fanosta CC BY-SA 4.0., 2019.

Given a matrix $A \in \mathbb{R}^{m\times n}$, vector $b \in \mathbb{R}^m$ and vector $c \in \mathbb{R}^n$, the *integer linear programming* problem is

$$\begin{aligned} \max \quad & c^\top x \\ \text{s.t.} \quad & Ax \leq b \\ & x \in \mathbb{Z}^n \end{aligned} \tag{2.2}$$

An even more general class is *Mixed-Integer Linear Programming (MILP)*. This is where we have n integer variables $x_1,\dots,x_n \in \mathbb{Z}$ and d continuous variables $x_{n+1},\dots,x_{n+d} \in \mathbb{R}$. Succinctly, we can write this as $x \in \mathbb{Z}^n \times \mathbb{R}^d$, where $\times$ stands for the *cross-product* between two spaces.

Below, the matrix A now has $n+d$ columns, that is, $A \in \mathbb{R}^{m\times n+d}$. Also note that we have not explicitly enforced non-negativity on the variables. If there are non-negativity restrictions, this can be assumed to be a part of the inequality description $Ax \leq b$.

Mixed-Integer Linear Programming (MILP):

NP-Complete

Given a matrix $A \in \mathbb{R}^{m\times(n+d)}$, vector $b \in \mathbb{R}^m$ and vector $c \in \mathbb{R}^{n+d}$, the *mixed-integer linear programming* problem is

$$\begin{aligned} \max \quad & c^\top x \\ \text{s.t.} \quad & Ax \leq b \\ & x \in \mathbb{Z}^n \times \mathbb{R}^d \end{aligned} \tag{2.3}$$

2.8 Non-Linear Programming (NLP)

NLP:

NP-Hard

Given a function $f(x)\colon \mathbb{R}^d \to \mathbb{R}$ and other functions $f_i(x)\colon \mathbb{R}^d \to \mathbb{R}$ for $i = 1,\dots,m$, the *nonlinear programming* problem is

$$\begin{aligned} \min \quad & f(x) \\ \text{s.t.} \quad & f_i(x) \leq 0 \quad \text{for } i = 1,\dots,m \\ & x \in \mathbb{R}^d \end{aligned} \tag{2.1}$$

Nonlinear programming can be separated into convex programming and non-convex programming. These two are very different beasts and it is important to distinguish between the two.

2.8.1. Convex Programming

Here the functions are all **convex!**

Convex Programming:

Polynomial time (P) (typically)

Given a convex function $f(x)\colon \mathbb{R}^d \to \mathbb{R}$ and convex functions $f_i(x)\colon \mathbb{R}^d \to \mathbb{R}$ for $i = 1, \dots, m$, the *convex programming* problem is

$$\begin{aligned} \min \quad & f(x) \\ \text{s.t.} \quad & f_i(x) \leq 0 \quad \text{for } i = 1, \dots, m \\ & x \in \mathbb{R}^d \end{aligned} \tag{2.2}$$

Observe that convex programming is a generalization of linear programming. This can be seen by letting $f(x) = c^\top x$ and $f_i(x) = A_i x - b_i$.

2.8.2. Non-Convex Non-linear Programming

When the function f or functions f_i are non-convex, this becomes a non-convex nonlinear programming problem. There are a few complexity issues with this.

IP AS NLP As seen above, quadratic constraints can be used to create a feasible region with discrete solutions. For example

$$x(1-x) = 0$$

has exactly two solutions: $x = 0, x = 1$. Thus, quadratic constraints can be used to model binary constraints.

Binary Integer programming (BIP) as a NLP:

NP-Hard

Given a matrix $A \in \mathbb{R}^{m \times n}$, vector $b \in \mathbb{R}^m$ and vector $c \in \mathbb{R}^n$, the *binary integer programming* problem is

$$
\begin{aligned}
\max \quad & c^\top x \\
\text{s.t.} \quad & Ax \le b \\
& \cancel{x \in \{0,1\}^n} \\
& x_i(1-x_i) = 0 \quad \text{for } i = 1,\dots,n
\end{aligned}
\tag{2.3}
$$

Alternatively, consider the transformation where $x_i \in \{-1,1\}$.

$$
\begin{aligned}
&\min c^\top x \\
&\text{s.t. } Ax \le b \\
&\quad x \in \{-1,1\}^n
\end{aligned}
$$

This can be reformulated with a single nonconvex constraint as

$$
\begin{aligned}
\min \quad & c^\top x \\
\text{s.t.} \quad & Ax \le b \\
& -1 \le x_j \le 1, \quad 1 \le j \le n, \\
& \|x\|^2 \ge n.
\end{aligned}
$$

2.8.3. Machine Learning

Machine learning problems are often cast as continuous optimization problems, which involve adjusting parameters to minimize or maximize a particular objective. Frequently they are convex optimization problems, but many turn out to be nonconvex. Here are two examples of how these problems arise at a glance. We will see examples in greater detail later in the book.

Loss Function Minimization

In supervised learning, this objective is typically a loss function L that quantifies the discrepancy between the predictions of a model and the true data labels. The aim is to adjust the parameters θ of the model to minimize this loss. Mathematically, this can be represented as:

$$
\min_\theta L(\theta) = \min_\theta \frac{1}{N} \sum_{i=1}^{N} l(y_i, f(x_i;\theta)) \tag{2.4}
$$

where N is the number of data points, l is a per-data-point loss (e.g., squared error for regression or cross-entropy for classification), y_i is the true label for the i-th data point, and $f(x_i;\theta)$ is the model's prediction for the i-th data point with parameters θ.

Clustering Formulation

Clustering, on the other hand, seeks to group or partition data points such that data points in the same group are more similar to each other than those in other groups. One popular method is the k-means clustering algorithm. The objective of k-means is to partition the data into k clusters by minimizing the within-cluster sum of squares (WCSS). The mathematical formulation can be given as:

$$\min_{\mathbf{c}_1,\ldots,\mathbf{c}_k} \sum_{j=1}^{k} \sum_{x_i \in C_j} \left\| x_i - \mathbf{c}_j \right\|^2 \tag{2.5}$$

where C_j represents the j-th cluster and $\mathbf{c}_j$ is the centroid of that cluster.

This encapsulation presents a glimpse into how ML problems are framed mathematically. In practice, numerous algorithms, constraints, and regularizations add complexity to these basic formulations.

2.9 Mixed Integer Non-Linear Programming (MINLP)

MINLP:

NP-Hard

Mixed Integer Nonlinear Programming (MINLP) combines elements of integer programming and nonlinear programming. In MINLP, some or all of the decision variables are constrained to be integers, and the objective function or constraints are nonlinear. A general MINLP problem can be formulated as:

$$\begin{aligned} \min \quad & f(x,y) \\ \text{s.t.} \quad & g_i(x,y) \leq 0 \quad \text{for } i = 1,\ldots,m \\ & h_j(x,y) = 0 \quad \text{for } j = 1,\ldots,p \\ & x \in \mathbb{R}^n, y \in \mathbb{Z}^k \end{aligned} \tag{2.1}$$

where f is the objective function, g_i and h_j are constraint functions, x is a vector of continuous variables, and y is a vector of integer variables.

MINLP is particularly challenging due to the nonlinearity in the objective and/or constraints and the discrete nature of some decision variables. It finds applications in various fields such as industry for process optimization, computational geometry, and machine learning for hyperparameter tuning.

2.9.1. Convex MINLP

In the convex case, both the objective function and the constraints are convex functions. This subclass is easier to solve compared to its non-convex counterpart.

Convex MINLP:

NP-Hard, but *Polynomial time (P)* in fixed dimension (typically)

For a convex MINLP, the problem is defined as:

$$\begin{aligned} \min \quad & f(x,y) \quad \text{(convex)} \\ \text{s.t.} \quad & g_i(x,y) \leq 0 \quad \text{(convex constraints)} \\ & x \in \mathbb{R}^n, y \in \mathbb{Z}^k \end{aligned} \tag{2.2}$$

Convex MINLPs, while still challenging, are typically more tractable due to the properties of convexity, which allow for more efficient solution methods.

2.9.2. Non-Convex MINLP

Non-convex MINLPs are significantly harder due to the presence of non-convex functions, which can lead to multiple local minima.

Non-Convex MINLP:

NP-Hard(in fact, undecidable)

The non-convex MINLP problem is formulated as:

$$\begin{aligned} \min \quad & f(x,y) \quad \text{(non-convex)} \\ \text{s.t.} \quad & g_i(x,y) \leq 0 \quad \text{(possibly non-convex constraints)} \\ & x \in \mathbb{R}^n, y \in \mathbb{Z}^k \end{aligned} \tag{2.3}$$

Non-convex MINLPs pose significant computational challenges due to the possibility of multiple local optima and the inherent complexity of integer constraints. These problems are common in real-world applications where decisions are discrete, and the system behavior is non-linear and complex.

Complexity and Applications

MINLP problems are known for their computational complexity, primarily due to the combination of non-linearity and integrality. The non-convex variants, in particular, are *NP-Hard*, making them some of the most challenging problems in optimization.

In practice, MINLP models find extensive applications across various domains. In industry, they are used for complex decision-making processes like supply chain optimization and production planning. In computational geometry, MINLP techniques help in solving problems like optimal packing or layout design. Furthermore, in the realm of machine learning, MINLPs are crucial for tasks like feature selection and hyperparameter optimization where discrete choices and nonlinear relationships are involved.

The versatility of MINLP models, combined with their inherent complexity, makes them a fascinating and essential area of study in the field of optimization.

Category	**Software (License)**
Linear programming (free)	CDD, CLP, GLPK, LPSOLVE, OSQP, SCIP
Mixed Integer Linear programming (free)	CBC, GLPK, LPSOLVE, SCIP
Linear programming (commercial)	COPT (free for academia), CPLEX (free for academia), GUROBI (free for academia), KNITRO, LINPROG, MOSEK (free for academia), XPRESS (free for academia)
Mixed Integer Linear programming (commercial)	COPT (free for academia), CPLEX (free for academia), GUROBI (free for academia), INTLINPROG, KNITRO, MOSEK (free for academia), XPRESS (free for academia)
Quadratic programming (free)	OSQP, CLP, DAQP (mixed-binary), OOQP, QPC, QPOASES, QUADPROGBB (nonconvex QP)
Quadratic programming (commercial)	COPT (free for academia), CPLEX (free for academia), GUROBI (free for academia), KNITRO, MOSEK (free for academia), QUADPROG, XPRESS (free for academia)
Mixed Integer Quadratic programming (commercial)	COPT (free for academia), CPLEX (free for academia), GUROBI (free for academia), KNITRO, MOSEK (free for academia), XPRESS (free for academia)
Second-order cone programming (free)	ECOS, SCS, SDPT3, SEDUMI
Second-order cone programming (commercial)	COPT (free for academia), CPLEX (free for academia), CONEPROG, GUROBI (free for academia), MOSEK (free for academia), XPRESS (free for academia)
Mixed Integer Second-order cone programming (commercial)	COPT (free for academia), CPLEX (free for academia), GUROBI (free for academia), MOSEK (free for academia), XPRESS (free for academia)
Semidefinite programming (free)	CSDP, DSDP, LOGDETPPA, PENLAB, SCS, SDPA, SDPLR, SDPT3, SDPNAL, SEDUMI
Semidefinite programming (commercial)	COPT (free for academia), LMIlab (not recommended), MOSEK (free for academia), PENBMI, PENSDP (free for academia)
General nonlinear programming and other solvers	BARON, FILTERSD, FMINCON, GPPSOY, IPOPT, KNITRO, LMIrank, MPT, NOMAD, PENLAB, SNOPT, SPARSEPOP

Table 2.1: Optimization Software and their Licenses

2.10 Optimization Under Uncertainty

Optimization problems often involve uncertainty in their parameters, whether due to measurement errors, variability in input data, or unpredictable future events. Addressing uncertainty is crucial for developing solutions that are effective in real-world settings. Two primary approaches to optimization under uncertainty are stochastic optimization and robust optimization.

2.10.1. Stochastic Optimization

Stochastic optimization deals with problems where some parameters are uncertain but can be described probabilistically. In this framework, the optimization process incorporates the expected values, variances, or other statistical characteristics of uncertain parameters. Solutions are often evaluated based on their performance over a distribution of possible scenarios.

For example, in supply chain management, uncertain demand can be modeled using probability distributions. A company may aim to minimize costs while ensuring a high service level under varying demand conditions. Similarly, in finance, portfolio optimization often accounts for uncertain future returns by maximizing expected returns while managing risk.

2.10.2. Robust Optimization

Robust optimization, on the other hand, focuses on creating solutions that perform well across the worst-case realizations of uncertainty. Rather than assuming a specific probability distribution, robust optimization works with uncertainty sets, which define the range of possible values that uncertain parameters can take. The goal is to find solutions that are feasible and effective for all possible scenarios within the uncertainty set.

Applications of robust optimization include power grid operations, where uncertainties in demand and renewable energy generation need to be managed, and scheduling problems, such as airline crew scheduling, where disruptions must be minimized under varying operational conditions.

2.10.3. Applications

Optimization under uncertainty has a wide array of applications across industries:

- **Healthcare:** Designing vaccination strategies to minimize the spread of diseases while accounting for uncertain infection rates and patient behaviors.
- **Transportation:** Managing traffic flow or public transit schedules under uncertain demand and travel times.
- **Energy:** Planning renewable energy investments and operations under uncertain weather conditions and energy prices.

- **Manufacturing:** Ensuring production schedules are resilient to supply chain disruptions and variability in raw material quality.

Addressing the complexity of uncertain data typically makes these problems much harder than their deterministic counterparts. While these methods are powerful and applicable to a wide range of real-world scenarios, we will not focus on optimization under uncertainty in this textbook.

Part I

Linear Programming

3. Modeling: Linear Programming

Outcomes

1. *Define what a linear program is*
2. *Understand how to model a linear program*
3. *View many examples and get a sense of what types of problems can be modeled as linear programs.*

Linear Programming, also known as Linear Optimization, is the starting point for most forms of optimization. It is the problem of optimization a linear function over linear constraints.

In this chapter, we will define what this means, how to setup a linear program, and discuss many examples. Examples will be connected with code in Excel and Python (using with PuLP or Gurobipy modeling tools) so that you can easily start solving optimization problems. Tutorials on these tools will come in later chapters.

As a warm up, consider the following word problem.
Question: Find the age of Justin and Brad, given that the sum of their ages is equal to 76 and after 7 years, the age of Brad will be twice the age of Justin.

Solution: Let x and y denote the age of Justin and Brad, respectively. Then

$$x + y = 76 \tag{1}$$
$$2(x+7) = y + 7. \tag{2}$$

On solving the foregoing equations we get

$$x = 23 \quad \text{and} \quad y = 53.$$

We will need to think about assigning variables to decisions and instead of equations guiding these variables, we may have inequalites that bound these variables in different ways. Thus, there can be many possible solutions.

3.1 Introductory LP Example

We begin with a simple example.

Example 3.1: Toy Maker Excel PuLP Gurobipy

Consider the problem of a toy company that produces toy planes and toy boats. The toy company can sell its planes for \$10 and its boats for \$8 dollars. It costs \$3 in raw materials to make a plane and \$2 in raw materials to make a boat. A plane requires 3 hours to make and 1 hour to finish while a boat requires 1 hour to make and 2 hours to finish. The toy company knows it will not sell anymore than 35 planes per week. Further, given the number of workers, the company cannot spend anymore than 160 hours per week finishing toys and 120 hours per week making toys. The company wishes to maximize the profit it makes by choosing how much of each toy to produce.

We can represent the profit maximization problem of the company as a linear programming problem. Let x_1 be the number of planes the company will produce and let x_2 be the number of boats the company will produce. The profit for each plane is $\$10 - \$3 = \$7$ per plane and the profit for each boat is $\$8 - \$2 = \$6$ per boat. Thus the total profit the company will make is:

$$z(x_1, x_2) = 7x_1 + 6x_2 \tag{3.1}$$

The company can spend no more than 120 hours per week making toys and since a plane takes 3 hours to make and a boat takes 1 hour to make we have:

$$3x_1 + x_2 \leq 120 \tag{3.2}$$

Likewise, the company can spend no more than 160 hours per week finishing toys and since it takes 1 hour to finish a plane and 2 hour to finish a boat we have:

$$x_1 + 2x_2 \leq 160 \tag{3.3}$$

Finally, we know that $x_1 \leq 35$, since the company will make no more than 35 planes per week. Thus the complete linear programming problem is given as:

$$\left\{\begin{aligned} \max \quad & z(x_1, x_2) = 7x_1 + 6x_2 \\ s.t. \quad & 3x_1 + x_2 \leq 120 \\ & x_1 + 2x_2 \leq 160 \\ & x_1 \leq 35 \\ & x_1 \geq 0 \\ & x_2 \geq 0 \end{aligned}\right. . \tag{3.4}$$

Exercise 3.1: Chemical Manufacturing

A chemical manufacturer produces three chemicals: A, B and C. These chemical are produced by two processes: 1 and 2.

- *Running process 1 for 1 hour costs \$4 and yields 3 units of chemical A, 1 unit of chemical B and 1 unit of chemical C.*
- *Running process 2 for 1 hour costs \$1 and produces 1 units of chemical A, and 1 unit of chemical B (but none of Chemical C).*

To meet customer demand,

- *at least 10 units of chemical A,*
- *at least 5 units of chemical B and*
- *at least 3 units of chemical C*

must be produced daily.

Assume that the chemical manufacturer wants to minimize the cost of production.
Develop a linear programming problem describing the constraints and objectives of the chemical manufacturer.

[Hint: Let x_1 be the amount of time Process 1 is executed and let x_2 be amount of time Process 2 is executed. Use the coefficients above to express the cost of running Process 1 for x_1 time and Process 2 for x_2 time. Do the same to compute the amount of chemicals A, B, and C that are produced.]

3.2 Modeling and Assumptions in Linear Programming

Outcomes

1. *Address crucial assumptions when chosing to model a problem with linear programming.*

A Generic Linear Program (LP)

Linear programming (LP) is a mathematical optimization framework widely used in decision-making and resource allocation. In its most basic form, an LP seeks to optimize (maximize or minimize) a linear objective function subject to a set of linear constraints. These constraints typically represent limitations or requirements in real-world scenarios, such as budgets, capacities, or resource availability.

Decision Variables:
The decision variables represent the quantities we want to determine. For example:

- x_i: Continuous variables ($x_i \in \mathbb{R}$, i.e., a real number), $\forall i = 1, \ldots, n$.

These variables could represent amounts of products to produce, resources to allocate, or investments to make. We can use any letters to represent these variables. Typically we use x_i, but frequently will use y_i or other letters depending on the context and what makes reading the problem most convenient.

Parameters (Known Input Parameters):
Parameters are the given data in the problem, which are assumed to be known in advance:

- c_i: Cost coefficients for each decision variable, $\forall i = 1, \ldots, n$. These values represent the contribution of each variable to the objective function.
- a_{ij}: Constraint coefficients, $\forall i = 1, \ldots, n$ and $j = 1, \ldots, m$. These define the relationships between decision variables and the constraints.
- b_j: Right-hand side values of the constraints, $\forall j = 1, \ldots, m$. These represent limits or requirements for each constraint.

The general form of a linear program can be written as follows:

$$\begin{aligned} \max \quad & z = c_1x_1 + \cdots + c_nx_n \\ s.t. \quad & a_{11}x_1 + \cdots + a_{1n}x_n \leq b_1 \\ & \vdots \\ & a_{m1}x_1 + \cdots + a_{mn}x_n \leq b_m \\ & x_i \geq 0 \quad \forall i. \end{aligned} \tag{3.1}$$

Features of a Linear Program

- **Objective Function**: The linear function $z = c_1x_1 + \cdots + c_nx_n$ represents the goal to optimize. This could involve maximizing profit, minimizing cost, or achieving some other measurable outcome.
- **Constraints**: The system of inequalities (or equalities) restricts the values of the decision variables. Each constraint reflects a real-world limitation, such as resource availability or capacity limits.
- **Feasibility**: The set of all possible solutions that satisfy the constraints is called the feasible region. The optimal solution must lie within this region.
- **Linear Relationships**: Both the objective function and constraints are linear, meaning the relationships between variables are additive and proportional.
- **Sets**: For specific optimization problem, it can be useful to define sets of variables or pramaters to help write down the problem in a generic forumlation.

Example of a Linear Program

To illustrate, consider a specific example with three decision variables and four constraints. This includes a mix of constraint types ($\leq$, $\geq$, and $=$), showcasing the versatility of linear programming:

Decision Variables:
x_i: Continuous variables ($x_i \in \mathbb{R}$, i.e., a real number), $\forall i = 1, \dots, 3$.

Parameters (Known Input Parameters):

- c_i: Cost coefficients, $\forall i = 1, \dots, 3$.
- a_{ij}: Constraint coefficients, $\forall i = 1, \dots, 3$ and $j = 1, \dots, 4$.
- b_j: Right-hand side values for each constraint, $\forall j = 1, \dots, 4$.

The linear program can be written as:

$$\begin{aligned}
\text{Minimize } & z = c_1x_1 + c_2x_2 + c_3x_3 && (3.2)\\
\text{Subject to } & a_{11}x_1 + a_{12}x_2 + a_{13}x_3 \geq b_1 && (3.3)\\
& a_{21}x_1 + a_{22}x_2 + a_{23}x_3 \leq b_2 && (3.4)\\
& a_{31}x_1 + a_{32}x_2 + a_{33}x_3 = b_3 && (3.5)\\
& a_{41}x_1 + a_{42}x_2 + a_{43}x_3 \geq b_4 && (3.6)\\
& x_1 \geq 0,\ x_2 \leq 0,\ x_3 \text{ unrestricted.} && (3.7)
\end{aligned}$$

A key feature of linear programming is the use of linear functions, defined as follows:

Definition 3.2: Linear Function

A function $z : \mathbb{R}^n \to \mathbb{R}$ is linear if there are constants $c_1, \dots, c_n \in \mathbb{R}$ such that:

$$z(x_1, \dots, x_n) = c_1x_1 + \cdots + c_nx_n. \tag{3.8}$$

Linear functions are characterized by their simplicity and additivity, making them well-suited for optimization problems where proportionality is assumed. This simplicity allows linear programs to be solved efficiently, even for large-scale problems.

3.2.1. Assumptions

Inspecting Example 3.1 (or the more general problem in Equation (3.1)) we can see there are several assumptions that must be satisfied when using a linear programming model. We enumerate these below:

Proportionality Assumption A problem can be phrased as a linear program only if the contribution to the objective function *and* the left-hand-side of each constraint by each decision variable $(x_1, \dots, x_n)$ is proportional to the value of the decision variable.

Additivity Assumption A problem can be phrased as a linear programming problem only if the contribution to the objective function *and* the left-hand-side of each constraint by any decision variable x_i $(i = 1, \dots, n)$ is completely independent of any other decision variable x_j $(j \neq i)$ and additive.

Divisibility Assumption A problem can be phrased as a linear programming problem only if the quantities represented by each decision variable are infinitely divisible (i.e., fractional answers make sense).

Certainty Assumption A problem can be phrased as a linear programming problem only if the coefficients in the objective function and constraints are known with certainty.

The first two assumptions simply assert (in English) that both the objective function and functions on the left-hand-side of the (in)equalities in the constraints are linear functions of the variables $x_1, \dots, x_n$.

The third assumption asserts that a valid optimal answer could contain fractional values for decision variables. It's important to understand how this assumption comes into play–even in the toy making example. Many quantities can be divided into non-integer values (ounces, pounds etc.) but many other quantities cannot be divided. For instance, can we really expect that it's reasonable to make $\frac{1}{2}$ a plane in the toy making example? When values must be constrained to true integer values, the linear programming problem is called an *integer programming problem*. There is a *vast* literature dealing with these problems. For many problems, particularly when the values of the decision variables may become large, a fractional optimal answer could be obtained and then rounded to the nearest integer to obtain a reasonable answer. For example, if our toy problem were re-written so that the optimal answer was to make 1045.3 planes, then we could round down to 1045.

The final assumption asserts that the coefficients (e.g., profit per plane or boat) is known with absolute certainty. In traditional linear programming, there is no lack of knowledge about the make up of the objective function, the coefficients in the left-hand-side of the constraints or the bounds on the right-hand-sides of the constraints. There is a literature on *stochastic programming* that relaxes some of these assumptions, but this too is outside the scope of the course.

Exercise 3.3

In a short sentence or two, discuss whether the problem given in Example 3.1 meets all of the assumptions of a scenario that can be modeled by a linear programming problem. Do the same for Exercise 3.1.
[Hint: Can you make $\frac{2}{3}$ of a toy? Can you run a process for $\frac{1}{3}$ of an hour?]

3.3 Examples

Outcomes

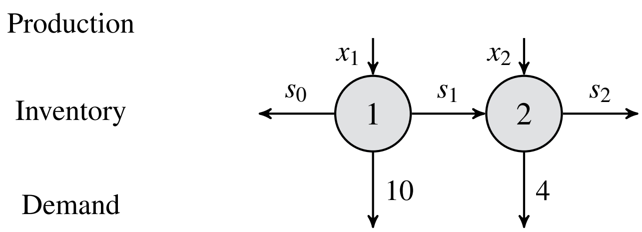
1. *Identify decision variables and paramters*
2. *Learn how to format a linear optimization problem.*
3. *Identify and understand common classes of linear optimization problems.*

We will begin with a few examples, and then discuss specific problem types that occur often.

Example 3.2: Production Planning Excel PuLP Gurobipy

A manufacturing firm seeks to optimize its production schedule over a two-day planning horizon to meet daily demand while minimizing costs. The firm starts with an initial inventory of s_0 units. The costs associated with production and inventory are as follows:

- **Production costs:** \$10 per unit on Day 1, and \$16 per unit on Day 2.
- **Inventory cost:** \$5 per unit for carrying excess inventory from one day to the next.



The goal is to determine the number of units to produce each day and the inventory levels to minimize total costs while satisfying the demand for each day.

Solution. Sets:

- Days of production $= \{1, 2\}$.

Parameters:

- c_{p1}: Per unit production cost on Day 1 (\$10).
- c_{p2}: Per unit production cost on Day 2 (\$16).
- c_{inv}: Per unit inventory cost (\$5).
- d_1: Demand on Day 1 (10 units).
- d_2: Demand on Day 2 (4 units).
- s_0: Inventory at the start of Day 1.

Decision variables:

- x_1: Number of units produced on Day 1.
- x_2: Number of units produced on Day 2.
- s_1: Inventory at the end of Day 1.
- s_2: Inventory at the end of Day 2.

Objective and Constraints:

$$\begin{aligned}
\min \quad & z = 10x_1 + 16x_2 + 5s_1 + 5s_2 && \text{(Total cost)}\\
\text{s.t.} \quad & s_0 + x_1 = 10 + s_1 && \text{(Day 1 balance)}\\
& s_1 + x_2 = 4 + s_2 && \text{(Day 2 balance)}\\
& x_1, x_2, s_0, s_1, s_2 \geq 0 && \text{(Non-negativity constraints).}
\end{aligned}$$

♠

Definition 3.4: Optimization model components

When defining an optimization model, it is essential to identify and clearly articulate the following five components:

- ***Sets:*** *What lists of data do you need to write down? These could include indices, categories, or groups relevant to the problem.*
- ***Parameters (Data):*** *What is the actual data for the problem? These are the fixed numerical values such as costs, capacities, or resource availabilities.*
- ***Decision Variables:*** *What decisions need to be made to provide a solution? These variables represent the quantities that can be controlled or adjusted, such as production levels or resource allocations.*
- ***Objective Function:*** *The expression that we want to maximize or minimize. This is the goal of the optimization, such as minimizing cost or maximizing profit.*
- ***Constraints:*** *Certain rules that those decision variables should satisfy. These include limitations or requirements, such as resource capacities or demand fulfillment.*

Each of these components plays a crucial role in formulating a well-structured optimization model that accurately represents the problem and provides actionable solutions. The role of Sets will become more important as we get into more complicated optimization models.

Example 3.3: Production with welding robot Excel PuLP Gurobipy

You have 21 units of transparent aluminum alloy (TAA), LazWeld1, a joining robot leased for 23 hours, and CrumCut1, a cutting robot leased for 17 hours of aluminum cutting. You also have production code for a bookcase, desk, and cabinet, along with commitments to buy any of these you can produce for \$18, \$16, and \$10 apiece, respectively. A bookcase requires 2 units of TAA, 3 hours of joining, and 1 hour of cutting, a desk requires 2 units of TAA, 2 hours of joining, and 2 hour of cutting, and a cabinet requires 1 unit of TAA, 2 hours of joining, and 1 hour of cutting.

Formulate an LP to maximize your revenue given your current resources.

Solution.

Sets:

- The types of objects = { bookcase, desk, cabinet}.

Parameters:

- Purchase cost of each object
- Units of TAA needed for each object
- Hours of joining needed for each object
- Hours of cutting needed for each object
- Hours of TAA, Joining, and Cutting available on robots

Decision variables:

x_i : number of units of product i to produce,
for all i =bookcase, desk, cabinet.

Objective and Constraints:

$$
\begin{aligned}
\max \quad & z = 18x_1 + 16x_2 + 10x_3 && \text{(profit)}\\
s.t. \quad & 2x_1 + 2x_2 + 1x_3 \leq 21 && (TAA)\\
& 3x_1 + 2x_2 + 2x_3 \leq 23 && (LazWeld1)\\
& 1x_1 + 2x_2 + 1x_3 \leq 17 && (CrumCut1)\\
& x_1, x_2, x_3 \geq 0.
\end{aligned}
$$

♠

Example 3.4: The Diet Problem Excel PuLP Gurobipy

In the future (as envisioned in a bad 70's science fiction film) all food is in tablet form, and there are four types, green, blue, yellow, and red. A balanced, futuristic diet requires, at least 20 units of Iron, 25 units of Vitamin B, 30 units of Vitamin C, and 15 units of Vitamin D. Formulate an LP that ensures a balanced diet at the minimum possible cost.

Tablet	Iron	B	C	D	Cost ($)
Chem 1	6	6	7	4	1.25
Chem 2	4	5	4	9	1.05
Chem 3	5	2	5	6	0.85
Chem 4	3	6	3	2	0.65

Solution. Sets:

- Set of tablets $\{1, 2, 3, 4\}$

Parameters:

- Iron in each tablet
- Vitamin B in each tablet
- Vitamin C in each tablet
- Vitamin D in each tablet
- Cost of each tablet

Decision variables:
x_i : number of tablet of type i to include in the diet, $\forall i \in \{1,2,3,4\}$.

Objective and Constraints:

$$\begin{array}{lr} \text{Min} & z = 1.25x_1 + 1.05x_2 + 0.85x_3 + 0.65x_4 \\ \text{s.t.} & 6x_1 + 4x_2 + 5x_3 + 3x_4 \geq 20 \\ & 6x_1 + 5x_2 + 2x_3 + 6x_4 \geq 25 \\ & 7x_1 + 4x_2 + 5x_3 + 3x_4 \geq 30 \\ & 4x_1 + 9x_2 + 6x_3 + 2x_4 \geq 15 \\ & x_1, x_2, x_3, x_4 \geq 0. \end{array}$$

♠

Example 3.5: The Next Diet Problem Excel PuLP Gurobipy

Progress is important, and our last problem had too many tablets, so we are going to produce a single, purple, 10 gram tablet for our futuristic diet requires, which are at least 20 units of Iron, 25 units of Vitamin B, 30 units of Vitamin C, and 15 units of Vitamin D, and 2000 calories. The tablet is made from blending 4 nutritious chemicals; the following table shows the units of our nutrients per, and cost of, grams of each chemical.

Formulate an LP that ensures a balanced diet at the minimum possible cost.

Tablet	Iron	B	C	D	Calories	Cost ($)
Chem 1	6	6	7	4	1000	1.25
Chem 2	4	5	4	9	250	1.05
Chem 3	5	2	5	6	850	0.85
Chem 4	3	6	3	2	750	0.65

Solution.

Sets:

- Set of chemicals $\{1,2,3,4\}$

Parameters:

- Iron in each chemical
- Vitamin B in each chemical
- Vitamin C in each chemical
- Vitamin D in each chemical
- Cost of each chemical

Decision variables:
x_i : grams of chemical i to include in the purple tablet, $\forall i = 1,2,3,4$.
Objective and Constraints:

$$\begin{aligned}
\min \quad & z = 1.25x_1 + 1.05x_2 + 0.85x_3 + 0.65x_4 \\
s.t. \quad & 6x_1 + 4x_2 + 5x_3 + 3x_4 \geq 20 \\
& 6x_1 + 5x_2 + 2x_3 + 6x_4 \geq 25 \\
& 7x_1 + 4x_2 + 5x_3 + 3x_4 \geq 30 \\
& 4x_1 + 9x_2 + 6x_3 + 2x_4 \geq 15 \\
& 1000x_1 + 250x_2 + 850x_3 + 750x_4 \geq 2000 \\
& x_1 + x_2 + x_3 + x_4 = 10 \\
& x_1, x_2, x_3, x_4 \geq 0.
\end{aligned}$$

♠

Example 3.6: Work Scheduling Problem Excel PuLP Gurobipy

You are the manager of LP Burger. The following table shows the minimum number of employees required to staff the restaurant on each day of the week. Each employees must work for five consecutive days. Formulate an LP to find the minimum number of employees required to staff the restaurant.

Day of Week	Workers Required
1 = Monday	6
2 = Tuesday	4
3 = Wednesday	5
4 = Thursday	4
5 = Friday	3
6 = Saturday	7
7 = Sunday	7

Solution. This problem has multiple optimal solutions for which days workers begin working on, all of which result in 8 total workers hired.

Decision variables:
x_i : the number of workers that start 5 consecutive days of work on day i, $i = 1, \ldots, 7$
Objective and Constraints:

$$
\begin{array}{lr}
\text{Min} & z = x_1 + x_2 + x_3 + x_4 + x_5 + x_6 + x_7 \\
\text{s.t.} & x_1 + x_4 + x_5 + x_6 + x_7 \geq 6 \\
& x_2 + x_5 + x_6 + x_7 + x_1 \geq 4 \\
& x_3 + x_6 + x_7 + x_1 + x_2 \geq 5 \\
& x_4 + x_7 + x_1 + x_2 + x_3 \geq 4 \\
& x_5 + x_1 + x_2 + x_3 + x_4 \geq 3 \\
& x_6 + x_2 + x_3 + x_4 + x_5 \geq 7 \\
& x_7 + x_3 + x_4 + x_5 + x_6 \geq 7 \\
& x_1, x_2, x_3, x_4, x_5, x_6, x_7 \geq 0.
\end{array}
$$

One solution is as follows:

LP Solution	IP Solution
$z_{LP} = 7.333$	$z_I = 8.0$
$x_1 = 0$	$x_1 = 0$
$x_2 = 0.333$	$x_2 = 0$
$x_3 = 1$	$x_3 = 0$
$x_4 = 2.333$	$x_4 = 3$
$x_5 = 0$	$x_5 = 0$
$x_6 = 3.333$	$x_6 = 4$
$x_7 = 0.333$	$x_7 = 1$

♠

Example 3.7: LP Burger - extended Excel PuLP Gurobipy

LP Burger has changed its policy, and allows, at most, two part time workers, who work for two consecutive days in a week. Formulate this problem.

Solution. Decision variables:
x_i : the number of workers that start 5 consecutive days of work on day i, $i = 1, \cdots, 7$
y_i : the number of workers that start 2 consecutive days of work on day i, $i = 1, \cdots, 7$.

Objective and Constraints:

$$\begin{aligned}
\text{Min} \quad & z = 5(x_1 + x_2 + x_3 + x_4 + x_5 + x_6 + x_7) \\
& \quad +2(y_1 + y_2 + y_3 + y_4 + y_5 + y_6 + y_7) \\
\text{s.t.} \quad & x_1 + x_4 + x_5 + x_6 + x_7 + y_1 + y_7 \geq 6 \\
& x_2 + x_5 + x_6 + x_7 + x_1 + y_2 + y_1 \geq 4 \\
& x_3 + x_6 + x_7 + x_1 + x_2 + y_3 + y_2 \geq 5 \\
& x_4 + x_7 + x_1 + x_2 + x_3 + y_4 + y_3 \geq 4 \\
& x_5 + x_1 + x_2 + x_3 + x_4 + y_5 + y_4 \geq 3 \\
& x_6 + x_2 + x_3 + x_4 + x_5 + y_6 + y_5 \geq 7 \\
& x_7 + x_3 + x_4 + x_5 + x_6 + y_7 + y_6 \geq 7 \\
& y_1 + y_2 + y_3 + y_4 + y_5 + y_6 + y_7 \leq 2 \\
& x_i \geq 0, y_i \geq 0, \forall i = 1, \cdots, 7.
\end{aligned}$$

♠

3.3.1. Knapsack Problem

We consider now a simple example where the variables can only take the values of 0 or 1. We call these *binary variables*.

Example 3.8: Camping Trip Excel PuLP Gurobipy

Imagine you are preparing for a week-long camping trip to the mountains. You have a backpack with a weight capacity of 20 kilograms. Your goal is to pack the most valuable items to ensure a comfortable and safe trip without exceeding the backpack's weight limit. Each item has a weight and a value associated with it, representing its importance and utility for the trip. The table below lists the potential items you can take, their weights in kilograms, and their values on a scale from 1 to 10 (with 10 being the most valuable).
Formulate an Integer Program to find the optimal items you should bring on your trip with you.

Item	Index	Weight (kg)	Value
Tent	A	4	10
Stove	B	2	9
Sleeping Bag	C	3	8
First Aid Kit	D	1	7
Food Supplies	E	5	9
Water Filter	F	1	6
Clothes	G	4	5
Map and Compass	H	0.5	7

Solution. Decision variables:
x_i : whether to include item i in the backpack, where $i \in \{A,B,C,D,E,F,G,H\}$, $x_i = 1$ if item i is included, and $x_i = 0$ otherwise.

Objective and Constraints:

$$\begin{aligned}
\text{Max} \quad & z = 10x_A + 9x_B + 8x_C + 7x_D + 9x_E + 6x_F + 5x_G + 7x_H \\
\text{s.t.} \quad & 4x_A + 2x_B + 3x_C + 1x_D + 5x_E + 1x_F + 4x_G + 0.5x_H \leq 20 \\
& x_A, x_B, x_C, x_D, x_E, x_F, x_G, x_H \in \{0,1\}.
\end{aligned}$$

♠

3.3.2. Capital Investment

We next see a similar application of binary variables.

Example 3.9: Capital Allocation Problem Excel PuLP Gurobipy

You are a financial planner for an investment firm. The firm has \$100,000 to invest in a portfolio of projects. Each project has a projected return and requires an initial investment. The goal is to maximize the total return of the portfolio while not exceeding the available capital. The following table lists potential projects, their required investments, and their projected returns.

Formulate an LP to maximize the total return of the portfolio while not exceeding the available investment capital.

Project	Investment Required (\$)	Projected Return (\$)
A	20,000	30,000
B	30,000	40,000
C	50,000	75,000
D	10,000	15,000
E	40,000	60,000

Solution.

Decision variables:
x_i : whether to include project i in the portfolio, where $i \in \{A,B,C,D,E\}$, $x_i = 1$ if project i is included, and $x_i = 0$ otherwise.

Objective and Constraints:

$$\begin{aligned}
\text{Max} \quad & z = 30,000x_A + 40,000x_B + 75,000x_C + 15,000x_D + 60,000x_E \\
\text{s.t.} \quad & 20,000x_A + 30,000x_B + 50,000x_C + 10,000x_D + 40,000x_E \leq 100,000 \\
& x_A, x_B, x_C, x_D, x_E \in \{0,1\}.
\end{aligned}$$

♠

3.3.3. Assignment Problem

Consider the assignment of 3 employees to 3 tasks, where each employee has a cost to do each ask. How to figure out the best assignment to the tasks?

Example 3.10: Hiring for tasks Excel PuLP Gurobipy

In this assignment problem, we need to hire three people (Person 1, Person 2, Person 3) to three tasks (Task 1, Task 2, Task 3). In the table below, we list the cost of hiring each person for each task, in dollars. Since each person has a different cost for each task, we must make an assignment to minimize our total cost.

Cost	Task 1	Task 2	Task 3
Person 1	40	47	80
Person 2	72	36	58
Person 3	24	61	71

Given the specific costs of assigning three people to three tasks, we can write out the mathematical model explicitly using the given numbers.

Solution. **Variables:**. Let $x_{ij} = 1$ if person i is assigned to task j and 0 otherwise.

Objective Function:

Minimize the total cost of assignments:

$$z = 40x_{11} + 47x_{12} + 80x_{13} + 72x_{21} + 36x_{22} + 58x_{23} + 24x_{31} + 61x_{32} + 71x_{33} \tag{3.1}$$

Subject to Constraints:

Each person is assigned to exactly one task:

$$x_{11} + x_{12} + x_{13} = 1 \tag{3.2}$$
$$x_{21} + x_{22} + x_{23} = 1 \tag{3.3}$$
$$x_{31} + x_{32} + x_{33} = 1 \tag{3.4}$$

Each task is assigned to exactly one person:

$$x_{11} + x_{21} + x_{31} = 1 \tag{3.5}$$
$$x_{12} + x_{22} + x_{32} = 1 \tag{3.6}$$
$$x_{13} + x_{23} + x_{33} = 1 \tag{3.7}$$

Binary constraints on the variables:

$$x_{ij} \in \{0,1\} \quad \forall i \in \{1,2,3\}, \forall j \in \{1,2,3\} \tag{3.8}$$

♠

4. Software - Excel

Outcomes

- *Learn relevant excel functions*
- *Setup linear programs in excel*
- *Solve linear programs in excel*

4.1 Using Excel Solver

Excel Solver is a powerful tool for solving optimization problems with linear and nonlinear constraints. Before diving into using Solver, it is essential to understand some basic Excel functions and techniques that are commonly used when setting up optimization problems. These functions help organize data, compute intermediate values, and define objective functions and constraints.

4.1.1. Useful Excel Functions and Techniques

USING FORMULAS IN CELLS In Excel, formulas begin with an equals sign (=). For example, typing `=A1+B1` in a cell will calculate the sum of the values in cells `A1` and `B1`.

`SUM()` The `SUM()` function adds a range of numbers. For example, `=SUM(A1:A10)` calculates the sum of all values in the range `A1` to `A10`.

`SUMPRODUCT()` The `SUMPRODUCT()` function multiplies corresponding elements in two or more arrays and returns the sum of those products. This is particularly useful for computing dot products or weighted sums. For example, `=SUMPRODUCT(A1:A10, B1:B10)` computes $\sum_{i=1}^{10} A_i B_i$.

COUNT() The COUNT() function counts the number of numeric entries in a range. For example, =COUNT(A1:A10) returns the number of numeric cells in the range A1 to A10.

COUNTIF() The COUNTIF() function counts the number of cells in a range that meet a specific condition. For example, =COUNTIF(A1:A10, ">5") counts the number of cells in A1:A10 with values greater than 5.

DUPLICATING FORMULAS ACROSS ROWS OR COLUMNS To apply a formula to multiple cells, you can drag the fill handle (a small square at the bottom-right corner of the selected cell) across the desired range. Excel automatically adjusts cell references based on the relative position (relative referencing). To keep a reference constant, use the dollar sign ($) in the cell reference (absolute referencing). For example, $A$1 always refers to cell A1.

CONDITIONAL FORMATTING Conditional formatting allows you to highlight cells that meet specific criteria. For example, you can format cells with values above a threshold in bold or a particular color to identify important data visually.

DATA VALIDATION Data validation helps restrict the type of data entered in a cell. For example, you can set a rule that only allows numbers within a specific range or text from a predefined list.

NAMING RANGES Naming ranges makes formulas easier to read and maintain. Instead of =SUM(A1:A10), you can name the range "Sales" and write =SUM(Sales).

USING LOGICAL FUNCTIONS Logical functions like IF(), AND(), and OR() are helpful for decision-making within formulas. For example, =IF(A1>10, Ḧigh,̈ L̈ow)̈ returns Ḧighïf A1 is greater than 10, and L̈owötherwise.

With these foundational tools, setting up optimization problems becomes much more manageable. In the next section, we will explore how to use these functions in combination with Excel Solver to define and solve optimization models.

Logical functions like IF(), AND(), and OR() are helpful for decision-making within formulas. For example, =IF(A1>10, Ḧigh,̈ L̈ow)̈ returns Ḧighïf A1 is greater than 10, and L̈owötherwise.

With these foundational tools, setting up optimization problems becomes much more manageable. In the next section, we will explore how to use these functions in combination with Excel Solver to define and solve optimization models.

4.1.2. How to Use Excel Solver

Excel Solver is a powerful add-in that allows users to define and solve optimization problems. This section provides a step-by-step guide on how to effectively layout data and use Solver for optimization.

STEP 1: LAYOUT THE DATA Organizing your data clearly is crucial for using Solver effectively. Use color coding to distinguish between different types of cells:

- **Data:** Cells containing fixed input values, such as coefficients or constants, should be highlighted in light blue.
- **Variables:** Cells representing decision variables should be highlighted in yellow.
- **Formulas:** Cells with formulas, such as the objective function or summarizing constraints values should be highlighted in light green.

Other color combinations are also acceptable, as long as they consistently distinguish between these components.

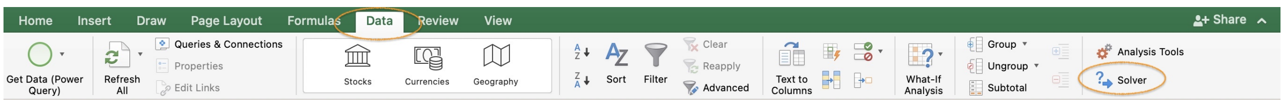
Objective	=SUMPRODUCT(B6:C6,B8:C8)				
Variables	x1	x2			
obj. coeff	7	6			
			Resources Used		
constraint 1	3	1	=SUMPRODUCT($B$6:$C$6,B11:C11)	<=	120
constraint 2	1	2	=SUMPRODUCT($B$6:$C$6,B12:C12)	<=	160
constraint 3	1	0	=SUMPRODUCT($B$6:$C$6,B13:C13)	<=	35

STEP 2: DEFINE THE OBJECTIVE FUNCTION Identify the cell that represents the objective function. This cell should contain a formula that calculates the value to be maximized or minimized, such as total profit or cost.

STEP 3: SET UP CONSTRAINTS Clearly specify the constraints of the problem. Constraints can be represented using formulas that involve the variables, coefficients, and right-hand side values. For example, a constraint ensuring that total production does not exceed capacity can be written as `=SUMPRODUCT(Production, Coefficients) <= Capacity`.

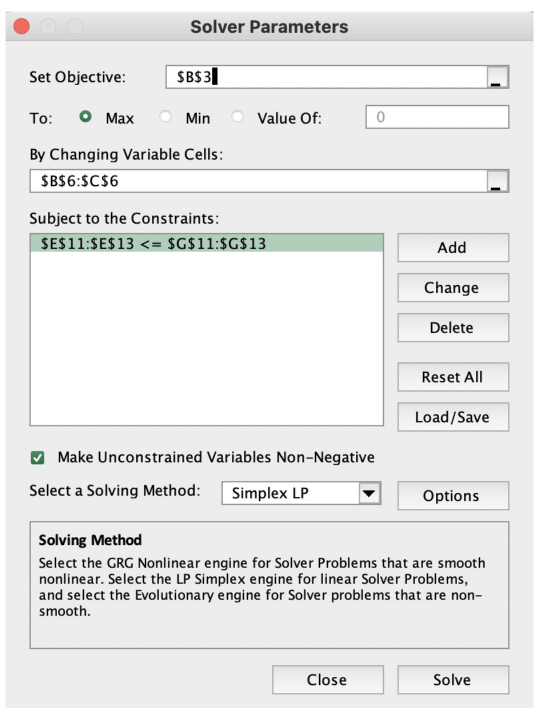
STEP 4: OPEN SOLVER Ensure Solver is enabled in Excel. If not, you can activate it by navigating to `File > Options > Add-Ins > Manage Excel Add-ins` and checking the box for Solver.

Once Solver is enabled, go to `Data > Solver` to open the Solver Parameters dialog box.



STEP 5: DEFINE SOLVER PARAMETERS In the Solver Parameters dialog box:

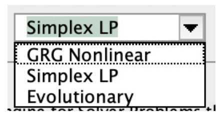
- Set the **Objective** to the cell containing the objective function.
- Choose **Max** or **Min** to indicate whether the objective is to be maximized or minimized.
- Specify the **Variable Cells** by selecting the range of cells representing decision variables.
- Add **Constraints** by specifying the relationship (e.g., <=, >=, =) and the cells involved.
- When the variables are non-negative, you can select `Make Unconstrainted Variables Non-Negative`.



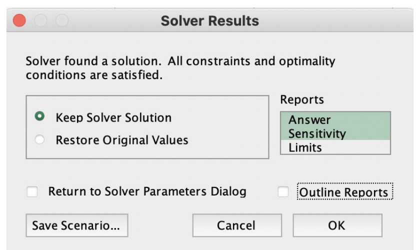
STEP 6: CHOOSE A SOLVING METHOD Solver provides three solving methods:

- **GRG Nonlinear:** For smooth nonlinear problems.
- **Simplex LP:** For linear programming problems.
- **Evolutionary:** For non-smooth problems or those with integer constraints.

Select the method that matches the problem type.

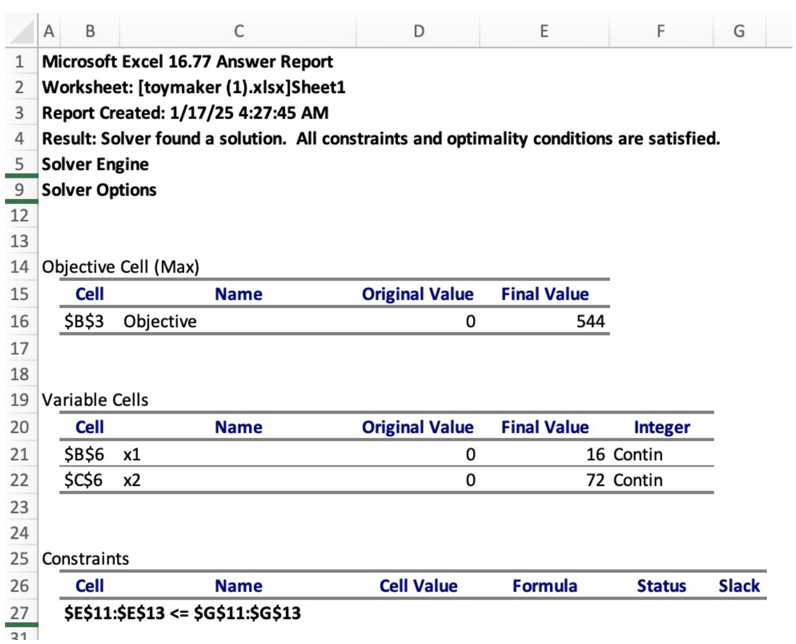


STEP 7: SOLVE THE PROBLEM Click `Solve` to run Solver. If Solver finds a solution, it will display the results and allow you to keep or discard them. Analyze the solution to ensure it meets the problem's requirements.



STEP 8: INTERPRET THE RESULTS Review the values of the decision variables and the objective function. Check that all constraints are satisfied and assess the solution's feasibility and quality.

- Answer report



Microsoft Excel 16.77 Answer Report
Worksheet: [toymaker (1).xlsx]Sheet1
Report Created: 1/17/25 4:27:45 AM
Result: Solver found a solution. All constraints and optimality conditions are satisfied.
Solver Engine
Solver Options

Objective Cell (Max)

Cell	Name	Original Value	Final Value
$B$3	Objective	0	544

Variable Cells

Cell	Name	Original Value	Final Value	Integer
$B$6	x1	0	16	Contin
$C$6	x2	0	72	Contin

Constraints

Cell	Name	Cell Value	Formula	Status	Slack
$E$11:$E$13 <= $G$11:$G$13					

- Sensitivity report

Microsoft Excel 16.77 Sensitivity Report
Worksheet: [toymaker (1).xlsx]Sheet1
Report Created: 1/17/25 4:27:46 AM

Variable Cells

Cell	Name	Final Value	Reduced Cost	Objective Coefficient	Allowable Increase	Allowable Decrease
$B$6	x1	16	0	7	11	4
$C$6	x2	72	0	6	8	3.666666667

Constraints

Cell	Name	Final Value	Shadow Price	Constraint R.H. Side	Allowable Increase	Allowable Decrease
$E$11:$E$13 <= $G$11:$G$13						

- Solution on spreadsheet

	A	B	C	D	E	F	G
1							
2							
3	Objective	544					
4							
5	Variables	x1	x2				
6		16	72				
7							
8	obj. coeff	7	6				
9							
10					Resources Used		
11	constraint 1	3	1		120	<=	120
12	constraint 2	1	2		160	<=	160
13	constraint 3	1	0		16	<=	35
14							

Resources

Examples and guides

- `https://www.excel-easy.com/data-analysis/solver.html`
- *Excel Solver - Introduction on Youtube*
- *Some notes from MIT*

Some videos

- *Solving a linear program*
- *Optimal product mix*
- *Set Cover*
- *Introduction to Designing Optimization Models Using Excel Solver*
- *Traveling Salesman Problem*
- *Also Travelin Salesman Problem*
- *Multiple Traveling Salesman Problem*
- *Shortest Path*

Some other links

- *Loan Example*
- *Several Examples including TSP*

A. Linear Equations

Outcomes

- *Graph linear equations.*
- *Find the slope of a line.*
- *Determine equations of lines.*
- *Solve linear systems.*
- *Apply linear equations to real-world problems.*

Chapter Overview

In this chapter[1], we focus on linear equations, their graphical representations, finding slopes, determining line equations from given information, and solving systems of linear equations. We also examine applications of these concepts in real-world scenarios, such as cost modeling, demand-supply equilibrium, and break-even analysis.

A.1 Graphing a Linear Equation

Outcomes

- *Graph a line from its equation.*
- *Graph a line given its equation in parametric form.*
- *Graph and find equations of vertical and horizontal lines.*

Equations whose graphs are straight lines are called *linear equations*. Examples include:

$$2x-3y=6, \quad 3x=4y-7, \quad y=2x-5, \quad 2y=3, \quad x-2=0.$$

[1]This section was modified and remixed from Applied Finite Mathematics (Sekhon and Bloom)/Section 1.1: Graphing a linear equation which was shared under a CC BY 4.0 license at `https://math.libretexts.org/Bookshelves/Applied_Mathematics/Applied_Finite_Mathematics_(Sekhon_and_Bloom)/01%3A_Linear_Equations/1.01%3A_Graphing_a_Linear_Equation`

A line is determined by two points. To graph a linear equation, we often choose values for one variable and solve for the other.

Example A.1: Graphing a Line

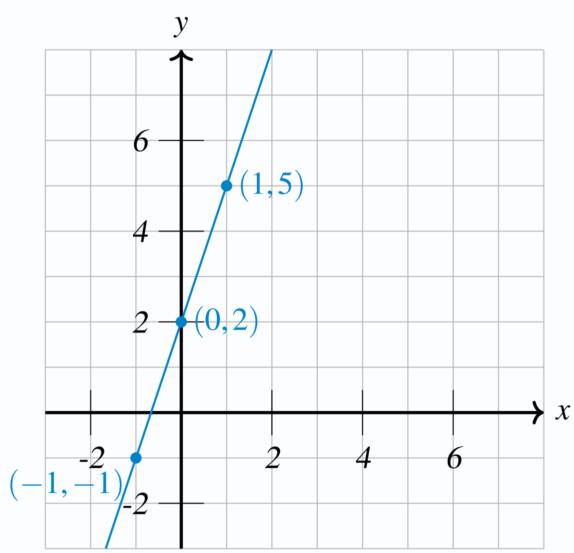
Graph the line: $y = 3x + 2$.
Solution: *Choose convenient* x*-values:*

$$x = -1 \implies y = 3(-1) + 2 = -1 \implies (-1, -1)$$

$$x = 0 \implies y = 3(0) + 2 = 2 \implies (0, 2)$$

$$x = 1 \implies y = 3(1) + 2 = 5 \implies (1, 5)$$

Plot these points and draw the line.



Example A.2: Graphing Another Line

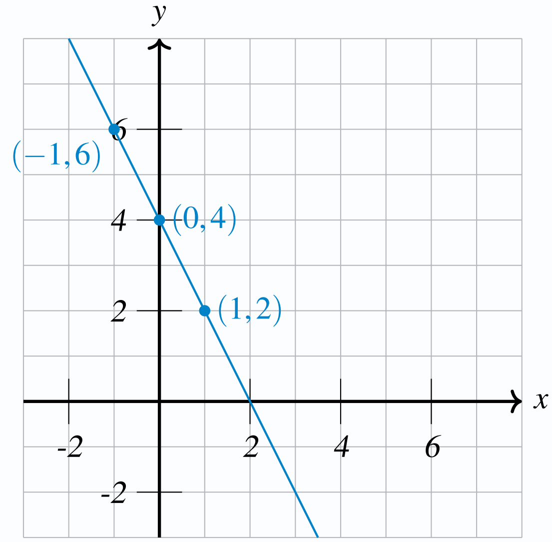
Graph the line: $2x + y = 4$.
Solution: *Choose points:*

$$x = -1 \implies 2(-1) + y = 4 \implies y = 6 \implies (-1, 6)$$

$$x = 0 \implies y = 4 \implies (0, 4)$$

$$y = 2 \implies 2x + 2 = 4 \implies x = 1 \implies (1, 2)$$

Plotting these points gives:



Intercepts

The x-intercept occurs where $y = 0$, and the y-intercept occurs where $x = 0$.

Example A.3: Finding Intercepts

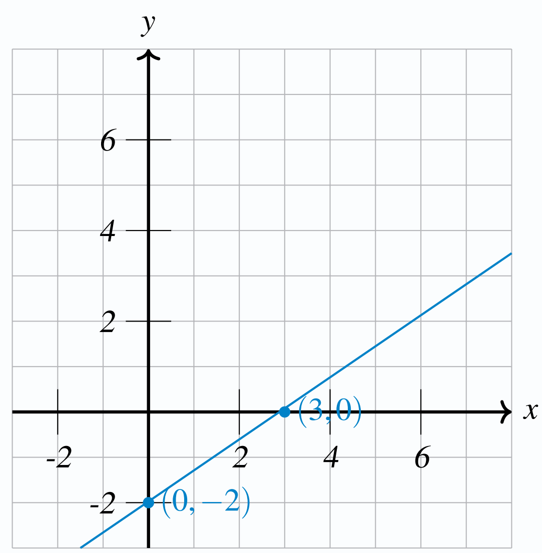
Find the intercepts of the line $2x - 3y = 6$ and graph it.
Solution: *For the x-intercept, set $y = 0$:*

$$2x - 3(0) = 6 \implies x = 3 \implies (3,0).$$

For the y-intercept, set $x = 0$:

$$2(0) - 3y = 6 \implies y = -2 \implies (0,-2).$$

Plot these intercepts and draw the line.



Parametric Form

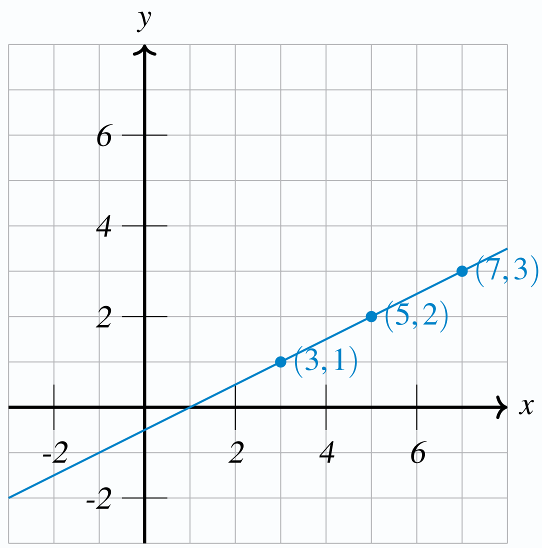
Lines can also be given in parametric form, e.g. $x = 3 + 2t,\ y = 1 + t$.

Example A.4: Parametric Equations

Graph the line $x = 3 + 2t, y = 1 + t$.
Solution: *Let* $t = 0, 1, 2$*:*

$$t = 0 \implies (x, y) = (3, 1), \quad t = 1 \implies (5, 2), \quad t = 2 \implies (7, 3).$$

Plot these points and draw the line.

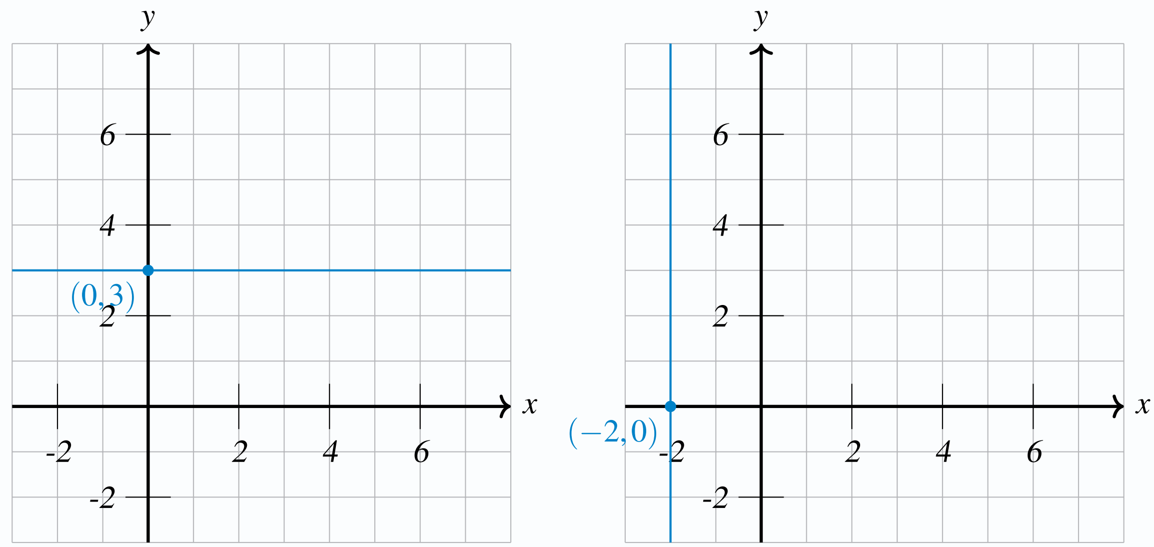


Horizontal and Vertical Lines

$x = a$ is a vertical line passing through $(a, 0)$, $\quad y = b$ is a horizontal line passing through $(0, b)$.

Example A.5: Horizontal and Vertical Lines

Graph $x = -2$ *and* $y = 3$.
Solution: *The line* $x = -2$ *is vertical through* $(-2, 0)$. *The line* $y = 3$ *is horizontal through* $(0, 3)$.



A.2 Slope of a Line

Outcomes

- *Find the slope of a line given two points.*
- *Graph a line if a point and slope are given.*

The slope m of a line passing through (x_1, y_1) and (x_2, y_2) is:

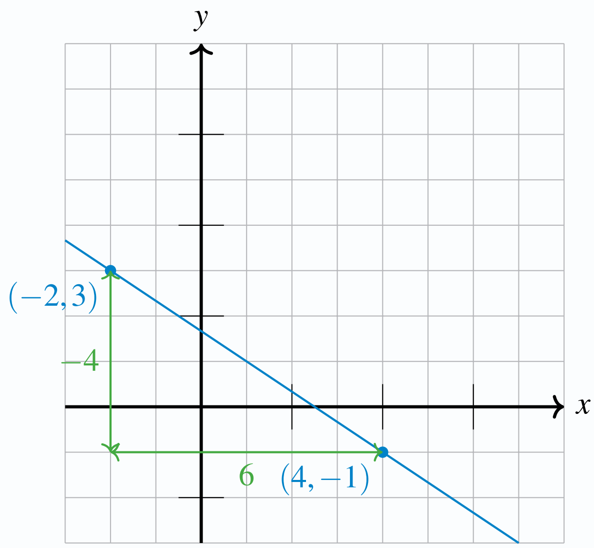
$$m = \frac{y_2 - y_1}{x_2 - x_1}.$$

Example A.6: Finding Slope

Find the slope of the line through $(-2, 3)$ and $(4, -1)$.

Solution:

$$m = \frac{-1-3}{4-(-2)} = \frac{-4}{6} = -\frac{2}{3}.$$



Vertical lines have undefined slope; horizontal lines have slope 0.

Example A.7: Vertical and Horizontal Slopes

a) For points $(2,3)$ *and* $(2,-1)$*:*

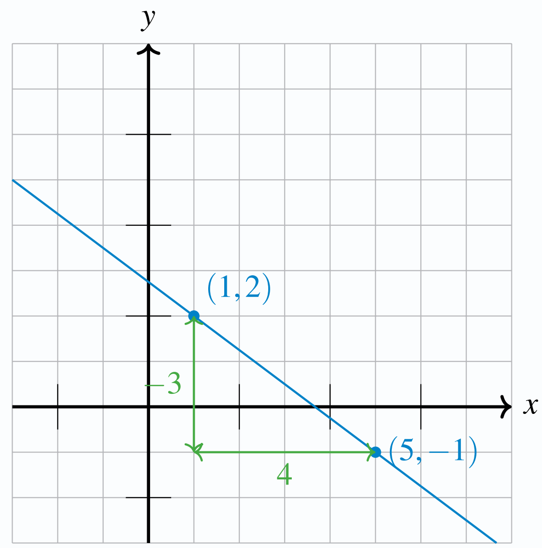
$$m = \frac{-1-3}{2-2} = \frac{-4}{0} \text{ (undefined), vertical line.}$$

b) For points $(-1,-4)$ *and* $(3,-4)$*:*

$$m = \frac{-4-(-4)}{3-(-1)} = \frac{0}{4} = 0 \text{ (horizontal line).}$$

Example A.8: Graphing a Line from a Point and Slope

Graph the line passing through $(1,2)$ *with slope* $-\frac{3}{4}$.
Solution: *Starting from* $(1,2)$*, a slope of* $-\frac{3}{4}$ *means down 3, right 4 to get another point* $(5,-1)$*. Plot and draw the line.*



If a line is written as $y = mx + b$, the coefficient m is the slope and b is the y-intercept.

A.3 Determining the Equation of a Line

Outcomes

- *Find an equation of a line given a point and slope.*
- *Find an equation of a line given two points.*

Forms of a Line

$$\text{Slope-Intercept: } y = mx + b, \quad \text{Point-Slope: } y - y_1 = m(x - x_1), \quad \text{Standard: } Ax + By = C.$$

Example A.9: Equation Given Slope and Intercept

Find the equation of a line with slope $m = 5$ and y-intercept $b = 3$.
Solution: $y = 5x + 3$.

Example A.10: Equation Given a Point and Slope

Find the equation of the line passing through $(2,7)$ with slope 3.
Solution:

$$y = 3x + b,\ 7 = 3(2) + b,\ b = 1 \implies y = 3x + 1.$$

Example A.11: Equation Given Two Points

Find the equation of the line passing through $(-1,2)$ and $(1,8)$.
Solution:

$$m = \frac{8-2}{1-(-1)} = \frac{6}{2} = 3, \; y = 3x + b.$$

Use one point:

$$2 = 3(-1) + b \implies b = 5, \text{ so } y = 3x + 5.$$

Example A.12: Using Intercepts

Find the equation of the line with x-intercept 3 and y-intercept 4.
Solution: *Intercepts give points $(3,0)$ and $(0,4)$:*

$$m = \frac{4-0}{0-3} = -\frac{4}{3}, \; y = -\frac{4}{3}x + 4.$$

A.4 Applications of Linear Equations

Outcomes

- *Use linear functions to model real-world applications.*

Linear equations commonly model cost, revenue, and population changes.

Example A.13: Cost Function

A taxi charges \$0.50 per mile plus a \$5 flat fee.
Solution: *Let $x =$ miles, $y =$ cost:*

$$y = 0.50x + 5.$$

At $x = 20$: $y = 0.50(20) + 5 = 15$.

Example A.14: Cost from Two Points

It costs \$750 to make 25 items and \$1000 to make 50 items. Assume linearity.
Solution: *Points $(25,750)$ and $(50,1000)$:*

$$m = \frac{1000-750}{50-25} = 10, \; y = 10x + b.$$

Using $(25,750)$*:*

$$750 = 10(25)+b \implies b = 500,\ y = 10x+500.$$

Cost of 100 items: $y = 10(100)+500 = 1500.$

Example A.15: Temperature Conversion

Freezing point of water: $(C,F) = (0,32)$*; boiling point:* $(100,212)$.

Solution:

$$m = \frac{212-32}{100-0} = \frac{180}{100} = 1.8,\ F = 1.8C+32.$$

For $C = 30$*:* $F = 1.8(30)+32 = 86.$

A.5 More Applications: Systems of Linear Equations

Outcomes

- *Solve linear systems in two variables.*
- *Find equilibrium points (supply = demand).*
- *Find break-even points (cost = revenue).*

Solving Systems of Equations

The intersection of two lines can be found by setting their equations equal or by using the elimination method.

Example A.16: Solving a System

Solve:

$$2x+y = 7, \quad 3x-y = 3.$$

Solution: *Add them:*

$$2x+y = 7,\ 3x-y = 3 \implies 5x = 10 \implies x = 2.$$

Substitute $x = 2$ *into* $2x+y = 7 \implies 4+y = 7 \implies y = 3.$

Solution: $(2,3)$.

Supply, Demand, and Equilibrium

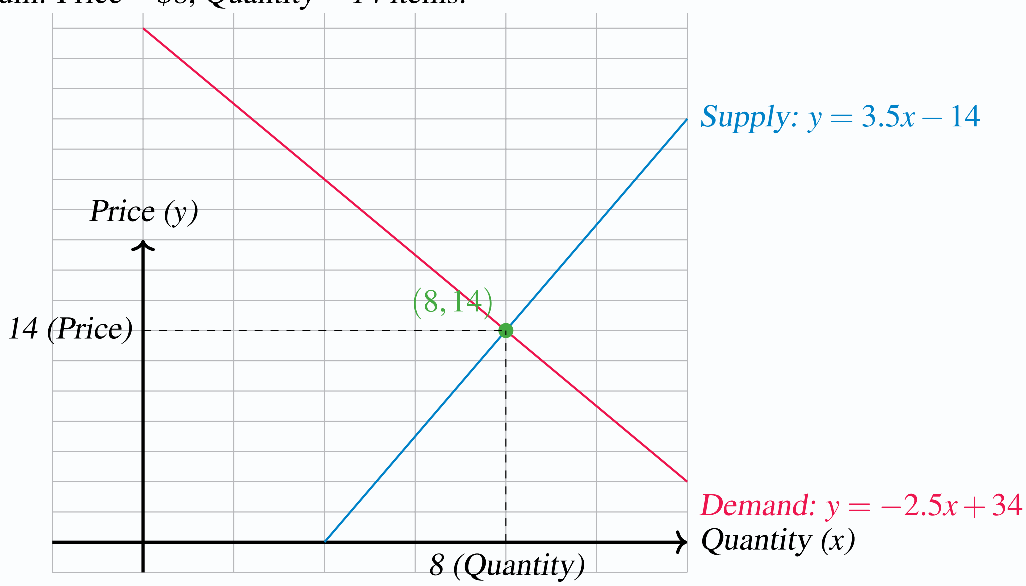
The *equilibrium price* occurs where supply equals demand.

Example A.17: Equilibrium Point

Supply: $y = 3.5x - 14$, *Demand:* $y = -2.5x + 34$.
Solution: *Set equal:*

$$3.5x - 14 = -2.5x + 34 \implies 6x = 48 \implies x = 8.$$

At $x = 8$: $y = 3.5(8) - 14 = 14$ *or* $y = -2.5(8) + 34 = 14$.
Equilibrium: Price = \$8, Quantity = 14 items.



Break-Even Point

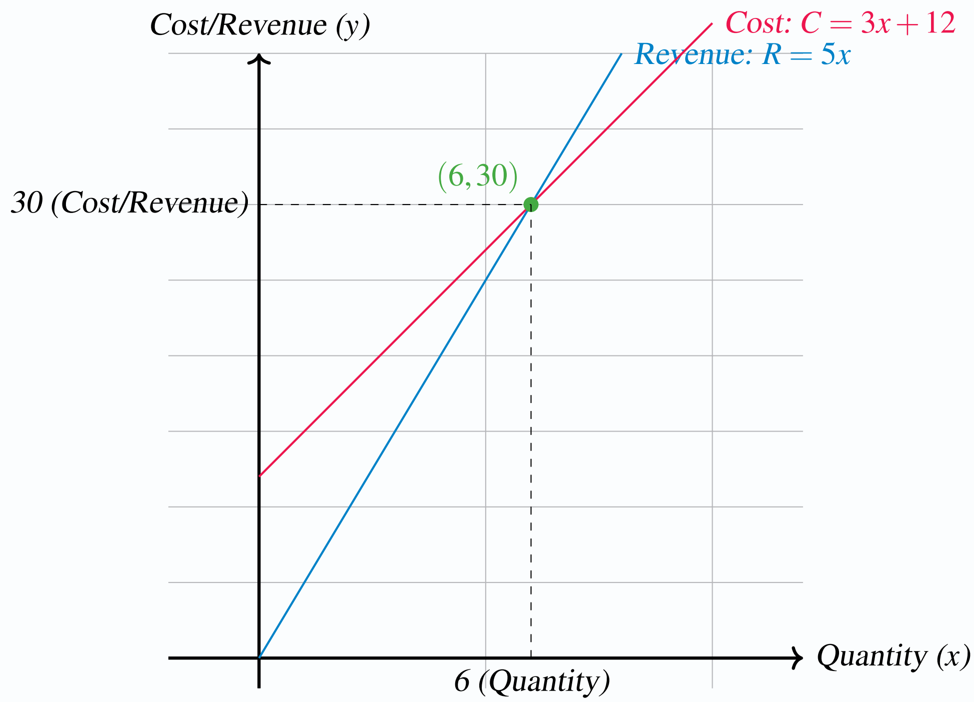
For cost C and revenue R, the break-even point is where $C = R$.

Example A.18: Break-Even Analysis

$R = 5x$, $C = 3x + 12$.
Solution:

$$5x = 3x + 12 \implies 2x = 12 \implies x = 6.$$

At $x = 6$, $R = C = 30$.
Break-even at $(6, 30)$.



Chapter Review

This chapter introduced the fundamental concepts of linear equations, their graphs, slopes, forms of equations, and their real-world applications. You can now model scenarios involving costs, pricing, and market equilibrium using linear models and solve these models for key insights.

B. Systems of Equations

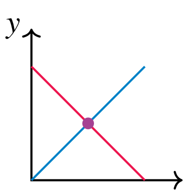
B.1 Systems of Equations, Geometry

Outcomes

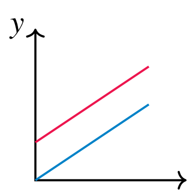
A. *Relate the types of solution sets of a system of two (three) variables to the intersections of lines in a plane (the intersection of planes in three space)*

As you may remember, linear equations like $2x+3y=6$ can be graphed as straight lines in the coordinate plane. We say that this equation is in two variables, in this case x and y. Suppose you have two such equations, each of which can be graphed as a straight line, and consider the resulting graph of two lines. What would it mean if there exists a point of intersection between the two lines? This point, which lies on *both* graphs, gives x and y values for which both equations are true. In other words, this point gives the ordered pair (x,y) that satisfy both equations. If the point (x,y) is a point of intersection, we say that (x,y) is a **solution** to the two equations. In linear algebra, we often are concerned with finding the solution(s) to a system of equations, if such solutions exist. First, we consider graphical representations of solutions and later we will consider the algebraic methods for finding solutions.

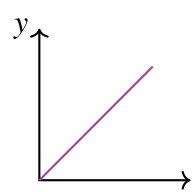
When looking for the intersection of two lines in a graph, several situations may arise. The following picture demonstrates the possible situations when considering two equations (two lines in the graph) involving two variables.



One Solution



No Solutions



Infinitely Many Solutions

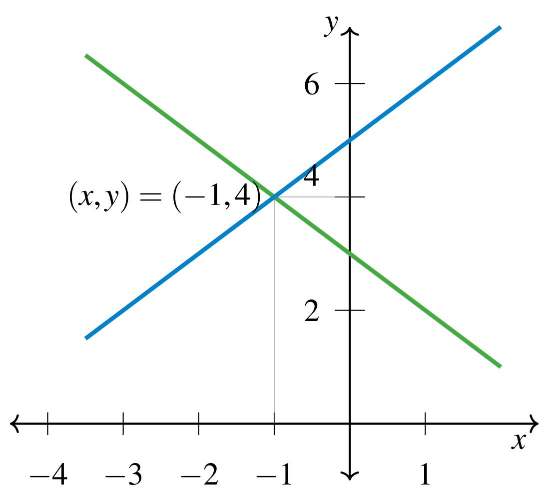
In the first diagram, there is a unique point of intersection, which means that there is only one (unique) solution to the two equations. In the second, there are no points of intersection and no solution. When no solution exists, this means that the two lines are parallel and they never intersect. The third situation which can occur, as demonstrated in diagram three, is that the two lines are really the same line. For example, $x+y=1$ and $2x+2y=2$ are equations which when graphed yield the same line. In this case there are infinitely many points which are solutions of these two equations, as every ordered pair which is on the graph of the line satisfies both equations. When considering linear systems of equations, there are always three types of solutions possible; exactly one (unique) solution, infinitely many solutions, or no solution.

Example B.1: A Graphical Solution

Use a graph to find the solution to the following system of equations

$$\begin{aligned} x+y &= 3 \\ y-x &= 5 \end{aligned}$$

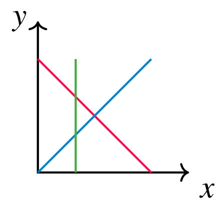
Solution. Through graphing the above equations and identifying the point of intersection, we can find the solution(s). Remember that we must have either one solution, infinitely many, or no solutions at all. The following graph shows the two equations, as well as the intersection. Remember, the point of intersection represents the solution of the two equations, or the (x,y) which satisfy both equations. In this case, there is one point of intersection at $(-1,4)$ which means we have one unique solution, $x=-1, y=4$.



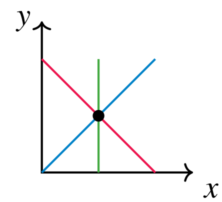
♠

In the above example, we investigated the intersection point of two equations in two variables, x and y. Now we will consider the graphical solutions of three equations in two variables.

Consider a system of three equations in two variables. Again, these equations can be graphed as straight lines in the plane, so that the resulting graph contains three straight lines. Recall the three possible types of solutions; no solution, one solution, and infinitely many solutions. There are now more complex ways of achieving these situations, due to the presence of the third line. For example, you can imagine the case of three intersecting lines having no common point of intersection. Perhaps you can also imagine three intersecting lines which do intersect at a single point. These two situations are illustrated below.



No Solution

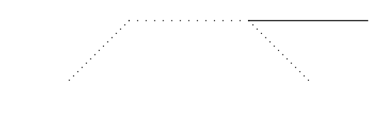


One Solution

Consider the first picture above. While all three lines intersect with one another, there is no common point of intersection where all three lines meet at one point. Hence, there is no solution to the three equations. Remember, a solution is a point (x,y) which satisfies **all** three equations. In the case of the second picture, the lines intersect at a common point. This means that there is one solution to the three equations whose graphs are the given lines. You should take a moment now to draw the graph of a system which results in three parallel lines. Next, try the graph of three identical lines. Which type of solution is represented in each of these graphs?

We have now considered the graphical solutions of systems of two equations in two variables, as well as three equations in two variables. However, there is no reason to limit our investigation to equations in two variables. We will now consider equations in three variables.

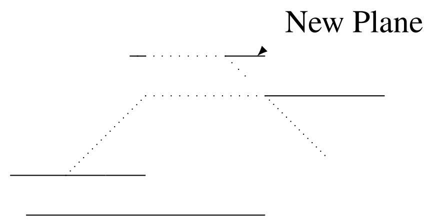
You may recall that equations in three variables, such as $2x+4y-5z=8$, form a plane. Above, we were looking for intersections of lines in order to identify any possible solutions. When graphically solving systems of equations in three variables, we look for intersections of planes. These points of intersection give the (x,y,z) that satisfy all the equations in the system. What types of solutions are possible when working with three variables? Consider the following picture involving two planes, which are given by two equations in three variables.



Notice how these two planes intersect in a line. This means that the points (x,y,z) on this line satisfy both equations in the system. Since the line contains infinitely many points, this system has infinitely many solutions.

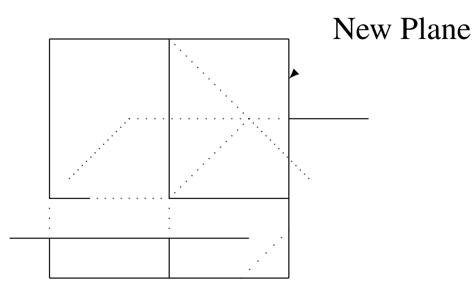
It could also happen that the two planes fail to intersect. However, is it possible to have two planes intersect at a single point? Take a moment to attempt drawing this situation, and convince yourself that it is not possible! This means that when we have only two equations in three variables, there is no way to have a unique solution! Hence, the types of solutions possible for two equations in three variables are no solution or infinitely many solutions.

Now imagine adding a third plane. In other words, consider three equations in three variables. What types of solutions are now possible? Consider the following diagram.

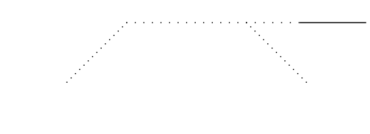


In this diagram, there is no point which lies in all three planes. There is no intersection between **all** planes so there is no solution. The picture illustrates the situation in which the line of intersection of the new plane with one of the original planes forms a line parallel to the line of intersection of the first two planes. However, in three dimensions, it is possible for two lines to fail to intersect even though they are not parallel. Such lines are called **skew lines**.

Recall that when working with two equations in three variables, it was not possible to have a unique solution. Is it possible when considering three equations in three variables? In fact, it is possible, and we demonstrate this situation in the following picture.



In this case, the three planes have a single point of intersection. Can you think of other types of solutions possible? Another is that the three planes could intersect in a line, resulting in infinitely many solutions, as in the following diagram.



We have now seen how three equations in three variables can have no solution, a unique solution, or intersect in a line resulting in infinitely many solutions. It is also possible that the three equations graph the same plane, which also leads to infinitely many solutions.

You can see that when working with equations in three variables, there are many more ways to achieve the different types of solutions than when working with two variables. It may prove enlightening to spend time imagining (and drawing) many possible scenarios, and you should take some time to try a few.

You should also take some time to imagine (and draw) graphs of systems in more than three variables. Equations like $x+y-2z+4w=8$ with more than three variables are often called **hyper-planes**. You may soon realize that it is tricky to draw the graphs of hyper-planes! Through the tools of linear algebra, we can algebraically examine these types of systems which are difficult to graph. In the following section, we will consider these algebraic tools.

B.2 Systems Of Equations, Algebraic Procedures

Outcomes

A. *Use elementary operations to find the solution to a linear system of equations.*
B. *Find the row-echelon form and reduced row-echelon form of a matrix.*
C. *Determine whether a system of linear equations has no solution, a unique solution or an infinite number of solutions from its row-echelon form.*
D. *Solve a system of equations using Gaussian Elimination and Gauss-Jordan Elimination.*
E. *Model a physical system with linear equations and then solve.*

We have taken an in depth look at graphical representations of systems of equations, as well as how to find possible solutions graphically. Our attention now turns to working with systems algebraically.

Definition B.2: System of Linear Equations

*A **system of linear equations** is a list of equations,*

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n &= b_1 \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n &= b_2 \\ &\vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n &= b_m \end{aligned}$$

where a_{ij} and b_j are real numbers. The above is a system of m equations in the n variables, $x_1, x_2 \cdots, x_n$. Written more simply in terms of summation notation, the above can be written in the form

$$\sum_{j=1}^{n} a_{ij}x_j = b_i,\ i = 1,2,3,\cdots,m$$

Finally, we can write this in matrix notaion as

$$A\mathbf{x} = \mathbf{b}$$

where

$$A = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & & & \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{bmatrix}, \quad \mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}, \quad \text{and} \quad \mathbf{b} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix}.$$

The relative size of m and n is not important here. Notice that we have allowed a_{ij} and b_j to be any real number. We can also call these numbers **scalars** . We will use this term throughout the text, so keep in mind that the term **scalar** just means that we are working with real numbers.

Now, suppose we have a system where $b_i = 0$ for all i. In other words every equation equals 0. This is a special type of system.

Definition B.3: Homogeneous System of Equations

*A system of equations is called **homogeneous** if each equation in the system is equal to 0. A homogeneous system has the form*

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n &= 0 \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n &= 0 \\ &\vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n &= 0 \end{aligned}$$

where a_{ij} are scalars and x_i are variables.

Recall from the previous section that our goal when working with systems of linear equations was to find the point of intersection of the equations when graphed. In other words, we looked for the solutions to the system. We now wish to find these solutions algebraically. We want to find values for $x_1, \cdots, x_n$ which solve all of the equations. If such a set of values exists, we call $(x_1, \cdots, x_n)$ the **solution set**.

Recall the above discussions about the types of solutions possible. We will see that systems of linear equations will have one unique solution, infinitely many solutions, or no solution. Consider the following definition.

Definition B.4: Consistent and Inconsistent Systems

*A system of linear equations is called **consistent** if there exists at least one solution. It is called **inconsistent** if there is no solution.*

If you think of each equation as a condition which must be satisfied by the variables, consistent would mean there is some choice of variables which can satisfy **all** the conditions. Inconsistent would mean there is no choice of the variables which can satisfy all of the conditions.

The following sections provide methods for determining if a system is consistent or inconsistent, and finding solutions if they exist.

B.2.1. Elementary Operations

We begin this section with an example. Recall from Example **??** that the solution to the given system was $(x,y) = (-1,4)$.

Example B.5: Verifying an Ordered Pair is a Solution

Algebraically verify that $(x,y) = (-1,4)$ is a solution to the following system of equations.

$$\begin{aligned} x+y &= 3 \\ y-x &= 5 \end{aligned}$$

Solution. By graphing these two equations and identifying the point of intersection, we previously found that $(x,y) = (-1,4)$ is the unique solution.

We can verify algebraically by substituting these values into the original equations, and ensuring that the equations hold. First, we substitute the values into the first equation and check that it equals 3.

$$x+y = (-1)+(4) = 3$$

This equals 3 as needed, so we see that $(-1,4)$ is a solution to the first equation. Substituting the values into the second equation yields

$$y-x = (4)-(-1) = 4+1 = 5$$

which is true. For $(x,y) = (-1,4)$ each equation is true and therefore, this is a solution to the system. ♠

Now, the interesting question is this: If you were not given these numbers to verify, how could you algebraically determine the solution? Linear algebra gives us the tools needed to answer this question. The following basic operations are important tools that we will utilize.

Definition B.6: Elementary Operations

Elementary operations *are those operations consisting of the following.*

1. *Interchange the order in which the equations are listed.*
2. *Multiply any equation by a nonzero number.*
3. *Replace any equation with itself added to a multiple of another equation.*

It is important to note that none of these operations will change the set of solutions of the system of equations. In fact, elementary operations are the *key tool* we use in linear algebra to find solutions to systems of equations.

Consider the following example.

Example B.7: Effects of an Elementary Operation

Show that the system

$$\begin{array}{ccccc} x & + & y & = & 7 \\ 2x & - & y & = & 8 \end{array}$$

has the same solution as the system

$$\begin{array}{ccccc} x & + & y & = & 7 \\ & & -3y & = & -6 \end{array}$$

Solution. Notice that the second system has been obtained by taking the second equation of the first system and adding -2 times the first equation, as follows:

$$2x - y + (-2)(x + y) = 8 + (-2)(7)$$

By simplifying, we obtain

$$-3y = -6$$

which is the second equation in the second system. Now, from here we can solve for y and see that $y = 2$. Next, we substitute this value into the first equation as follows

$$x + y = x + 2 = 7$$

Hence $x = 5$ and so $(x, y) = (5, 2)$ is a solution to the second system. We want to check if $(5, 2)$ is also a solution to the first system. We check this by substituting $(x, y) = (5, 2)$ into the system and ensuring the equations are true.

$$\begin{array}{c} x + y = (5) + (2) = 7 \\ 2x - y = 2(5) - (2) = 8 \end{array}$$

Hence, $(5,2)$ is also a solution to the first system. ♠

This example illustrates how an elementary operation applied to a system of two equations in two variables does not affect the solution set. However, a linear system may involve many equations and many variables and there is no reason to limit our study to small systems. For any size of system in any number of variables, the solution set is still the collection of solutions to the equations. In every case, the above operations of Definition B.6 do not change the set of solutions to the system of linear equations.

In the following theorem, we use the notation E_i to represent an equation, while b_i denotes a constant.

Theorem B.8: Elementary Operations and Solutions

Suppose you have a system of two linear equations

$$\begin{array}{c} E_1 = b_1 \\ E_2 = b_2 \end{array} \tag{2.1}$$

Then the following systems have the same solution set as 2.1:

1. $$\begin{array}{c} E_2 = b_2 \\ E_1 = b_1 \end{array} \tag{2.2}$$

2. $$\begin{array}{c} E_1 = b_1 \\ kE_2 = kb_2 \end{array} \tag{2.3}$$

   for any scalar k, provided $k \neq 0$.

3. $$\begin{array}{c} E_1 = b_1 \\ E_2 + kE_1 = b_2 + kb_1 \end{array} \tag{2.4}$$

   for any scalar k (including $k = 0$).

Before we proceed with the proof of Theorem B.8, let us consider this theorem in context of Example **??**. Then,

$$\begin{array}{cc} E_1 = x + y, & b_1 = 7 \\ E_2 = 2x - y, & b_2 = 8 \end{array}$$

Recall the elementary operations that we used to modify the system in the solution to the example. First, we added (-2) times the first equation to the second equation. In terms of Theorem B.8, this action is given by

$$E_2 + (-2)E_1 = b_2 + (-2)b_1$$

or

$$2x - y + (-2)(x + y) = 8 + (-2)7$$

This gave us the second system in Example **??**, given by

$$\begin{array}{c} E_1 = b_1 \\ E_2 + (-2)E_1 = b_2 + (-2)b_1 \end{array}$$

From this point, we were able to find the solution to the system. Theorem B.8 tells us that the solution we found is in fact a solution to the original system.

Stated simply, the above theorem shows that the elementary operations do not change the solution set of a system of equations.

We will now look at an example of a system of three equations and three variables. Similarly to the previous examples, the goal is to find values for x,y,z such that each of the given equations are satisfied when these values are substituted in.

Example B.9: Solving a System of Equations with Elementary Operations

Find the solutions to the system,

$$\begin{array}{ccccccc} x & + & 3y & + & 6z & = & 25 \\ 2x & + & 7y & + & 14z & = & 58 \\ & & 2y & + & 5z & = & 19 \end{array} \tag{2.5}$$

Solution. We can relate this system to Theorem B.8 above. In this case, we have

$$\begin{array}{ll} E_1 = x+3y+6z, & b_1 = 25 \\ E_2 = 2x+7y+14z, & b_2 = 58 \\ E_3 = 2y+5z, & b_3 = 19 \end{array}$$

Theorem B.8 claims that if we do elementary operations on this system, we will not change the solution set. Therefore, we can solve this system using the elementary operations given in Definition B.6. First, replace the second equation by (-2) times the first equation added to the second. This yields the system

$$\begin{array}{c} x+3y+6z = 25 \\ y+2z = 8 \\ 2y+5z = 19 \end{array} \tag{2.6}$$

Now, replace the third equation with (-2) times the second added to the third. This yields the system

$$\begin{array}{c} x+3y+6z = 25 \\ y+2z = 8 \\ z = 3 \end{array} \tag{2.7}$$

At this point, we can easily find the solution. Simply take $z=3$ and substitute this back into the previous equation to solve for y, and similarly to solve for x.

$$\begin{array}{c} x+3y+6(3) = x+3y+18 = 25 \\ y+2(3) = y+6 = 8 \\ z = 3 \end{array}$$

The second equation is now

$$y+6=8$$

You can see from this equation that $y = 2$. Therefore, we can substitute this value into the first equation as follows:

$$x + 3(2) + 18 = 25$$

By simplifying this equation, we find that $x = 1$. Hence, the solution to this system is $(x,y,z) = (1,2,3)$. This process is called **back substitution**.

Alternatively, in 2.7 you could have continued as follows. Add (-2) times the third equation to the second and then add (-6) times the second to the first. This yields

$$\begin{aligned} x + 3y &= 7 \\ y &= 2 \\ z &= 3 \end{aligned}$$

Now add (-3) times the second to the first. This yields

$$\begin{aligned} x &= 1 \\ y &= 2 \\ z &= 3 \end{aligned}$$

a system which has the same solution set as the original system. This avoided back substitution and led to the same solution set. It is your decision which you prefer to use, as both methods lead to the correct solution, $(x,y,z) = (1,2,3)$. ♠

B.2.2. Gaussian Elimination

The work we did in the previous section will always find the solution to the system. In this section, we will explore a less cumbersome way to find the solutions. First, we will represent a linear system with an **augmented matrix.** A **matrix** is simply a rectangular array of numbers. The size or dimension of a matrix is defined as $m \times n$ where m is the number of rows and n is the number of columns. In order to construct an augmented matrix from a linear system, we create a **coefficient matrix** from the coefficients of the variables in the system, as well as a **constant matrix** from the constants. The coefficients from one equation of the system create one row of the augmented matrix.

For example, consider the linear system in Example **??**

$$\begin{array}{ccccccc} x & + & 3y & + & 6z & = & 25 \\ 2x & + & 7y & + & 14z & = & 58 \\ & & 2y & + & 5z & = & 19 \end{array}$$

This system can be written as an augmented matrix, as follows

$$\left[\begin{array}{rrr|r} 1 & 3 & 6 & 25 \\ 2 & 7 & 14 & 58 \\ 0 & 2 & 5 & 19 \end{array}\right]$$

Notice that it has exactly the same information as the original system. Here it is understood that the first column contains the coefficients from x in each equation, in order, $\left[\begin{array}{c}1\\2\\0\end{array}\right]$. Similarly, we create a column from the coefficients on y in each equation, $\left[\begin{array}{c}3\\7\\2\end{array}\right]$ and a column from the coefficients on z in each equation, $\left[\begin{array}{c}6\\14\\5\end{array}\right]$. For a system of more than three variables, we would continue in this way constructing a column for each variable. Similarly, for a system of less than three variables, we simply construct a column for each variable.

Finally, we construct a column from the constants of the equations, $\left[\begin{array}{c}25\\58\\19\end{array}\right]$.

The rows of the augmented matrix correspond to the equations in the system. For example, the top row in the augmented matrix, $\left[\begin{array}{ccc|c}1 & 3 & 6 & 25\end{array}\right]$ corresponds to the equation

$$x+3y+6z=25.$$

Consider the following definition.

Definition B.10: Augmented Matrix of a Linear System

For a linear system of the form

$$\begin{gathered}a_{11}x_1+\cdots+a_{1n}x_n=b_1\\ \vdots\\ a_{m1}x_1+\cdots+a_{mn}x_n=b_m\end{gathered}$$

where the x_i are variables and the a_{ij} and b_i are constants, the augmented matrix of this system is given by

$$\left[\begin{array}{ccc|c}a_{11} & \cdots & a_{1n} & b_1\\ \vdots & & \vdots & \vdots\\ a_{m1} & \cdots & a_{mn} & b_m\end{array}\right]$$

Now, consider elementary operations in the context of the augmented matrix. The elementary operations in Definition B.6 can be used on the rows just as we used them on equations previously. Changes to a system of equations in as a result of an elementary operation are equivalent to changes in the augmented matrix resulting from the corresponding row operation. Note that Theorem B.8 implies that any elementary row operations used on an augmented matrix will not change the solution to the corresponding system of equations. We now formally define elementary row operations. These are the *key tool* we will use to find solutions to systems of equations.

Definition B.11: Elementary Row Operations

The ***elementary row operations*** *(also known as* ***row operations****) consist of the following*

1. *Switch two rows.*
2. *Multiply a row by a nonzero number.*
3. *Replace a row by any multiple of another row added to it.*

Recall how we solved Example **??**. We can do the exact same steps as above, except now in the context of an augmented matrix and using row operations. The augmented matrix of this system is

$$\left[\begin{array}{rrr|r} 1 & 3 & 6 & 25 \\ 2 & 7 & 14 & 58 \\ 0 & 2 & 5 & 19 \end{array}\right]$$

Thus the first step in solving the system given by 2.5 would be to take (-2) times the first row of the augmented matrix and add it to the second row,

$$\left[\begin{array}{rrr|r} 1 & 3 & 6 & 25 \\ 0 & 1 & 2 & 8 \\ 0 & 2 & 5 & 19 \end{array}\right]$$

Note how this corresponds to 2.6. Next take (-2) times the second row and add to the third,

$$\left[\begin{array}{rrr|r} 1 & 3 & 6 & 25 \\ 0 & 1 & 2 & 8 \\ 0 & 0 & 1 & 3 \end{array}\right]$$

This augmented matrix corresponds to the system

$$\begin{array}{ccccccc} x & + & 3y & + & 6z & = & 25 \\ & & y & + & 2z & = & 8 \\ & & & & z & = & 3 \end{array}$$

which is the same as 2.7. By back substitution you obtain the solution $x = 1, y = 2$, and $z = 3$.

Through a systematic procedure of row operations, we can simplify an augmented matrix and carry it to **row-echelon form** or **reduced row-echelon form**, which we define next. These forms are used to find the solutions of the system of equations corresponding to the augmented matrix.

In the following definitions, the term **leading entry** refers to the first nonzero entry of a row when scanning the row from left to right.

Definition B.12: Row-Echelon Form

An augmented matrix is in ***row-echelon form*** *if*

1. *All nonzero rows are above any rows of zeros.*
2. *Each leading entry of a row is in a column to the right of the leading entries of any row above it.*
3. *Each leading entry of a row is equal to* 1.

We also consider another reduced form of the augmented matrix which has one further condition.

Definition B.13: Reduced Row-Echelon Form

An augmented matrix is in ***reduced row-echelon form*** *if*

1. *All nonzero rows are above any rows of zeros.*
2. *Each leading entry of a row is in a column to the right of the leading entries of any rows above it.*
3. *Each leading entry of a row is equal to* 1.
4. *All entries in a column above and below a leading entry are zero.*

Notice that the first three conditions on a reduced row-echelon form matrix are the same as those for row-echelon form.

Hence, every reduced row-echelon form matrix is also in row-echelon form. The converse is not necessarily true; we cannot assume that every matrix in row-echelon form is also in reduced row-echelon form. However, it often happens that the row-echelon form is sufficient to provide information about the solution of a system.

The following examples describe matrices in these various forms. As an exercise, take the time to carefully verify that they are in the specified form.

Example B.14: Not in Row-Echelon Form

The following augmented matrices are not in row-echelon form (and therefore also not in reduced row-echelon form).

$$\left[\begin{array}{ccc|c} 0 & 0 & 0 & 0 \\ 1 & 2 & 3 & 3 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{array}\right], \left[\begin{array}{cc|r} 1 & 2 & 3 \\ 2 & 4 & -6 \\ 4 & 0 & 7 \end{array}\right], \left[\begin{array}{ccc|c} 0 & 2 & 3 & 3 \\ 1 & 5 & 0 & 2 \\ 7 & 5 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{array}\right]$$

Example B.15: Matrices in Row-Echelon Form

The following augmented matrices are in row-echelon form, but not in reduced row-echelon form.

$$\left[\begin{array}{ccccc|c} 1 & 0 & 6 & 5 & 8 & 2 \\ 0 & 0 & 1 & 2 & 7 & 3 \\ 0 & 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array}\right], \left[\begin{array}{ccc|c} 1 & 3 & 5 & 4 \\ 0 & 1 & 0 & 7 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{array}\right], \left[\begin{array}{ccc|c} 1 & 0 & 6 & 0 \\ 0 & 1 & 4 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

Notice that we could apply further row operations to these matrices to carry them to reduced row-echelon form. Take the time to try that on your own. Consider the following matrices, which are in reduced row-echelon form.

Example B.16: Matrices in Reduced Row-Echelon Form

The following augmented matrices are in reduced row-echelon form.

$$\left[\begin{array}{ccccc|c} 1 & 0 & 0 & 5 & 0 & 0 \\ 0 & 0 & 1 & 2 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array}\right], \left[\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{array}\right], \left[\begin{array}{ccc|c} 1 & 0 & 0 & 4 \\ 0 & 1 & 0 & 3 \\ 0 & 0 & 1 & 2 \end{array}\right]$$

One way in which the row-echelon form of a matrix is useful is in identifying the pivot positions and pivot columns of the matrix.

Definition B.17: Pivot Position and Pivot Column

A ***pivot position*** *in a matrix is the location of a leading entry in the row-echelon formof a matrix. A* ***pivot column*** *is a column that contains a pivot position.*

For example consider the following.

Example B.18: Pivot Position

Let

$$A = \left[\begin{array}{ccc|c} 1 & 2 & 3 & 4 \\ 3 & 2 & 1 & 6 \\ 4 & 4 & 4 & 10 \end{array}\right]$$

Where are the pivot positions and pivot columns of the augmented matrix A?

Solution. The row-echelon form of this matrix is

$$\left[\begin{array}{ccc|c} 1 & 2 & 3 & 4 \\ 0 & 1 & 2 & \frac{3}{2} \\ 0 & 0 & 0 & 0 \end{array}\right]$$

This is all we need in this example, but note that this matrix is not in reduced row-echelon form.

In order to identify the pivot positions in the original matrix, we look for the leading entries in the row-echelon form of the matrix. Here, the entry in the first row and first column, as well as the entry in the second row and second column are the leading entries. Hence, these locations are the pivot positions. We identify the pivot positions in the original matrix, as in the following:

$$\left[\begin{array}{rrr|r} \boxed{1} & 2 & 3 & 4 \\ 3 & \boxed{2} & 1 & 6 \\ 4 & 4 & 4 & 10 \end{array}\right]$$

Thus the pivot columns in the matrix are the first two columns. ♠

The following is an algorithm for carrying a matrix to row-echelon form and reduced row-echelon form. You may wish to use this algorithm to carry the above matrix to row-echelon form or reduced row-echelon form yourself for practice.

Algorithm 1 Reduced Row-Echelon Form Algorithm

This algorithm provides a method for using row operations to take a matrix to its reduced row-echelon form. We begin with the matrix in its original form.

1. Starting from the left, find the first nonzero column. This is the first pivot column, and the position at the top of this column is the first pivot position. Switch rows if necessary to place a nonzero number in the first pivot position.
2. Use row operations to make the entries below the first pivot position (in the first pivot column) equal to zero.
3. Ignoring the row containing the first pivot position, repeat steps 1 and 2 with the remaining rows. Repeat the process until there are no more rows to modify.
4. Divide each nonzero row by the value of the leading entry, so that the leading entry becomes 1. The matrix will then be in row-echelon form.

   The following step will carry the matrix from row-echelon form to reduced row-echelon form.
5. Moving from right to left, use row operations to create zeros in the entries of the pivot columns which are above the pivot positions. The result will be a matrix in reduced row-echelon form.

Most often we will apply this algorithm to an augmented matrix in order to find the solution to a system of linear equations. However, we can use this algorithm to compute the reduced row-echelon form of any matrix which could be useful in other applications.

Consider the following example of Algorithm 1.

Example B.19: Finding Row-Echelon Form and Reduced Row-Echelon Form of a Matrix

Let

$$A = \left[\begin{array}{rrr} 0 & -5 & -4 \\ 1 & 4 & 3 \\ 5 & 10 & 7 \end{array}\right]$$

Find the row-echelon form of A. Then complete the process until A is in reduced row-echelon form.

Solution. In working through this example, we will use the steps outlined in Algorithm 1.

1. The first pivot column is the first column of the matrix, as this is the first nonzero column from the left. Hence the first pivot position is the one in the first row and first column. Switch the first two rows to obtain a nonzero entry in the first pivot position, outlined in a box below.

$$\left[\begin{array}{rrr} \boxed{1} & 4 & 3 \\ 0 & -5 & -4 \\ 5 & 10 & 7 \end{array}\right]$$

2. Step two involves creating zeros in the entries below the first pivot position. The first entry of the second row is already a zero. All we need to do is subtract 5 times the first row from the third row. The resulting matrix is

$$\left[\begin{array}{rrr} 1 & 4 & 3 \\ 0 & -5 & -4 \\ 0 & 10 & 8 \end{array}\right]$$

3. Now ignore the top row. Apply steps 1 and 2 to the smaller matrix

$$\left[\begin{array}{rr} -5 & -4 \\ 10 & 8 \end{array}\right]$$

   In this matrix, the first column is a pivot column, and -5 is in the first pivot position. Therefore, we need to create a zero below it. To do this, add 2 times the first row (of this matrix) to the second. The resulting matrix is

$$\left[\begin{array}{rr} -5 & -4 \\ 0 & 0 \end{array}\right]$$

   Our original matrix now looks like

$$\left[\begin{array}{rrr} 1 & 4 & 3 \\ 0 & -5 & -4 \\ 0 & 0 & 0 \end{array}\right]$$

   We can see that there are no more rows to modify.

4. Now, we need to create leading 1s in each row. The first row already has a leading 1 so no work is needed here. Divide the second row by -5 to create a leading 1. The resulting matrix is

$$\left[\begin{array}{ccc} 1 & 4 & 3 \\ 0 & 1 & \frac{4}{5} \\ 0 & 0 & 0 \end{array}\right]$$

   This matrix is now in row-echelon form.

5. Now create zeros in the entries above pivot positions in each column, in order to carry this matrix all the way to reduced row-echelon form. Notice that there is no pivot position in the third column so we do not need to create any zeros in this column! The column in which we need to create zeros is the second. To do so, subtract 4 times the second row from the first row. The resulting matrix is

$$\left[\begin{array}{ccr} 1 & 0 & -\frac{1}{5} \\ 0 & 1 & \frac{4}{5} \\ 0 & 0 & 0 \end{array}\right]$$

This matrix is now in reduced row-echelon form. ♠

The above algorithm gives you a simple way to obtain the row-echelon form and reduced row-echelon form of a matrix. The main idea is to do row operations in such a way as to end up with a matrix in row-echelon form or reduced row-echelon form. This process is important because the resulting matrix will allow you to describe the solutions to the corresponding linear system of equations in a meaningful way.

In the next example, we look at how to solve a system of equations using the corresponding augmented matrix.

Example B.20: Finding the Solution to a System

Give the complete solution to the following system of equations

$$\begin{array}{ccccccc} 2x & + & 4y & - & 3z & = & -1 \\ 5x & + & 10y & - & 7z & = & -2 \\ 3x & + & 6y & + & 5z & = & 9 \end{array}$$

Solution. The augmented matrix for this system is

$$\left[\begin{array}{rrr|r} 2 & 4 & -3 & -1 \\ 5 & 10 & -7 & -2 \\ 3 & 6 & 5 & 9 \end{array}\right]$$

In order to find the solution to this system, we wish to carry the augmented matrix to reduced row-echelon form. We will do so using Algorithm 1. Notice that the first column is nonzero, so this is our first pivot column. The first entry in the first row, 2, is the first leading entry and it is in the first pivot position. We will use row operations to create zeros in the entries below the 2. First, replace the second row with -5 times the first row plus 2 times the second row. This yields

$$\left[\begin{array}{rrr|r} 2 & 4 & -3 & -1 \\ 0 & 0 & 1 & 1 \\ 3 & 6 & 5 & 9 \end{array}\right]$$

Now, replace the third row with -3 times the first row plus to 2 times the third row. This yields

$$\left[\begin{array}{rrr|r} 2 & 4 & -3 & -1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 21 \end{array}\right]$$

Now the entries in the first column below the pivot position are zeros. We now look for the second pivot column, which in this case is column three. Here, the 1 in the second row and third column is in the pivot position. We need to do just one row operation to create a zero below the 1.

Taking -1 times the second row and adding it to the third row yields

$$\left[\begin{array}{ccc|c} 2 & 4 & -3 & -1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 20 \end{array}\right]$$

We could proceed with the algorithm to carry this matrix to row-echelon form or reduced row-echelon form. However, remember that we are looking for the solutions to the system of equations. Take another look at the third row of the matrix. Notice that it corresponds to the equation

$$0x+0y+0z=20$$

There is no solution to this equation because for all x,y,z, the left side will equal 0 and $0 \neq 20$. This shows there is no solution to the given system of equations. In other words, this system is inconsistent. ♠

The following is another example of how to find the solution to a system of equations by carrying the corresponding augmented matrix to reduced row-echelon form.

Example B.21: An Infinite Set of Solutions

Give the complete solution to the system of equations

$$\begin{array}{rcrcrcr} 3x & - & y & - & 5z & = & 9 \\ & & y & - & 10z & = & 0 \\ -2x & + & y & & & = & -6 \end{array} \tag{2.8}$$

Solution. The augmented matrix of this system is

$$\left[\begin{array}{rrr|r} 3 & -1 & -5 & 9 \\ 0 & 1 & -10 & 0 \\ -2 & 1 & 0 & -6 \end{array}\right]$$

In order to find the solution to this system, we will carry the augmented matrix to reduced row-echelon form, using Algorithm 1. The first column is the first pivot column. We want to use row operations to create zeros beneath the first entry in this column, which is in the first pivot position. Replace the third row with 2 times the first row added to 3 times the third row. This gives

$$\left[\begin{array}{rrr|r} 3 & -1 & -5 & 9 \\ 0 & 1 & -10 & 0 \\ 0 & 1 & -10 & 0 \end{array}\right]$$

Now, we have created zeros beneath the 3 in the first column, so we move on to the second pivot column (which is the second column) and repeat the procedure. Take -1 times the second row and add to the third row.

$$\left[\begin{array}{rrr|r} 3 & -1 & -5 & 9 \\ 0 & 1 & -10 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

The entry below the pivot position in the second column is now a zero. Notice that we have no more pivot columns because we have only two leading entries.

At this stage, we also want the leading entries to be equal to one. To do so, divide the first row by 3.

$$\left[\begin{array}{rrr|r} 1 & -\frac{1}{3} & -\frac{5}{3} & 3 \\ 0 & 1 & -10 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

This matrix is now in row-echelon form.

Let's continue with row operations until the matrix is in reduced row-echelon form. This involves creating zeros above the pivot positions in each pivot column. This requires only one step, which is to add $\frac{1}{3}$ times the second row to the first row.

$$\left[\begin{array}{rrr|r} 1 & 0 & -5 & 3 \\ 0 & 1 & -10 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

This is in reduced row-echelon form, which you should verify using Definition B.13. The equations corresponding to this reduced row-echelon form are

$$\begin{array}{ccccccc} x & - & & & 5z & = & 3 \\ & & y & - & 10z & = & 0 \end{array}$$

or

$$\begin{array}{ccccc} x & = & 3 & + & 5z \\ y & = & & & 10z \end{array}$$

Observe that z is not restrained by any equation. In fact, z can equal any number. For example, we can let $z = t$, where we can choose t to be any number. In this context t is called a **parameter** . Therefore, the solution set of this system is

$$\begin{aligned} x &= 3 + 5t \\ y &= 10t \\ z &= t \end{aligned}$$

where t is arbitrary. The system has an infinite set of solutions which are given by these equations. For any value of t we select, x, y, and z will be given by the above equations. For example, if we choose $t = 4$ then the corresponding solution would be

$$\begin{aligned} x &= 3 + 5(4) = 23 \\ y &= 10(4) = 40 \\ z &= 4 \end{aligned}$$

♠

In Example **??** the solution involved one parameter. It may happen that the solution to a system involves more than one parameter, as shown in the following example.

Example B.22: A Two Parameter Set of Solutions

Find the solution to the system

$$\begin{array}{ccccccccc} x & + & 2y & - & z & + & w & = & 3 \\ x & + & y & - & z & + & w & = & 1 \\ x & + & 3y & - & z & + & w & = & 5 \end{array}$$

Solution. The augmented matrix is

$$\left[\begin{array}{rrrr|r} 1 & 2 & -1 & 1 & 3 \\ 1 & 1 & -1 & 1 & 1 \\ 1 & 3 & -1 & 1 & 5 \end{array}\right]$$

We wish to carry this matrix to row-echelon form. Here, we will outline the row operations used. However, make sure that you understand the steps in terms of Algorithm 1.

Take -1 times the first row and add to the second. Then take -1 times the first row and add to the third. This yields

$$\left[\begin{array}{rrrr|r} 1 & 2 & -1 & 1 & 3 \\ 0 & -1 & 0 & 0 & -2 \\ 0 & 1 & 0 & 0 & 2 \end{array}\right]$$

Now add the second row to the third row and divide the second row by -1.

$$\left[\begin{array}{rrrr|r} 1 & 2 & -1 & 1 & 3 \\ 0 & 1 & 0 & 0 & 2 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right] \tag{2.9}$$

This matrix is in row-echelon form and we can see that x and y correspond to pivot columns, while z and w do not. Therefore, we will assign parameters to the variables z and w. Assign the parameter s to z and the parameter t to w. Then the first row yields the equation $x+2y-s+t=3$, while the second row yields the equation $y=2$. Since $y=2$, the first equation becomes $x+4-s+t=3$ showing that the solution is given by

$$\begin{aligned} x &= -1+s-t \\ y &= 2 \\ z &= s \\ w &= t \end{aligned}$$

It is customary to write this solution in the form

$$\left[\begin{array}{c} x \\ y \\ z \\ w \end{array}\right] = \left[\begin{array}{c} -1+s-t \\ 2 \\ s \\ t \end{array}\right] \tag{2.10}$$

♠

This example shows a system of equations with an infinite solution set which depends on two parameters. It can be less confusing in the case of an infinite solution set to first place the augmented matrix in reduced row-echelon form rather than just row-echelon form before seeking to write down the description of the solution.

In the above steps, this means we don't stop with the row-echelon form in equation 2.9. Instead we first place it in reduced row-echelon form as follows.

$$\left[\begin{array}{rrrr|r} 1 & 0 & -1 & 1 & -1 \\ 0 & 1 & 0 & 0 & 2 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right]$$

Then the solution is $y=2$ from the second row and $x=-1+z-w$ from the first. Thus letting $z=s$ and $w=t$, the solution is given by 2.10.

You can see here that there are two paths to the correct answer, which both yield the same answer. Hence, either approach may be used. The process which we first used in the above solution is called **Gaussian Elimination** This process involves carrying the matrix to row-echelon form, converting back to equations, and using back substitution to find the solution. When you do row operations until you obtain reduced row-echelon form, the process is called **Gauss-Jordan Elimination**.

We have now found solutions for systems of equations with no solution and infinitely many solutions, with one parameter as well as two parameters. Recall the three types of solution sets which we discussed in the previous section; no solution, one solution, and infinitely many solutions. Each of these types of solutions could be identified from the graph of the system. It turns out that we can also identify the type of solution from the reduced row-echelon form of the augmented matrix.

- *No Solution:* In the case where the system of equations has no solution, the row-echelon form of the augmented matrix will have a row of the form

$$\left[\begin{array}{ccc|c} 0 & 0 & 0 & 1 \end{array}\right]$$

  This row indicates that the system is inconsistent and has no solution.

- *One Solution:* In the case where the system of equations has one solution, every column of the coefficient matrix is a pivot column. The following is an example of an augmented matrix in reduced row-echelon form for a system of equations with one solution.

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & 5 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 2 \end{array}\right]$$

- *Infinitely Many Solutions:* In the case where the system of equations has infinitely many solutions, the solution contains parameters. There will be columns of the coefficient matrix which are not pivot columns. The following are examples of augmented matrices in reduced row-echelon form for systems of equations with infinitely many solutions.

$$\left[\begin{array}{ccc|r} 1 & 0 & 0 & 5 \\ 0 & 1 & 2 & -3 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

or

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & 5 \\ 0 & 1 & 0 & -3 \end{array}\right]$$

B.2.3. Uniqueness of the Reduced Row-Echelon Form

As we have seen in earlier sections, we know that every matrix can be brought into reduced row-echelon form by a sequence of elementary row operations. Here we will prove that the resulting matrix is unique; in other words, the resulting matrix in reduced row-echelon form does not depend upon the particular sequence of elementary row operations or the order in which they were performed.

Let A be the augmented matrix of a homogeneous system of linear equations in the variables $x_1, x_2, \cdots, x_n$ which is also in reduced row-echelon form. The matrix A divides the set of variables in two different types. We say that x_i is a *basic variable* whenever A has a leading 1 in column number i, in other words, when column i is a pivot column. Otherwise we say that x_i is a *free variable*.

Recall Example **??**.

Example B.23: Basic and Free Variables

Find the basic and free variables in the system

$$\begin{aligned} x+2y-z+w &= 3 \\ x+y-z+w &= 1 \\ x+3y-z+w &= 5 \end{aligned}$$

Solution. Recall from the solution of Example **??** that the row-echelon form of the augmented matrix of this system is given by

$$\left[\begin{array}{cccc|c} 1 & 2 & -1 & 1 & 3 \\ 0 & 1 & 0 & 0 & 2 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right]$$

You can see that columns 1 and 2 are pivot columns. These columns correspond to variables x and y, making these the basic variables. Columns 3 and 4 are not pivot columns, which means that z and w are free variables.

We can write the solution to this system as

$$\begin{aligned} x &= -1+s-t \\ y &= 2 \\ z &= s \\ w &= t \end{aligned}$$

Here the free variables are written as parameters, and the basic variables are given by linear functions of these parameters. ♠

In general, all solutions can be written in terms of the free variables. In such a description, the free variables can take any values (they become parameters), while the basic variables become simple linear functions of these parameters. Indeed, a basic variable x_i is a linear function of *only* those free variables x_j with $j > i$. This leads to the following observation.

Proposition B.24: Basic and Free Variables

If x_i is a basic variable of a homogeneous system of linear equations, then any solution of the system with $x_j = 0$ for all those free variables x_j with $j > i$ must also have $x_i = 0$.

Using this proposition, we prove a lemma which will be used in the proof of the main result of this section below.

Lemma B.25: Solutions and the Reduced Row-Echelon Form of a Matrix

Let A and B be two distinct augmented matrices for two homogeneous systems of m equations in n variables, such that A and B are each in reduced row-echelon form. Then, the two systems do not have exactly the same solutions.

Now, we say that the matrix B is **equivalent** to the matrix A provided that B can be obtained from A by performing a sequence of elementary row operations beginning with A. The importance of this concept lies in the following result.

Theorem B.26: Equivalent Matrices

The two linear systems of equations corresponding to two equivalent augmented matrices have exactly the same solutions.

The proof of this theorem is left as an exercise.

Now, we can use Lemma B.25 and Theorem B.26 to prove the main result of this section.

Theorem B.27: Uniqueness of the Reduced Row-Echelon Form

Every matrix A is equivalent to a unique matrix in reduced row-echelon form.

According to this theorem we can say that each matrix A has a unique reduced row-echelon form.

B.2.4. Rank and Homogeneous Systems

There is a special type of system which requires additional study. This type of system is called a homogeneous system of equations, which we defined above in Definition B.3. Our focus in this section is to consider what types of solutions are possible for a homogeneous system of equations.

Consider the following definition.

Definition B.28: Trivial Solution

Consider the homogeneous system of equations given by

$$\begin{gathered}
a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n = 0 \\
a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n = 0 \\
\vdots \\
a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n = 0
\end{gathered}$$

Then, $x_1 = 0, x_2 = 0, \cdots, x_n = 0$ is always a solution to this system. We call this the ***trivial solution*** *.*

If the system has a solution in which not all of the $x_1, \cdots, x_n$ are equal to zero, then we call this solution **nontrivial** . The trivial solution does not tell us much about the system, as it says that $0 = 0$! Therefore, when working with homogeneous systems of equations, we want to know when the system has a nontrivial solution.

Suppose we have a homogeneous system of m equations, using n variables, and suppose that $n > m$. In other words, there are more variables than equations. Then, it turns out that this system always has a nontrivial solution. Not only will the system have a nontrivial solution, but it also will have infinitely many solutions. It is also possible, but not required, to have a nontrivial solution if $n = m$ and $n < m$.

Consider the following example.

Example B.29: Solutions to a Homogeneous System of Equations

Find the nontrivial solutions to the following homogeneous system of equations

$$\begin{array}{ccccccc}
2x & + & y & - & z & = & 0 \\
x & + & 2y & - & 2z & = & 0
\end{array}$$

Solution. Notice that this system has $m = 2$ equations and $n = 3$ variables, so $n > m$. Therefore by our previous discussion, we expect this system to have infinitely many solutions.

The process we use to find the solutions for a homogeneous system of equations is the same process we used in the previous section. First, we construct the augmented matrix, given by

$$\left[\begin{array}{ccc|c} 2 & 1 & -1 & 0 \\ 1 & 2 & -2 & 0 \end{array}\right]$$

Then, we carry this matrix to its reduced row-echelon form, given below.

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 0 & 1 & -1 & 0 \end{array}\right]$$

The corresponding system of equations is

$$\begin{aligned} x &= 0 \\ y - z &= 0 \end{aligned}$$

Since z is not restrained by any equation, we know that this variable will become our parameter. Let $z = t$ where t is any number. Therefore, our solution has the form

$$\begin{aligned} x &= 0 \\ y = z &= t \\ z &= t \end{aligned}$$

Hence this system has infinitely many solutions, with one parameter t. ♠

Suppose we were to write the solution to the previous example in another form. Specifically,

$$\begin{aligned} x &= 0 \\ y &= 0 + t \\ z &= 0 + t \end{aligned}$$

can be written as

$$\left[\begin{array}{c} x \\ y \\ z \end{array}\right] = \left[\begin{array}{c} 0 \\ 0 \\ 0 \end{array}\right] + t \left[\begin{array}{c} 0 \\ 1 \\ 1 \end{array}\right]$$

Notice that we have constructed a column from the constants in the solution (all equal to 0), as well as a column corresponding to the coefficients on t in each equation. While we will discuss this form of solution more in further chapters, for now consider the column of coefficients of the parameter t. In this case, this is the column $\left[\begin{array}{c} 0 \\ 1 \\ 1 \end{array}\right]$.

There is a special name for this column, which is **basic solution**. The basic solutions of a system are columns constructed from the coefficients on parameters in the solution. We often denote basic solutions by X_1, X_2 etc., depending on how many solutions occur. Therefore, Example **??** has the basic solution $X_1 = \left[\begin{array}{c} 0 \\ 1 \\ 1 \end{array}\right]$.

We explore this further in the following example.

Example B.30: Basic Solutions of a Homogeneous System

Consider the following homogeneous system of equations.

$$\begin{array}{ccccccc} x & + & 4y & + & 3z & = & 0 \\ 3x & + & 12y & + & 9z & = & 0 \end{array}$$

Find the basic solutions to this system.

Solution. The augmented matrix of this system and the resulting reduced row-echelon form are

$$\left[\begin{array}{rrr|r} 1 & 4 & 3 & 0 \\ 3 & 12 & 9 & 0 \end{array}\right] \rightarrow \cdots \rightarrow \left[\begin{array}{rrr|r} 1 & 4 & 3 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

When written in equations, this system is given by

$$x+4y+3z=0$$

Notice that only x corresponds to a pivot column. In this case, we will have two parameters, one for y and one for z. Let $y=s$ and $z=t$ for any numbers s and t. Then, our solution becomes

$$\begin{aligned} x &= -4s-3t \\ y &= s \\ z &= t \end{aligned}$$

which can be written as

$$\left[\begin{array}{c} x \\ y \\ z \end{array}\right] = \left[\begin{array}{c} 0 \\ 0 \\ 0 \end{array}\right] + s\left[\begin{array}{r} -4 \\ 1 \\ 0 \end{array}\right] + t\left[\begin{array}{r} -3 \\ 0 \\ 1 \end{array}\right]$$

You can see here that we have two columns of coefficients corresponding to parameters, specifically one for s and one for t. Therefore, this system has two basic solutions! These are

$$X_1 = \left[\begin{array}{r} -4 \\ 1 \\ 0 \end{array}\right], X_2 = \left[\begin{array}{r} -3 \\ 0 \\ 1 \end{array}\right]$$

♠

We now present a new definition.

Definition B.31: Linear Combination

Let $X_1, \cdots, X_n, V$ be column matrices. Then V is said to be a ***linear combination*** *of the columns $X_1, \cdots, X_n$ if there exist scalars, $a_1, \cdots, a_n$ such that*

$$V = a_1X_1 + \cdots + a_nX_n$$

A remarkable result of this section is that a linear combination of the basic solutions is again a solution to the system. Even more remarkable is that every solution can be written as a linear combination of these solutions. Therefore, if we take a linear combination of the two solutions to Example **??**, this would also be a solution. For example, we could take the following linear combination

$$3\left[\begin{array}{r} -4 \\ 1 \\ 0 \end{array}\right] + 2\left[\begin{array}{r} -3 \\ 0 \\ 1 \end{array}\right] = \left[\begin{array}{r} -18 \\ 3 \\ 2 \end{array}\right]$$

You should take a moment to verify that

$$\left[\begin{array}{c} x \\ y \\ z \end{array}\right] = \left[\begin{array}{r} -18 \\ 3 \\ 2 \end{array}\right]$$

is in fact a solution to the system in Example **??**.

Another way in which we can find out more information about the solutions of a homogeneous system is to consider the **rank** of the associated coefficient matrix. We now define what is meant by the rank of a matrix.

Definition B.32: Rank of a Matrix

Let A be a matrix and consider any row-echelon form of A. Then, the number r of leading entries of A does not depend on the row-echelon form you choose, and is called the **rank** *of A. We denote it by* $\operatorname{rank}(A)$.

Similarly, we could count the number of pivot positions (or pivot columns) to determine the rank of A.

Example B.33: Finding the Rank of a Matrix

Consider the matrix

$$\left[\begin{array}{ccc} 1 & 2 & 3 \\ 1 & 5 & 9 \\ 2 & 4 & 6 \end{array}\right]$$

What is its rank?

Solution. First, we need to find the reduced row-echelon form of A. Through the usual algorithm, we find that this is

$$\left[\begin{array}{rrr} \boxed{1} & 0 & -1 \\ 0 & \boxed{1} & 2 \\ 0 & 0 & 0 \end{array}\right]$$

Here we have two leading entries, or two pivot positions, shown above in boxes.The rank of A is $r = 2$.

♠

Notice that we would have achieved the same answer if we had found the row-echelon form of A instead of the reduced row-echelon form.

Suppose we have a homogeneous system of m equations in n variables, and suppose that $n > m$. From our above discussion, we know that this system will have infinitely many solutions. If we consider the rank of the coefficient matrix of this system, we can find out even more about the solution. Note that we are looking at just the coefficient matrix, not the entire augmented matrix.

Theorem B.34: Rank and Solutions to a Homogeneous System

Let A be the $m \times n$ coefficient matrix corresponding to a homogeneous system of equations, and suppose A has rank r. Then, the solution to the corresponding system has $n - r$ parameters.

Consider our above Example **??** in the context of this theorem. The system in this example has $m = 2$ equations in $n = 3$ variables. First, because $n > m$, we know that the system has a nontrivial solution, and therefore infinitely many solutions. This tells us that the solution will contain at least one parameter. The rank of the coefficient matrix can tell us even more about the solution! The rank of the coefficient matrix of the system is 1, as it has one leading entry in row-echelon form. Theorem B.34 tells us that the solution will have $n - r = 3 - 1 = 2$ parameters. You can check that this is true in the solution to Example **??**.

Notice that if $n = m$ or $n < m$, it is possible to have either a unique solution (which will be the trivial solution) or infinitely many solutions.

We are not limited to homogeneous systems of equations here. The rank of a matrix can be used to learn about the solutions of any system of linear equations. In the previous section, we discussed that a system of equations can have no solution, a unique solution, or infinitely many solutions. Suppose the system is consistent, whether it is homogeneous or not. The following theorem tells us how we can use the rank to learn about the type of solution we have.

Theorem B.35: Rank and Solutions to a Consistent System of Equations

Let A be the $m \times (n+1)$ augmented matrix corresponding to a consistent system of equations in n variables, and suppose A has rank r. Then

1. *the system has a unique solution if $r = n$*
2. *the system has infinitely many solutions if $r < n$*

We will not present a formal proof of this, but consider the following discussions.

1. *No Solution* The above theorem assumes that the system is consistent, that is, that it has a solution. It turns out that it is possible for the augmented matrix of a system with no solution to have any rank r as long as $r > 1$. Therefore, we must know that the system is consistent in order to use this theorem!

2. *Unique Solution* Suppose $r = n$. Then, there is a pivot position in every column of the coefficient matrix of A. Hence, there is a unique solution.

3. *Infinitely Many Solutions* Suppose $r < n$. Then there are infinitely many solutions. There are less pivot positions (and hence less leading entries) than columns, meaning that not every column is a pivot column. The columns which are *not* pivot columns correspond to parameters. In fact, in this case we have $n - r$ parameters.

C. Matrices

C.1 Matrix Arithmetic

Outcomes

A. *Perform the matrix operations of matrix addition, scalar multiplication, transposition and matrix multiplication. Identify when these operations are not defined. Represent these operations in terms of the entries of a matrix.*
B. *Prove algebraic properties for matrix addition, scalar multiplication, transposition, and matrix multiplication. Apply these properties to manipulate an algebraic expression involving matrices.*
C. *Compute the inverse of a matrix using row operations, and prove identities involving matrix inverses.*
E. *Solve a linear system using matrix algebra.*
F. *Use multiplication by an elementary matrix to apply row operations.*
G. *Write a matrix as a product of elementary matrices.*

You have now solved systems of equations by writing them in terms of an augmented matrix and then doing row operations on this augmented matrix. It turns out that matrices are important not only for systems of equations but also in many applications.

Recall that a **matrix** is a rectangular array of numbers. Several of them are referred to as **matrices**. For example, here is a matrix.

$$\left[\begin{array}{rrrr} 1 & 2 & 3 & 4 \\ 5 & 2 & 8 & 7 \\ 6 & -9 & 1 & 2 \end{array}\right] \tag{3.1}$$

Recall that the size or dimension of a matrix is defined as $m \times n$ where m is the number of rows and n is the number of columns. The above matrix is a 3×4 matrix because there are three rows and four columns. You can remember the columns are like columns in a Greek temple. They stand upright while the rows lay flat like rows made by a tractor in a plowed field.

When specifying the size of a matrix, you always list the number of rows before the number of columns.You might remember that you always list the rows before the columns by using the phrase **Row**man **C**atholic.

Consider the following definition.

Definition C.1: Square Matrix

*A matrix A which has size $n \times n$ is called a **square matrix**. In other words, A is a square matrix if it has the same number of rows and columns.*

There is some notation specific to matrices which we now introduce. We denote the columns of a matrix A by A_j as follows

$$A = \left[\begin{array}{cccc} A_1 & A_2 & \cdots & A_n \end{array} \right]$$

Therefore, A_j is the j^{th} column of A, when counted from left to right.

The individual elements of the matrix are called **entries** or **components** of A. Elements of the matrix are identified according to their position. The $(\mathbf{i},\mathbf{j})$**-entry** of a matrix is the entry in the i^{th} row and j^{th} column. For example, in the matrix 3.1 above, 8 is in position $(2,3)$ (and is called the $(2,3)$-entry) because it is in the second row and the third column.

In order to remember which matrix we are speaking of, we will denote the entry in the i^{th} row and the j^{th} column of matrix A by a_{ij}. Then, we can write A in terms of its entries, as $A = \left[a_{ij} \right]$. Using this notation on the matrix in 3.1, $a_{23} = 8, a_{32} = -9, a_{12} = 2$, etc.

There are various operations which are done on matrices of appropriate sizes. Matrices can be added to and subtracted from other matrices, multiplied by a scalar, and multiplied by other matrices. We will never divide a matrix by another matrix, but we will see later how matrix inverses play a similar role.

In doing arithmetic with matrices, we often define the action by what happens in terms of the entries (or components) of the matrices. Before looking at these operations in depth, consider a few general definitions.

Definition C.2: The Zero Matrix

*The $m \times n$ **zero matrix** is the $m \times n$ matrix having every entry equal to zero. It is denoted by 0.*

One possible zero matrix is shown in the following example.

Example C.3: The Zero Matrix

The 2×3 zero matrix is $0 = \left[\begin{array}{ccc} 0 & 0 & 0 \\ 0 & 0 & 0 \end{array} \right]$.

Note there is a 2×3 zero matrix, a 3×4 zero matrix, etc. In fact there is a zero matrix for every size!

Definition C.4: Equality of Matrices

Let A and B be two $m \times n$ matrices. Then $A = B$ means that for $A = \left[a_{ij} \right]$ and $B = \left[b_{ij} \right]$, $a_{ij} = b_{ij}$ for all $1 \leq i \leq m$ and $1 \leq j \leq n$.

In other words, two matrices are equal exactly when they are the same size and the corresponding entries are identical. Thus

$$\begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix} \neq \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

because they are different sizes. Also,

$$\begin{bmatrix} 0 & 1 \\ 3 & 2 \end{bmatrix} \neq \begin{bmatrix} 1 & 0 \\ 2 & 3 \end{bmatrix}$$

because, although they are the same size, their corresponding entries are not identical.

In the following section, we explore addition of matrices.

C.1.1. Addition of Matrices

When adding matrices, all matrices in the sum need have the same size. For example,

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 2 \end{bmatrix}$$

and

$$\begin{bmatrix} -1 & 4 & 8 \\ 2 & 8 & 5 \end{bmatrix}$$

cannot be added, as one has size 3×2 while the other has size 2×3.

However, the addition

$$\begin{bmatrix} 4 & 6 & 3 \\ 5 & 0 & 4 \\ 11 & -2 & 3 \end{bmatrix} + \begin{bmatrix} 0 & 5 & 0 \\ 4 & -4 & 14 \\ 1 & 2 & 6 \end{bmatrix}$$

is possible.

The formal definition is as follows.

Definition C.5: Addition of Matrices

Let $A = \left[a_{ij}\right]$ and $B = \left[b_{ij}\right]$ be two $m \times n$ matrices. Then $A + B = C$ where C is the $m \times n$ matrix $C = \left[c_{ij}\right]$ defined by

$$c_{ij} = a_{ij} + b_{ij}$$

This definition tells us that when adding matrices, we simply add corresponding entries of the matrices. This is demonstrated in the next example.

Example C.6: Addition of Matrices of Same Size

Add the following matrices, if possible.

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 1 & 0 & 4 \end{bmatrix}, B = \begin{bmatrix} 5 & 2 & 3 \\ -6 & 2 & 1 \end{bmatrix}$$

Solution. Notice that both A and B are of size 2×3. Since A and B are of the same size, the addition is possible. Using Definition C.5, the addition is done as follows.

$$A+B = \begin{bmatrix} 1 & 2 & 3 \\ 1 & 0 & 4 \end{bmatrix} + \begin{bmatrix} 5 & 2 & 3 \\ -6 & 2 & 1 \end{bmatrix} = \begin{bmatrix} 1+5 & 2+2 & 3+3 \\ 1+-6 & 0+2 & 4+1 \end{bmatrix} = \begin{bmatrix} 6 & 4 & 6 \\ -5 & 2 & 5 \end{bmatrix}$$

♠

Addition of matrices obeys very much the same properties as normal addition with numbers. Note that when we write for example $A+B$ then we assume that both matrices are of equal size so that the operation is indeed possible.

Proposition C.7: Properties of Matrix Addition

Let A, B and C be matrices. Then, the following properties hold.

- *Commutative Law of Addition*
$$A+B = B+A \tag{3.2}$$
- *Associative Law of Addition*
$$(A+B)+C = A+(B+C) \tag{3.3}$$
- *Existence of an Additive Identity*
$$\begin{gathered}\text{There exists a zero matrix } 0 \text{ such that} \\ A+0 = A\end{gathered} \tag{3.4}$$
- *Existence of an Additive Inverse*
$$\begin{gathered}\text{There exists a matrix } -A \text{ such that} \\ A+(-A) = 0\end{gathered} \tag{3.5}$$

We call the zero matrix in 3.4 the **additive identity**. Similarly, we call the matrix $-A$ in 3.5 the **additive inverse**. $-A$ is defined to equal $(-1)A = [-a_{ij}]$. In other words, every entry of A is multiplied by -1. In the next section we will study scalar multiplication in more depth to understand what is meant by $(-1)A$.

C.1.2. Scalar Multiplication of Matrices

Recall that we use the word *scalar* when referring to numbers. Therefore, *scalar multiplication of a matrix* is the multiplication of a matrix by a number. To illustrate this concept, consider the following example in which a matrix is multiplied by the scalar 3.

$$3\left[\begin{array}{rrrr} 1 & 2 & 3 & 4 \\ 5 & 2 & 8 & 7 \\ 6 & -9 & 1 & 2 \end{array}\right] = \left[\begin{array}{rrrr} 3 & 6 & 9 & 12 \\ 15 & 6 & 24 & 21 \\ 18 & -27 & 3 & 6 \end{array}\right]$$

The new matrix is obtained by multiplying every entry of the original matrix by the given scalar.

The formal definition of scalar multiplication is as follows.

Definition C.8: Scalar Multiplication of Matrices

If $A = \left[a_{ij}\right]$ *and* k *is a scalar, then* $kA = \left[ka_{ij}\right]$.

Consider the following example.

Example C.9: Effect of Multiplication by a Scalar

Find the result of multiplying the following matrix A *by* 7.

$$A = \left[\begin{array}{rr} 2 & 0 \\ 1 & -4 \end{array}\right]$$

Solution. By Definition C.8, we multiply each element of A by 7. Therefore,

$$7A = 7\left[\begin{array}{rr} 2 & 0 \\ 1 & -4 \end{array}\right] = \left[\begin{array}{rr} 7(2) & 7(0) \\ 7(1) & 7(-4) \end{array}\right] = \left[\begin{array}{rr} 14 & 0 \\ 7 & -28 \end{array}\right]$$

♠

Similarly to addition of matrices, there are several properties of scalar multiplication which hold.

Proposition C.10: Properties of Scalar Multiplication

Let A, B be matrices, and k, p be scalars. Then, the following properties hold.

- *Distributive Law over Matrix Addition*

$$k(A+B) = kA + kB$$

- *Distributive Law over Scalar Addition*

$$(k+p)A = kA + pA$$

- *Associative Law for Scalar Multiplication*

$$k(pA) = (kp)A$$

- *Rule for Multiplication by* 1

$$1A = A$$

The proof of this proposition is similar to the proof of Proposition C.7 and is left an exercise to the reader.

C.1.3. Multiplication of Matrices

The next important matrix operation we will explore is multiplication of matrices. The operation of matrix multiplication is one of the most important and useful of the matrix operations. Throughout this section, we will also demonstrate how matrix multiplication relates to linear systems of equations.

First, we provide a formal definition of row and column vectors.

Definition C.11: Row and Column Vectors

Matrices of size $n \times 1$ or $1 \times n$ are called **vectors**. *If X is such a matrix, then we write x_i to denote the entry of X in the i^{th} row of a column matrix, or the i^{th} column of a row matrix.*
The $n \times 1$ matrix

$$X = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}$$

is called a **column vector.** *The $1 \times n$ matrix*

$$X = \begin{bmatrix} x_1 & \cdots & x_n \end{bmatrix}$$

is called a **row vector**.

We may simply use the term **vector** throughout this text to refer to either a column or row vector. If we do so, the context will make it clear which we are referring to.

In this chapter, we will again use the notion of linear combination of vectors as in Definition D.7. In this context, a linear combination is a sum consisting of vectors multiplied by scalars. For example,

$$\begin{bmatrix} 50 \\ 122 \end{bmatrix} = 7\begin{bmatrix} 1 \\ 4 \end{bmatrix} + 8\begin{bmatrix} 2 \\ 5 \end{bmatrix} + 9\begin{bmatrix} 3 \\ 6 \end{bmatrix}$$

is a linear combination of three vectors.

It turns out that we can express any system of linear equations as a linear combination of vectors. In fact, the vectors that we will use are just the columns of the corresponding augmented matrix!

Definition C.12: The Vector Form of a System of Linear Equations

Suppose we have a system of equations given by

$$\begin{gathered} a_{11}x_1 + \cdots + a_{1n}x_n = b_1 \\ \vdots \\ a_{m1}x_1 + \cdots + a_{mn}x_n = b_m \end{gathered}$$

We can express this system in ***vector form*** *which is as follows:*

$$x_1 \begin{bmatrix} a_{11} \\ a_{21} \\ \vdots \\ a_{m1} \end{bmatrix} + x_2 \begin{bmatrix} a_{12} \\ a_{22} \\ \vdots \\ a_{m2} \end{bmatrix} + \cdots + x_n \begin{bmatrix} a_{1n} \\ a_{2n} \\ \vdots \\ a_{mn} \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix}$$

Notice that each vector used here is one column from the corresponding augmented matrix. There is one vector for each variable in the system, along with the constant vector.

The first important form of matrix multiplication is multiplying a matrix by a vector. Consider the product given by

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} \begin{bmatrix} 7 \\ 8 \\ 9 \end{bmatrix}$$

We will soon see that this equals

$$7\begin{bmatrix} 1 \\ 4 \end{bmatrix} + 8\begin{bmatrix} 2 \\ 5 \end{bmatrix} + 9\begin{bmatrix} 3 \\ 6 \end{bmatrix} = \begin{bmatrix} 50 \\ 122 \end{bmatrix}$$

In general terms,

$$\begin{aligned} \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} &= x_1 \begin{bmatrix} a_{11} \\ a_{21} \end{bmatrix} + x_2 \begin{bmatrix} a_{12} \\ a_{22} \end{bmatrix} + x_3 \begin{bmatrix} a_{13} \\ a_{23} \end{bmatrix} \\ &= \begin{bmatrix} a_{11}x_1 + a_{12}x_2 + a_{13}x_3 \\ a_{21}x_1 + a_{22}x_2 + a_{23}x_3 \end{bmatrix} \end{aligned}$$

Thus you take x_1 times the first column, add to x_2 times the second column, and finally x_3 times the third column. The above sum is a linear combination of the columns of the matrix. When you multiply a matrix on the left by a vector on the right, the numbers making up the vector are just the scalars to be used in the linear combination of the columns as illustrated above.

Here is the formal definition of how to multiply an $m \times n$ matrix by an $n \times 1$ column vector.

Definition C.13: Multiplication of Vector by Matrix

Let $A = \left[a_{ij}\right]$ be an $m \times n$ matrix and let X be an $n \times 1$ matrix given by

$$A = [A_1 \cdots A_n], X = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}$$

Then the product AX is the $m \times 1$ column vector which equals the following linear combination of the columns of A:

$$x_1 A_1 + x_2 A_2 + \cdots + x_n A_n = \sum_{j=1}^{n} x_j A_j$$

If we write the columns of A in terms of their entries, they are of the form

$$A_j = \begin{bmatrix} a_{1j} \\ a_{2j} \\ \vdots \\ a_{mj} \end{bmatrix}$$

Then, we can write the product AX as

$$AX = x_1 \begin{bmatrix} a_{11} \\ a_{21} \\ \vdots \\ a_{m1} \end{bmatrix} + x_2 \begin{bmatrix} a_{12} \\ a_{22} \\ \vdots \\ a_{m2} \end{bmatrix} + \cdots + x_n \begin{bmatrix} a_{1n} \\ a_{2n} \\ \vdots \\ a_{mn} \end{bmatrix}$$

Note that multiplication of an $m \times n$ matrix and an $n \times 1$ vector produces an $m \times 1$ vector.

Here is an example.

Example C.14: A Vector Multiplied by a Matrix

Compute the product AX *for*

$$A = \begin{bmatrix} 1 & 2 & 1 & 3 \\ 0 & 2 & 1 & -2 \\ 2 & 1 & 4 & 1 \end{bmatrix}, X = \begin{bmatrix} 1 \\ 2 \\ 0 \\ 1 \end{bmatrix}$$

Solution. We will use Definition C.13 to compute the product. Therefore, we compute the product AX as follows.

$$\begin{aligned} & 1 \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix} + 2 \begin{bmatrix} 2 \\ 2 \\ 1 \end{bmatrix} + 0 \begin{bmatrix} 1 \\ 1 \\ 4 \end{bmatrix} + 1 \begin{bmatrix} 3 \\ -2 \\ 1 \end{bmatrix} \\ = & \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix} + \begin{bmatrix} 4 \\ 4 \\ 2 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} 3 \\ -2 \\ 1 \end{bmatrix} \\ = & \begin{bmatrix} 8 \\ 2 \\ 5 \end{bmatrix} \end{aligned}$$

♠

Using the above operation, we can also write a system of linear equations in **matrix form**. In this form, we express the system as a matrix multiplied by a vector. Consider the following definition.

Definition C.15: The Matrix Form of a System of Linear Equations

Suppose we have a system of equations given by

$$\begin{gathered} a_{11}x_1 + \cdots + a_{1n}x_n = b_1 \\ a_{21}x_1 + \cdots + a_{2n}x_n = b_2 \\ \vdots \\ a_{m1}x_1 + \cdots + a_{mn}x_n = b_m \end{gathered}$$

Then we can express this system in ***matrix form*** *as follows.*

$$\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix}$$

The expression $AX = B$ is also known as the **Matrix Form** of the corresponding system of linear equations. The matrix A is simply the coefficient matrix of the system, the vector X is the column vector constructed from the variables of the system, and finally the vector B is the column vector constructed from the constants of the system. It is important to note that any system of linear equations can be written in this form.

Notice that if we write a homogeneous system of equations in matrix form, it would have the form $AX = 0$, for the zero vector 0.

You can see from this definition that a vector

$$X = \left[\begin{array}{c} x_1 \\ x_2 \\ \vdots \\ x_n \end{array} \right]$$

will satisfy the equation $AX = B$ only when the entries $x_1, x_2, \cdots, x_n$ of the vector X are solutions to the original system.

Now that we have examined how to multiply a matrix by a vector, we wish to consider the case where we multiply two matrices of more general sizes, although these sizes still need to be appropriate as we will see. For example, in Example **??**, we multiplied a 3×4 matrix by a 4×1 vector. We want to investigate how to multiply other sizes of matrices.

We have not yet given any conditions on when matrix multiplication is possible! For matrices A and B, in order to form the product AB, the number of columns of A must equal the number of rows of B. Consider a product AB where A has size $m \times n$ and B has size $n \times p$. Then, the product in terms of size of matrices is given by

$$(m \times \overbrace{n)\,(n}^{\text{these must match!}} \times p) = m \times p$$

Note the two outside numbers give the size of the product. One of the most important rules regarding matrix multiplication is the following. If the two middle numbers don't match, you can't multiply the matrices!

When the number of columns of A equals the number of rows of B the two matrices are said to be **conformable** and the product AB is obtained as follows.

Definition C.16: Multiplication of Two Matrices

Let A be an $m \times n$ matrix and let B be an $n \times p$ matrix of the form

$$B = [B_1 \cdots B_p]$$

where $B_1, ..., B_p$ are the $n \times 1$ columns of B. Then the $m \times p$ matrix AB is defined as follows:

$$AB = A\,[B_1 \cdots B_p] = [(AB)_1 \cdots (AB)_p]$$

where $(AB)_k$ is an $m \times 1$ matrix or column vector which gives the k^{th} column of AB.

Consider the following example.

Example C.17: Multiplying Two Matrices

Find AB if possible.

$$A = \begin{bmatrix} 1 & 2 & 1 \\ 0 & 2 & 1 \end{bmatrix}, B = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 3 & 1 \\ -2 & 1 & 1 \end{bmatrix}$$

Solution. The first thing you need to verify when calculating a product is whether the multiplication is possible. The first matrix has size 2×3 and the second matrix has size 3×3. The inside numbers are equal, so A and B are conformable matrices. According to the above discussion AB will be a 2×3 matrix. Definition C.16 gives us a way to calculate each column of AB, as follows.

$$\left[\overbrace{\begin{bmatrix} 1 & 2 & 1 \\ 0 & 2 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ -2 \end{bmatrix}}^{\text{First column}}, \overbrace{\begin{bmatrix} 1 & 2 & 1 \\ 0 & 2 & 1 \end{bmatrix} \begin{bmatrix} 2 \\ 3 \\ 1 \end{bmatrix}}^{\text{Second column}}, \overbrace{\begin{bmatrix} 1 & 2 & 1 \\ 0 & 2 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}}^{\text{Third column}} \right]$$

You know how to multiply a matrix times a vector, using Definition C.13 for each of the three columns. Thus

$$\begin{bmatrix} 1 & 2 & 1 \\ 0 & 2 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 0 \\ 0 & 3 & 1 \\ -2 & 1 & 1 \end{bmatrix} = \begin{bmatrix} -1 & 9 & 3 \\ -2 & 7 & 3 \end{bmatrix}$$

♠

Since vectors are simply $n \times 1$ or $1 \times m$ matrices, we can also multiply a vector by another vector.

Example C.18: Vector Times Vector Multiplication

Multiply if possible $\begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 1 & 0 \end{bmatrix}$.

Solution. In this case we are multiplying a matrix of size 3×1 by a matrix of size 1×4. The inside numbers match so the product is defined. Note that the product will be a matrix of size 3×4. Using Definition C.16, we can compute this product as follows

$$\begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 1 & 0 \end{bmatrix} = \left[\overbrace{\begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix} \begin{bmatrix} 1 \end{bmatrix}}^{\text{First column}}, \overbrace{\begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix} \begin{bmatrix} 2 \end{bmatrix}}^{\text{Second column}}, \overbrace{\begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix} \begin{bmatrix} 1 \end{bmatrix}}^{\text{Third column}}, \overbrace{\begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix} \begin{bmatrix} 0 \end{bmatrix}}^{\text{Fourth column}} \right]$$

You can use Definition C.13 to verify that this product is

$$\left[\begin{array}{cccc} 1 & 2 & 1 & 0 \\ 2 & 4 & 2 & 0 \\ 1 & 2 & 1 & 0 \end{array}\right]$$

♠

Example C.19: A Multiplication Which is Not Defined

Find BA if possible.

$$B = \left[\begin{array}{ccc} 1 & 2 & 0 \\ 0 & 3 & 1 \\ -2 & 1 & 1 \end{array}\right], A = \left[\begin{array}{ccc} 1 & 2 & 1 \\ 0 & 2 & 1 \end{array}\right]$$

Solution. First check if it is possible. This product is of the form $(3\times 3)(2\times 3)$. The inside numbers do not match and so you can't do this multiplication. ♠

In this case, we say that the multiplication is not defined. Notice that these are the same matrices which we used in Example **??**. In this example, we tried to calculate BA instead of AB. This demonstrates another property of matrix multiplication. While the product AB maybe be defined, we cannot assume that the product BA will be possible. Therefore, it is important to always check that the product is defined before carrying out any calculations.

Earlier, we defined the zero matrix 0 to be the matrix (of appropriate size) containing zeros in all entries. Consider the following example for multiplication by the zero matrix.

Example C.20: Multiplication by the Zero Matrix

Compute the product $A0$ for the matrix

$$A = \left[\begin{array}{cc} 1 & 2 \\ 3 & 4 \end{array}\right]$$

and the 2×2 zero matrix given by

$$0 = \left[\begin{array}{cc} 0 & 0 \\ 0 & 0 \end{array}\right]$$

Solution. In this product, we compute

$$\left[\begin{array}{cc} 1 & 2 \\ 3 & 4 \end{array}\right]\left[\begin{array}{cc} 0 & 0 \\ 0 & 0 \end{array}\right] = \left[\begin{array}{cc} 0 & 0 \\ 0 & 0 \end{array}\right]$$

Hence, $A0 = 0$. ♠

Notice that we could also multiply A by the 2×1 zero vector given by $\begin{bmatrix} 0 \\ 0 \end{bmatrix}$. The result would be the 2×1 zero vector. Therefore, it is always the case that $A0 = 0$, for an appropriately sized zero matrix or vector.

C.1.4. The ij^{th} Entry of a Product

In previous sections, we used the entries of a matrix to describe the action of matrix addition and scalar multiplication. We can also study matrix multiplication using the entries of matrices.

What is the ij^{th} entry of AB? It is the entry in the i^{th} row and the j^{th} column of the product AB.

Now if A is $m\times n$ and B is $n\times p$, then we know that the product AB has the form

$$\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix} \begin{bmatrix} b_{11} & b_{12} & \cdots & b_{1j} & \cdots & b_{1p} \\ b_{21} & b_{22} & \cdots & b_{2j} & \cdots & b_{2p} \\ \vdots & \vdots & & \vdots & & \vdots \\ b_{n1} & b_{n2} & \cdots & b_{nj} & \cdots & b_{np} \end{bmatrix}$$

The j^{th} column of AB is of the form

$$\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix} \begin{bmatrix} b_{1j} \\ b_{2j} \\ \vdots \\ b_{nj} \end{bmatrix}$$

which is an $m\times 1$ column vector. It is calculated by

$$b_{1j}\begin{bmatrix} a_{11} \\ a_{21} \\ \vdots \\ a_{m1} \end{bmatrix} + b_{2j}\begin{bmatrix} a_{12} \\ a_{22} \\ \vdots \\ a_{m2} \end{bmatrix} + \cdots + b_{nj}\begin{bmatrix} a_{1n} \\ a_{2n} \\ \vdots \\ a_{mn} \end{bmatrix}$$

Therefore, the ij^{th} entry is the entry in row i of this vector. This is computed by

$$a_{i1}b_{1j} + a_{i2}b_{2j} + \cdots + a_{in}b_{nj} = \sum_{k=1}^{n} a_{ik}b_{kj}$$

The following is the formal definition for the ij^{th} entry of a product of matrices.

Definition C.21: The ij^{th} Entry of a Product

Let $A = [a_{ij}]$ be an $m \times n$ matrix and let $B = [b_{ij}]$ be an $n \times p$ matrix. Then AB is an $m \times p$ matrix and the (i, j)-entry of AB is defined as

$$(AB)_{ij} = \sum_{k=1}^{n} a_{ik} b_{kj}$$

Another way to write this is

$$(AB)_{ij} = \left[\begin{array}{cccc} a_{i1} & a_{i2} & \cdots & a_{in} \end{array} \right] \left[\begin{array}{c} b_{1j} \\ b_{2j} \\ \vdots \\ b_{nj} \end{array} \right] = a_{i1}b_{1j} + a_{i2}b_{2j} + \cdots + a_{in}b_{nj}$$

In other words, to find the (i, j)-entry of the product AB, or $(AB)_{ij}$, you multiply the i^{th} row of A, on the left by the j^{th} column of B. To express AB in terms of its entries, we write $AB = [(AB)_{ij}]$.

Consider the following example.

Example C.22: The Entries of a Product

Compute AB if possible. If it is, find the $(3, 2)$-entry of AB using Definition C.21.

$$A = \left[\begin{array}{cc} 1 & 2 \\ 3 & 1 \\ 2 & 6 \end{array} \right], B = \left[\begin{array}{ccc} 2 & 3 & 1 \\ 7 & 6 & 2 \end{array} \right]$$

Solution. First check if the product is possible. It is of the form $(3 \times 2)(2 \times 3)$ and since the inside numbers match, it is possible to do the multiplication. The result should be a 3×3 matrix. We can first compute AB:

$$\left[\left[\begin{array}{cc} 1 & 2 \\ 3 & 1 \\ 2 & 6 \end{array} \right] \left[\begin{array}{c} 2 \\ 7 \end{array} \right], \left[\begin{array}{cc} 1 & 2 \\ 3 & 1 \\ 2 & 6 \end{array} \right] \left[\begin{array}{c} 3 \\ 6 \end{array} \right], \left[\begin{array}{cc} 1 & 2 \\ 3 & 1 \\ 2 & 6 \end{array} \right] \left[\begin{array}{c} 1 \\ 2 \end{array} \right] \right]$$

where the commas separate the columns in the resulting product. Thus the above product equals

$$\left[\begin{array}{ccc} 16 & 15 & 5 \\ 13 & 15 & 5 \\ 46 & 42 & 14 \end{array} \right]$$

which is a 3×3 matrix as desired. Thus, the $(3, 2)$-entry equals 42.

Now using Definition C.21, we can find that the $(3,2)$-entry equals

$$\begin{aligned}\sum_{k=1}^{2} a_{3k}b_{k2} &= a_{31}b_{12} + a_{32}b_{22} \\ &= 2\times 3 + 6\times 6 = 42\end{aligned}$$

Consulting our result for AB above, this is correct!

You may wish to use this method to verify that the rest of the entries in AB are correct. ♠

Here is another example.

Example C.23: Finding the Entries of a Product

Determine if the product AB is defined. If it is, find the $(2,1)$-entry of the product.

$$A = \begin{bmatrix} 2 & 3 & 1 \\ 7 & 6 & 2 \\ 0 & 0 & 0 \end{bmatrix}, B = \begin{bmatrix} 1 & 2 \\ 3 & 1 \\ 2 & 6 \end{bmatrix}$$

Solution. This product is of the form $(3\times 3)(3\times 2)$. The middle numbers match so the matrices are conformable and it is possible to compute the product.

We want to find the $(2,1)$-entry of AB, that is, the entry in the second row and first column of the product. We will use Definition C.21, which states

$$(AB)_{ij} = \sum_{k=1}^{n} a_{ik}b_{kj}$$

In this case, $n = 3$, $i = 2$ and $j = 1$. Hence the $(2,1)$-entry is found by computing

$$(AB)_{21} = \sum_{k=1}^{3} a_{2k}b_{k1} = \begin{bmatrix} a_{21} & a_{22} & a_{23} \end{bmatrix} \begin{bmatrix} b_{11} \\ b_{21} \\ b_{31} \end{bmatrix}$$

Substituting in the appropriate values, this product becomes

$$\begin{bmatrix} a_{21} & a_{22} & a_{23} \end{bmatrix} \begin{bmatrix} b_{11} \\ b_{21} \\ b_{31} \end{bmatrix} = \begin{bmatrix} 7 & 6 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 3 \\ 2 \end{bmatrix} = 1\times 7 + 3\times 6 + 2\times 2 = 29$$

Hence, $(AB)_{21} = 29$.

You should take a moment to find a few other entries of AB. You can multiply the matrices to check that your answers are correct. The product AB is given by

$$AB = \begin{bmatrix} 13 & 13 \\ 29 & 32 \\ 0 & 0 \end{bmatrix}$$

♠

C.1.5. Properties of Matrix Multiplication

As pointed out above, it is sometimes possible to multiply matrices in one order but not in the other order. However, even if both AB and BA are defined, they may not be equal.

Example C.24: Matrix Multiplication is Not Commutative

Compare the products AB and BA, for matrices $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, B = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$

Solution. First, notice that A and B are both of size 2×2. Therefore, both products AB and BA are defined. The first product, AB is

$$AB = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 2 & 1 \\ 4 & 3 \end{bmatrix}$$

The second product, BA is

$$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} = \begin{bmatrix} 3 & 4 \\ 1 & 2 \end{bmatrix}$$

Therefore, $AB \neq BA$. ♠

This example illustrates that you cannot assume $AB = BA$ even when multiplication is defined in both orders. If for some matrices A and B it is true that $AB = BA$, then we say that A and B **commute**. This is one important property of matrix multiplication.

The following are other important properties of matrix multiplication. Notice that these properties hold only when the size of matrices are such that the products are defined.

Proposition C.25: Properties of Matrix Multiplication

The following hold for matrices A, B, and C and for scalars r and s,

$$A(rB + sC) = r(AB) + s(AC) \tag{3.6}$$

$$(B + C)A = BA + CA \tag{3.7}$$

$$A(BC) = (AB)C \tag{3.8}$$

C.1.6. The Transpose

Another important operation on matrices is that of taking the **transpose**. For a matrix A, we denote the **transpose** of A by A^T. Before formally defining the transpose, we explore this operation on the following matrix.

$$\left[\begin{array}{cc} 1 & 4 \\ 3 & 1 \\ 2 & 6 \end{array}\right]^T = \left[\begin{array}{ccc} 1 & 3 & 2 \\ 4 & 1 & 6 \end{array}\right]$$

What happened? The first column became the first row and the second column became the second row. Thus the 3×2 matrix became a 2×3 matrix. The number 4 was in the first row and the second column and it ended up in the second row and first column.

The definition of the transpose is as follows.

Definition C.26: The Transpose of a Matrix

*Let A be an $m \times n$ matrix. Then A^T, the **transpose** of A, denotes the $n \times m$ matrix given by*

$$A^T = \left[a_{ij}\right]^T = \left[a_{ji}\right]$$

The (i, j)-entry of A becomes the (j, i)-entry of A^T.

Consider the following example.

Example C.27: The Transpose of a Matrix

Calculate A^T for the following matrix

$$A = \left[\begin{array}{rrr} 1 & 2 & -6 \\ 3 & 5 & 4 \end{array}\right]$$

Solution. By Definition C.26, we know that for $A = \left[a_{ij}\right]$, $A^T = \left[a_{ji}\right]$. In other words, we switch the row and column location of each entry. The $(1, 2)$-entry becomes the $(2, 1)$-entry.

Thus,

$$A^T = \left[\begin{array}{rr} 1 & 3 \\ 2 & 5 \\ -6 & 4 \end{array}\right]$$

Notice that A is a 2×3 matrix, while A^T is a 3×2 matrix. ♠

The transpose of a matrix has the following important properties .

Lemma C.28: Properties of the Transpose of a Matrix

Let A be an $m \times n$ matrix, B an $n \times p$ matrix, and r and s scalars. Then

1. $\left(A^T\right)^T = A$
2. $(AB)^T = B^T A^T$
3. $(rA + sB)^T = rA^T + sB^T$

The transpose of a matrix is related to other important topics. Consider the following definition.

Definition C.29: Symmetric and Skew Symmetric Matrices

An $n \times n$ matrix A is said to be ***symmetric*** *if $A = A^T$. It is said to be* ***skew symmetric*** *if $A = -A^T$.*

We will explore these definitions in the following examples.

Example C.30: Symmetric Matrices

Let

$$A = \begin{bmatrix} 2 & 1 & 3 \\ 1 & 5 & -3 \\ 3 & -3 & 7 \end{bmatrix}$$

Use Definition C.29 to show that A is symmetric.

Solution. By Definition C.29, we need to show that $A = A^T$. Now, using Definition C.26,

$$A^T = \begin{bmatrix} 2 & 1 & 3 \\ 1 & 5 & -3 \\ 3 & -3 & 7 \end{bmatrix}$$

Hence, $A = A^T$, so A is symmetric. ♠

Example C.31: A Skew Symmetric Matrix

Let

$$A = \begin{bmatrix} 0 & 1 & 3 \\ -1 & 0 & 2 \\ -3 & -2 & 0 \end{bmatrix}$$

Show that A is skew symmetric.

Solution. By Definition C.29,

$$A^T = \begin{bmatrix} 0 & -1 & -3 \\ 1 & 0 & -2 \\ 3 & 2 & 0 \end{bmatrix}$$

You can see that each entry of A^T is equal to -1 times the same entry of A. Hence, $A^T = -A$ and so by Definition C.29, A is skew symmetric. ♠

C.1.7. The Identity and Inverses

There is a special matrix, denoted I, which is called to as the **identity matrix**. The identity matrix is always a square matrix, and it has the property that there are ones down the main diagonal and zeroes elsewhere. Here are some identity matrices of various sizes.

$$[1], \left[\begin{array}{cc} 1 & 0 \\ 0 & 1 \end{array}\right], \left[\begin{array}{ccc} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{array}\right], \left[\begin{array}{cccc} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{array}\right]$$

The first is the 1×1 identity matrix, the second is the 2×2 identity matrix, and so on. By extension, you can likely see what the $n \times n$ identity matrix would be. When it is necessary to distinguish which size of identity matrix is being discussed, we will use the notation I_n for the $n \times n$ identity matrix.

The identity matrix is so important that there is a special symbol to denote the ij^{th} entry of the identity matrix. This symbol is given by $I_{ij} = \delta_{ij}$ where δ_{ij} is the **Kronecker symbol** defined by

$$\delta_{ij} = \begin{cases} 1 \text{ if } i = j \\ 0 \text{ if } i \neq j \end{cases}$$

I_n is called the **identity matrix** because it is a **multiplicative identity** in the following sense.

Lemma C.32: Multiplication by the Identity Matrix

Suppose A is an $m \times n$ matrix and I_n is the $n \times n$ identity matrix. Then $AI_n = A$. If I_m is the $m \times m$ identity matrix, it also follows that $I_m A = A$.

We now define the matrix operation which in some ways plays the role of division.

Definition C.33: The Inverse of a Matrix

*A square $n \times n$ matrix A is said to have an **inverse** A^{-1} if and only if*

$$AA^{-1} = A^{-1}A = I_n$$

*In this case, the matrix A is called **invertible**.*

Such a matrix A^{-1} will have the same size as the matrix A. It is very important to observe that the inverse of a matrix, if it exists, is unique. Another way to think of this is that if it acts like the inverse, then it **is** the inverse.

Theorem C.34: Uniqueness of Inverse

Suppose A is an $n \times n$ matrix such that an inverse A^{-1} exists. Then there is only one such inverse matrix. That is, given any matrix B such that $AB = BA = I$, $B = A^{-1}$.

The next example demonstrates how to check the inverse of a matrix.

Example C.35: Verifying the Inverse of a Matrix

Let $A = \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix}$. *Show* $\begin{bmatrix} 2 & -1 \\ -1 & 1 \end{bmatrix}$ *is the inverse of* A.

Solution. To check this, multiply

$$\begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} 2 & -1 \\ -1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I$$

and

$$\begin{bmatrix} 2 & -1 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I$$

showing that this matrix is indeed the inverse of A. ♠

Unlike ordinary multiplication of numbers, it can happen that $A \neq 0$ but A may fail to have an inverse. This is illustrated in the following example.

Example C.36: A Nonzero Matrix With No Inverse

Let $A = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$. *Show that* A *does not have an inverse.*

Solution. One might think A would have an inverse because it does not equal zero. However, note that

$$\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} -1 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

If A^{-1} existed, we would have the following

$$\begin{aligned}
\begin{bmatrix} 0 \\ 0 \end{bmatrix} &= A^{-1}\left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}\right) \\
&= A^{-1}\left(A\begin{bmatrix} -1 \\ 1 \end{bmatrix}\right) \\
&= \left(A^{-1}A\right)\begin{bmatrix} -1 \\ 1 \end{bmatrix} \\
&= I\begin{bmatrix} -1 \\ 1 \end{bmatrix} \\
&= \begin{bmatrix} -1 \\ 1 \end{bmatrix}
\end{aligned}$$

This says that

$$\begin{bmatrix} 0 \\ 0 \end{bmatrix} = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

which is impossible! Therefore, A does not have an inverse. ♠

In the next section, we will explore how to find the inverse of a matrix, if it exists.

C.1.8. Finding the Inverse of a Matrix

In Example **??**, we were given A^{-1} and asked to verify that this matrix was in fact the inverse of A. In this section, we explore how to find A^{-1}.

Let

$$A = \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix}$$

as in Example **??**. In order to find A^{-1}, we need to find a matrix $\begin{bmatrix} x & z \\ y & w \end{bmatrix}$ such that

$$\begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix}\begin{bmatrix} x & z \\ y & w \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

We can multiply these two matrices, and see that in order for this equation to be true, we must find the solution to the systems of equations,

$$\begin{aligned} x + y &= 1 \\ x + 2y &= 0 \end{aligned}$$

and

$$\begin{aligned} z + w &= 0 \\ z + 2w &= 1 \end{aligned}$$

Writing the augmented matrix for these two systems gives

$$\left[\begin{array}{cc|c} 1 & 1 & 1 \\ 1 & 2 & 0 \end{array}\right]$$

for the first system and

$$\left[\begin{array}{cc|c} 1 & 1 & 0 \\ 1 & 2 & 1 \end{array}\right] \tag{3.9}$$

for the second.

Let's solve the first system. Take -1 times the first row and add to the second to get

$$\left[\begin{array}{cc|r} 1 & 1 & 1 \\ 0 & 1 & -1 \end{array}\right]$$

Now take -1 times the second row and add to the first to get

$$\left[\begin{array}{cc|r} 1 & 0 & 2 \\ 0 & 1 & -1 \end{array}\right]$$

Writing in terms of variables, this says $x = 2$ and $y = -1$.

Now solve the second system, 3.9 to find z and w. You will find that $z = -1$ and $w = 1$.

If we take the values found for x, y, z, and w and put them into our inverse matrix, we see that the inverse is

$$A^{-1} = \left[\begin{array}{cc} x & z \\ y & w \end{array}\right] = \left[\begin{array}{rr} 2 & -1 \\ -1 & 1 \end{array}\right]$$

After taking the time to solve the second system, you may have noticed that exactly the same row operations were used to solve both systems. In each case, the end result was something of the form $[I|X]$ where I is the identity and X gave a column of the inverse. In the above,

$$\left[\begin{array}{c} x \\ y \end{array}\right]$$

the first column of the inverse was obtained by solving the first system and then the second column

$$\left[\begin{array}{c} z \\ w \end{array}\right]$$

To simplify this procedure, we could have solved both systems at once! To do so, we could have written

$$\left[\begin{array}{cc|cc} 1 & 1 & 1 & 0 \\ 1 & 2 & 0 & 1 \end{array}\right]$$

and row reduced until we obtained

$$\left[\begin{array}{cc|rr} 1 & 0 & 2 & -1 \\ 0 & 1 & -1 & 1 \end{array}\right]$$

and read off the inverse as the 2×2 matrix on the right side.

This exploration motivates the following important algorithm.

This algorithm shows how to find the inverse if it exists. It will also tell you if A does not have an inverse.

Consider the following example.

Matrix Inverse Algorithmmatrixinversionalgorithm Suppose A is an $n \times n$ matrix. To find A^{-1} if it exists, form the augmented $n \times 2n$ matrix

$$[A|I]$$

If possible do row operations until you obtain an $n \times 2n$ matrix of the form

$$[I|B]$$

When this has been done, $B = A^{-1}$. In this case, we say that A is **invertible**. If it is impossible to row reduce to a matrix of the form $[I|B]$, then A has no inverse.

Example C.37: Finding the Inverse

Let $A = \left[\begin{array}{rrr} 1 & 2 & 2 \\ 1 & 0 & 2 \\ 3 & 1 & -1 \end{array}\right]$. *Find* A^{-1} *if it exists.*

Solution. Set up the augmented matrix

$$[A|I] = \left[\begin{array}{rrr|rrr} 1 & 2 & 2 & 1 & 0 & 0 \\ 1 & 0 & 2 & 0 & 1 & 0 \\ 3 & 1 & -1 & 0 & 0 & 1 \end{array}\right]$$

Now we row reduce, with the goal of obtaining the 3×3 identity matrix on the left hand side. First, take -1 times the first row and add to the second followed by -3 times the first row added to the third row. This yields

$$\left[\begin{array}{rrr|rrr} 1 & 2 & 2 & 1 & 0 & 0 \\ 0 & -2 & 0 & -1 & 1 & 0 \\ 0 & -5 & -7 & -3 & 0 & 1 \end{array}\right]$$

Then take 5 times the second row and add to -2 times the third row.

$$\left[\begin{array}{rrr|rrr} 1 & 2 & 2 & 1 & 0 & 0 \\ 0 & -10 & 0 & -5 & 5 & 0 \\ 0 & 0 & 14 & 1 & 5 & -2 \end{array}\right]$$

Next take the third row and add to -7 times the first row. This yields

$$\left[\begin{array}{rrr|rrr} -7 & -14 & 0 & -6 & 5 & -2 \\ 0 & -10 & 0 & -5 & 5 & 0 \\ 0 & 0 & 14 & 1 & 5 & -2 \end{array}\right]$$

Now take $-\frac{7}{5}$ times the second row and add to the first row.

$$\left[\begin{array}{rrr|rrr} -7 & 0 & 0 & 1 & -2 & -2 \\ 0 & -10 & 0 & -5 & 5 & 0 \\ 0 & 0 & 14 & 1 & 5 & -2 \end{array}\right]$$

Finally divide the first row by -7, the second row by -10 and the third row by 14 which yields

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & -\frac{1}{7} & \frac{2}{7} & \frac{2}{7} \\ 0 & 1 & 0 & \frac{1}{2} & -\frac{1}{2} & 0 \\ 0 & 0 & 1 & \frac{1}{14} & \frac{5}{14} & -\frac{1}{7} \end{array}\right]$$

Notice that the left hand side of this matrix is now the 3×3 identity matrix I_3. Therefore, the inverse is the 3×3 matrix on the right hand side, given by

$$\left[\begin{array}{ccc} -\frac{1}{7} & \frac{2}{7} & \frac{2}{7} \\ \frac{1}{2} & -\frac{1}{2} & 0 \\ \frac{1}{14} & \frac{5}{14} & -\frac{1}{7} \end{array}\right]$$

♠

It may happen that through this algorithm, you discover that the left hand side cannot be row reduced to the identity matrix. Consider the following example of this situation.

Example C.38: A Matrix Which Has No Inverse

Let $A = \left[\begin{array}{ccc} 1 & 2 & 2 \\ 1 & 0 & 2 \\ 2 & 2 & 4 \end{array}\right]$. *Find* A^{-1} *if it exists.*

Solution. Write the augmented matrix $[A|I]$

$$\left[\begin{array}{ccc|ccc} 1 & 2 & 2 & 1 & 0 & 0 \\ 1 & 0 & 2 & 0 & 1 & 0 \\ 2 & 2 & 4 & 0 & 0 & 1 \end{array}\right]$$

and proceed to do row operations attempting to obtain $\left[I|A^{-1}\right]$. Take -1 times the first row and add to the second. Then take -2 times the first row and add to the third row.

$$\left[\begin{array}{ccc|ccc} 1 & 2 & 2 & 1 & 0 & 0 \\ 0 & -2 & 0 & -1 & 1 & 0 \\ 0 & -2 & 0 & -2 & 0 & 1 \end{array}\right]$$

Next add -1 times the second row to the third row.

$$\left[\begin{array}{ccc|ccc} 1 & 2 & 2 & 1 & 0 & 0 \\ 0 & -2 & 0 & -1 & 1 & 0 \\ 0 & 0 & 0 & -1 & -1 & 1 \end{array}\right]$$

At this point, you can see there will be no way to obtain I on the left side of this augmented matrix. Hence, there is no way to complete this algorithm, and therefore the inverse of A does not exist. In this case, we say that A is not invertible. ♠

If the algorithm provides an inverse for the original matrix, it is always possible to check your answer. To do so, use the method demonstrated in Example **??**. Check that the products AA^{-1} and $A^{-1}A$ both equal the identity matrix. Through this method, you can always be sure that you have calculated A^{-1} properly!

One way in which the inverse of a matrix is useful is to find the solution of a system of linear equations. Recall from Definition C.15 that we can write a system of equations in matrix form, which is of the form $AX = B$. Suppose you find the inverse of the matrix A^{-1}. Then you could multiply both sides of this equation on the left by A^{-1} and simplify to obtain

$$\begin{aligned}
\left(A^{-1}\right)AX &= A^{-1}B \\
\left(A^{-1}A\right)X &= A^{-1}B \\
IX &= A^{-1}B \\
X &= A^{-1}B
\end{aligned}$$

Therefore we can find X, the solution to the system, by computing $X = A^{-1}B$. Note that once you have found A^{-1}, you can easily get the solution for different right hand sides (different B). It is always just $A^{-1}B$.

We will explore this method of finding the solution to a system in the following example.

Example C.39: Using the Inverse to Solve a System of Equations

Consider the following system of equations. Use the inverse of a suitable matrix to give the solutions to this system.

$$\begin{aligned}
x+z &= 1 \\
x-y+z &= 3 \\
x+y-z &= 2
\end{aligned}$$

Solution. First, we can write the system of equations in matrix form

$$AX = \begin{bmatrix} 1 & 0 & 1 \\ 1 & -1 & 1 \\ 1 & 1 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 3 \\ 2 \end{bmatrix} = B \tag{3.10}$$

The inverse of the matrix

$$A = \begin{bmatrix} 1 & 0 & 1 \\ 1 & -1 & 1 \\ 1 & 1 & -1 \end{bmatrix}$$

is

$$A^{-1} = \begin{bmatrix} 0 & \frac{1}{2} & \frac{1}{2} \\ 1 & -1 & 0 \\ 1 & -\frac{1}{2} & -\frac{1}{2} \end{bmatrix}$$

Verifying this inverse is left as an exercise.

From here, the solution to the given system 3.10 is found by

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = A^{-1}B = \begin{bmatrix} 0 & \frac{1}{2} & \frac{1}{2} \\ 1 & -1 & 0 \\ 1 & -\frac{1}{2} & -\frac{1}{2} \end{bmatrix} \begin{bmatrix} 1 \\ 3 \\ 2 \end{bmatrix} = \begin{bmatrix} \frac{5}{2} \\ -2 \\ -\frac{3}{2} \end{bmatrix}$$

♠

What if the right side, B, of 3.10 had been $\begin{bmatrix} 0 \\ 1 \\ 3 \end{bmatrix}$? In other words, what would be the solution to

$$\begin{bmatrix} 1 & 0 & 1 \\ 1 & -1 & 1 \\ 1 & 1 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 3 \end{bmatrix} ?$$

By the above discussion, the solution is given by

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = A^{-1}B = \begin{bmatrix} 0 & \frac{1}{2} & \frac{1}{2} \\ 1 & -1 & 0 \\ 1 & -\frac{1}{2} & -\frac{1}{2} \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ 3 \end{bmatrix} = \begin{bmatrix} 2 \\ -1 \\ -2 \end{bmatrix}$$

This illustrates that for a system $AX = B$ where A^{-1} exists, it is easy to find the solution when the vector B is changed.

D. $\mathbb{R}^n$

D.1 Vectors in $\mathbb{R}^n$

Outcomes

A. *Find the position vector of a point in* $\mathbb{R}^n$.

The notation $\mathbb{R}^n$ refers to the collection of ordered lists of n real numbers, that is

$$\mathbb{R}^n = \left\{ (x_1 \cdots x_n) : x_j \in \mathbb{R} \text{ for } j = 1, \cdots, n \right\}$$

In this chapter, we take a closer look at vectors in $\mathbb{R}^n$. First, we will consider what $\mathbb{R}^n$ looks like in more detail. Recall that the point given by $0 = (0, \cdots, 0)$ is called the **origin**.

Now, consider the case of $\mathbb{R}^n$ for $n = 1$. Then from the definition we can identify $\mathbb{R}$ with points in $\mathbb{R}^1$ as follows:

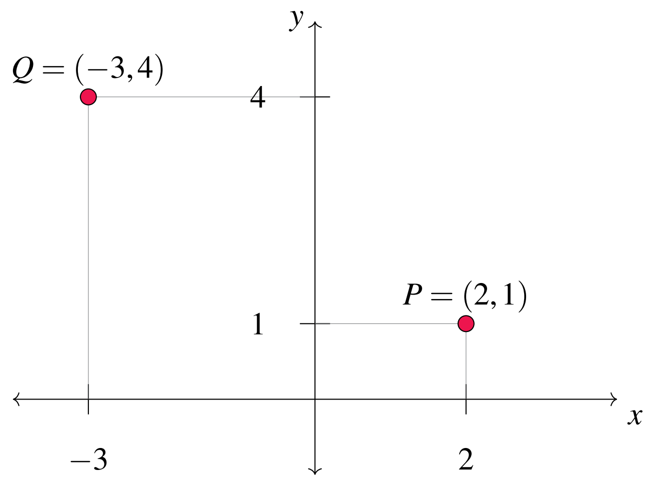
$$\mathbb{R} = \mathbb{R}^1 = \{(x_1) : x_1 \in \mathbb{R}\}$$

Hence, $\mathbb{R}$ is defined as the set of all real numbers and geometrically, we can describe this as all the points on a line.

Now suppose $n = 2$. Then, from the definition,

$$\mathbb{R}^2 = \left\{ (x_1, x_2) : x_j \in \mathbb{R} \text{ for } j = 1, 2 \right\}$$

Consider the familiar coordinate plane, with an x axis and a y axis. Any point within this coordinate plane is identified by where it is located along the x axis, and also where it is located along the y axis. Consider as an example the following diagram.



Hence, every element in $\mathbb{R}^2$ is identified by two components, x and y, in the usual manner. The coordinates x, y (or x_1,x_2) uniquely determine a point in the plan. Note that while the definition uses x_1 and x_2 to label the coordinates and you may be used to x and y, these notations are equivalent.

Now suppose $n = 3$. You may have previously encountered the 3-dimensional coordinate system, given by

$$\mathbb{R}^3 = \left\{ (x_1, x_2, x_3) : x_j \in \mathbb{R} \text{ for } j = 1,2,3 \right\}$$

Points in $\mathbb{R}^3$ will be determined by three coordinates, often written (x, y, z) which correspond to the x, y, and z axes. We can think as above that the first two coordinates determine a point in a plane. The third component determines the height above or below the plane, depending on whether this number is positive or negative, and all together this determines a point in space. You see that the ordered triples correspond to points in space just as the ordered pairs correspond to points in a plane and single real numbers correspond to points on a line.

The idea behind the more general $\mathbb{R}^n$ is that we can extend these ideas beyond $n = 3$. This discussion regarding points in $\mathbb{R}^n$ leads into a study of vectors in $\mathbb{R}^n$. While we consider $\mathbb{R}^n$ for all n, we will largely focus on $n = 2, 3$ in this section.

Consider the following definition.

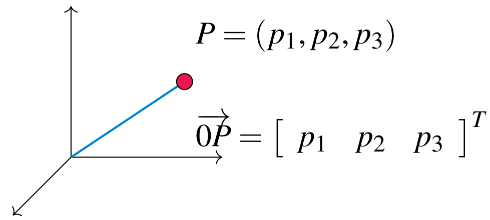
Definition D.1: The Position Vector

*Let $P = (p_1, \cdots, p_n)$ be the coordinates of a point in $\mathbb{R}^n$. Then the vector $\overrightarrow{0P}$ with its tail at $0 = (0, \cdots, 0)$ and its tip at P is called the **position vector** of the point P. We write*

$$\overrightarrow{0P} = \begin{bmatrix} p_1 \\ \vdots \\ p_n \end{bmatrix}$$

For this reason we may write both $P = (p_1, \cdots, p_n) \in \mathbb{R}^n$ and $\overrightarrow{0P} = [p_1 \cdots p_n]^T \in \mathbb{R}^n$.

This definition is illustrated in the following picture for the special case of $\mathbb{R}^3$.

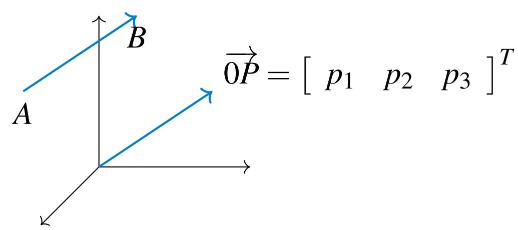


Thus every point P in $\mathbb{R}^n$ determines its position vector $\overrightarrow{0P}$. Conversely, every such position vector $\overrightarrow{0P}$ which has its tail at 0 and point at P determines the point P of $\mathbb{R}^n$.

Now suppose we are given two points, P,Q whose coordinates are $(p_1,\cdots,p_n)$ and $(q_1,\cdots,q_n)$ respectively. We can also determine the **position vector from** P **to** Q (also called the **vector from** P **to** Q) defined as follows.

$$\overrightarrow{PQ} = \begin{bmatrix} q_1 - p_1 \\ \vdots \\ q_n - p_n \end{bmatrix} = \overrightarrow{0Q} - \overrightarrow{0P}$$

Now, imagine taking a vector in $\mathbb{R}^n$ and moving it around, always keeping it pointing in the same direction as shown in the following picture.



After moving it around, it is regarded as the same vector. Each vector, $\overrightarrow{0P}$ and $\overrightarrow{AB}$ has the same length (or magnitude) and direction. Therefore, they are equal.

Consider now the general definition for a vector in $\mathbb{R}^n$.

Definition D.2: Vectors in $\mathbb{R}^n$

Let $\mathbb{R}^n = \left\{(x_1,\cdots,x_n) : x_j \in \mathbb{R} \text{ for } j = 1,\cdots,n\right\}$. *Then,*

$$\mathbf{x} = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}$$

*is called a **vector**. Vectors have both size (magnitude) and direction. The numbers x_j are called the **components** of* $\mathbf{x}$.

Using this notation, we may use $\mathbf{p}$ to denote the position vector of point P. Notice that in this context, $\mathbf{p} = \overrightarrow{0P}$. These notations may be used interchangeably.

You can think of the components of a vector as directions for obtaining the vector. Consider $n = 3$. Draw a vector with its tail at the point $(0,0,0)$ and its tip at the point (a,b,c). This vector it is obtained by starting at $(0,0,0)$, moving parallel to the x axis to $(a,0,0)$ and then from here, moving parallel to the y axis to $(a,b,0)$ and finally parallel to the z axis to (a,b,c). Observe that the same vector would result if you began at the point (d,e,f), moved parallel to the x axis to $(d+a,e,f)$, then parallel to the y axis to $(d+a,e+b,f)$, and finally parallel to the z axis to $(d+a,e+b,f+c)$. Here, the vector would have its tail sitting at the point determined by $A=(d,e,f)$ and its point at $B=(d+a,e+b,f+c)$. It is the **same vector** because it will point in the same direction and have the same length. It is like you took an actual arrow, and moved it from one location to another keeping it pointing the same direction.

We conclude this section with a brief discussion regarding notation. In previous sections, we have written vectors as columns, or $n\times 1$ matrices. For convenience in this chapter we may write vectors as the transpose of row vectors, or $1\times n$ matrices. These are of course equivalent and we may move between both notations. Therefore, recognize that

$$\left[\begin{array}{c} 2 \\ 3 \end{array}\right] = \left[\begin{array}{cc} 2 & 3 \end{array}\right]^T$$

Notice that two vectors $\mathbf{u}=[u_1 \cdots u_n]^T$ and $\mathbf{v}=[v_1 \cdots v_n]^T$ are equal if and only if all corresponding components are equal. Precisely,

$$\begin{array}{c} \mathbf{u}=\mathbf{v} \text{ if and only if} \\ u_j = v_j \text{ for all } j=1,\cdots,n \end{array}$$

Thus $\left[\begin{array}{ccc} 1 & 2 & 4 \end{array}\right]^T \in \mathbb{R}^3$ and $\left[\begin{array}{ccc} 2 & 1 & 4 \end{array}\right]^T \in \mathbb{R}^3$ but $\left[\begin{array}{ccc} 1 & 2 & 4 \end{array}\right]^T \neq \left[\begin{array}{ccc} 2 & 1 & 4 \end{array}\right]^T$ because, even though the same numbers are involved, the order of the numbers is different.

For the specific case of $\mathbb{R}^3$, there are three special vectors which we often use. They are given by

$$\mathbf{i} = \left[\begin{array}{ccc} 1 & 0 & 0 \end{array}\right]^T$$

$$\mathbf{j} = \left[\begin{array}{ccc} 0 & 1 & 0 \end{array}\right]^T$$

$$\mathbf{k} = \left[\begin{array}{ccc} 0 & 0 & 1 \end{array}\right]^T$$

We can write any vector $\mathbf{u}=\left[\begin{array}{ccc} u_1 & u_2 & u_3 \end{array}\right]^T$ as a linear combination of these vectors, written as $\mathbf{u}=u_1\mathbf{i}+u_2\mathbf{j}+u_3\mathbf{k}$. This notation will be used throughout this chapter.

D.2 Algebra in $\mathbb{R}^n$

Outcomes

A. *Understand vector addition and scalar multiplication, algebraically.*
B. *Introduce the notion of linear combination of vectors.*

Addition and scalar multiplication are two important algebraic operations done with vectors. Notice that these operations apply to vectors in $\mathbb{R}^n$, for any value of n. We will explore these operations in more detail in the following sections.

D.2.1. Addition of Vectors in $\mathbb{R}^n$

Addition of vectors in $\mathbb{R}^n$ is defined as follows.

Definition D.3: Addition of Vectors in $\mathbb{R}^n$

If $\mathbf{u} = \begin{bmatrix} u_1 \\ \vdots \\ u_n \end{bmatrix}$, $\mathbf{v} = \begin{bmatrix} v_1 \\ \vdots \\ v_n \end{bmatrix} \in \mathbb{R}^n$ *then* $\mathbf{u}+\mathbf{v} \in \mathbb{R}^n$ *and is defined by*

$$\begin{aligned} \mathbf{u}+\mathbf{v} &= \begin{bmatrix} u_1 \\ \vdots \\ u_n \end{bmatrix} + \begin{bmatrix} v_1 \\ \vdots \\ v_n \end{bmatrix} \\ &= \begin{bmatrix} u_1+v_1 \\ \vdots \\ u_n+v_n \end{bmatrix} \end{aligned}$$

To add vectors, we simply add corresponding components. Therefore, in order to add vectors, they must be the same size.

Addition of vectors satisfies some important properties which are outlined in the following theorem.

Theorem D.4: Properties of Vector Addition

The following properties hold for vectors $\mathbf{u},\mathbf{v},\mathbf{w} \in \mathbb{R}^n$.

- *The Commutative Law of Addition*

$$\mathbf{u}+\mathbf{v} = \mathbf{v}+\mathbf{u}$$

- *The Associative Law of Addition*

$$(\mathbf{u}+\mathbf{v})+\mathbf{w} = \mathbf{u}+(\mathbf{v}+\mathbf{w})$$

- *The Existence of an Additive Identity*

$$\mathbf{u}+0 = \mathbf{u} \tag{4.1}$$

- *The Existence of an Additive Inverse*

$$\mathbf{u}+(-\mathbf{u}) = 0$$

The additive identity shown in equation 4.1 is also called the **zero vector**, the $n \times 1$ vector in which all components are equal to 0. Further, $-\mathbf{u}$ is simply the vector with all components having same value as those of $\mathbf{u}$ but opposite sign; this is just $(-1)\mathbf{u}$. This will be made more explicit in the next section when we explore scalar multiplication of vectors. Note that subtraction is defined as $\mathbf{u}-\mathbf{v}=\mathbf{u}+(-\mathbf{v})$.

D.2.2. Scalar Multiplication of Vectors in $\mathbb{R}^n$

Scalar multiplication of vectors in $\mathbb{R}^n$ is defined as follows.

Definition D.5: Scalar Multiplication of Vectors in $\mathbb{R}^n$

If $\mathbf{u} \in \mathbb{R}^n$ and $k \in \mathbb{R}$ is a scalar, then $k\mathbf{u} \in \mathbb{R}^n$ is defined by

$$k\mathbf{u} = k\begin{bmatrix} u_1 \\ \vdots \\ u_n \end{bmatrix} = \begin{bmatrix} ku_1 \\ \vdots \\ ku_n \end{bmatrix}$$

Just as with addition, scalar multiplication of vectors satisfies several important properties. These are outlined in the following theorem.

Theorem D.6: Properties of Scalar Multiplication

The following properties hold for vectors $\mathbf{u},\mathbf{v} \in \mathbb{R}^n$ and k,p scalars.

- *The Distributive Law over Vector Addition*

$$k(\mathbf{u}+\mathbf{v}) = k\mathbf{u}+k\mathbf{v}$$

- *The Distributive Law over Scalar Addition*

$$(k+p)\mathbf{u} = k\mathbf{u}+p\mathbf{u}$$

- *The Associative Law for Scalar Multiplication*

$$k(p\mathbf{u}) = (kp)\mathbf{u}$$

- *Rule for Multiplication by 1*

$$1\mathbf{u} = \mathbf{u}$$

We now present a useful notion you may have seen earlier combining vector addition and scalar multiplication

Definition D.7: Linear Combination

A vector $\mathbf{v}$ is said to be a ***linear combination*** *of the vectors $\mathbf{u}_1,\cdots,\mathbf{u}_n$ if there exist scalars, $a_1,\cdots,a_n$ such that*

$$\mathbf{v} = a_1\mathbf{u}_1+\cdots+a_n\mathbf{u}_n$$

For example,

$$3\begin{bmatrix} -4 \\ 1 \\ 0 \end{bmatrix} + 2\begin{bmatrix} -3 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} -18 \\ 3 \\ 2 \end{bmatrix}.$$

Thus we can say that

$$\mathbf{v} = \begin{bmatrix} -18 \\ 3 \\ 2 \end{bmatrix}$$

is a linear combination of the vectors

$$\mathbf{u}_1 = \begin{bmatrix} -4 \\ 1 \\ 0 \end{bmatrix} \text{ and } \mathbf{u}_2 = \begin{bmatrix} -3 \\ 0 \\ 1 \end{bmatrix}$$

D.3 Geometric Meaning of Vector Addition

Outcomes

A. *Understand vector addition, geometrically.*

Recall that an element of $\mathbb{R}^n$ is an ordered list of numbers. For the specific case of $n = 2,3$ this can be used to determine a point in two or three dimensional space. This point is specified relative to some coordinate axes.

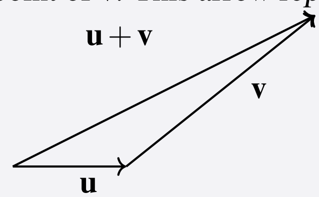
Consider the case $n = 3$. Recall that taking a vector and moving it around without changing its length or direction does not change the vector. This is important in the geometric representation of vector addition.

Suppose we have two vectors, $\mathbf{u}$ and $\mathbf{v}$ in $\mathbb{R}^3$. Each of these can be drawn geometrically by placing the tail of each vector at 0 and its point at (u_1,u_2,u_3) and (v_1,v_2,v_3) respectively. Suppose we slide the vector $\mathbf{v}$ so that its tail sits at the point of $\mathbf{u}$. We know that this does not change the vector $\mathbf{v}$. Now, draw a new vector from the tail of $\mathbf{u}$ to the point of $\mathbf{v}$. This vector is $\mathbf{u}+\mathbf{v}$.

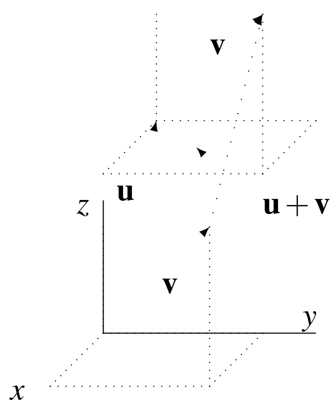
The geometric significance of vector addition in $\mathbb{R}^n$ for any n is given in the following definition.

Definition D.8: Geometry of Vector Addition

Let $\mathbf{u}$ and $\mathbf{v}$ be two vectors. Slide $\mathbf{v}$ so that the tail of $\mathbf{v}$ is on the point of $\mathbf{u}$. Then draw the arrow which goes from the tail of $\mathbf{u}$ to the point of $\mathbf{v}$. This arrow represents the vector $\mathbf{u}+\mathbf{v}$.



This definition is illustrated in the following picture in which $\mathbf{u}+\mathbf{v}$ is shown for the special case $n=3$.

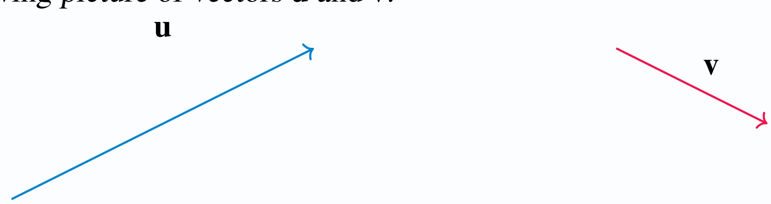


Notice the parallelogram created by $\mathbf{u}$ and $\mathbf{v}$ in the above diagram. Then $\mathbf{u}+\mathbf{v}$ is the directed diagonal of the parallelogram determined by the two vectors $\mathbf{u}$ and $\mathbf{v}$.

When you have a vector $\mathbf{v}$, its additive inverse $-\mathbf{v}$ will be the vector which has the same magnitude as $\mathbf{v}$ but the opposite direction. When one writes $\mathbf{u}-\mathbf{v}$, the meaning is $\mathbf{u}+(-\mathbf{v})$ as with real numbers. The following example illustrates these definitions and conventions.

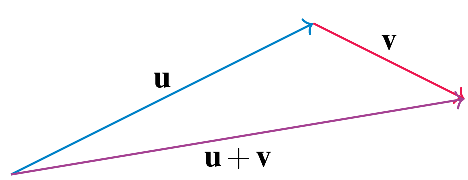
Example D.9: Graphing Vector Addition

Consider the following picture of vectors $\mathbf{u}$ *and* $\mathbf{v}$.

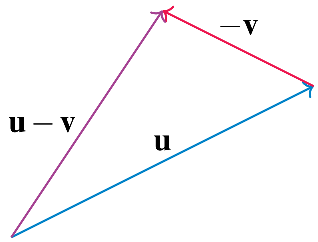


Sketch a picture of $\mathbf{u}+\mathbf{v}, \mathbf{u}-\mathbf{v}$.

Solution. We will first sketch $\mathbf{u}+\mathbf{v}$. Begin by drawing $\mathbf{u}$ and then at the point of $\mathbf{u}$, place the tail of $\mathbf{v}$ as shown. Then $\mathbf{u}+\mathbf{v}$ is the vector which results from drawing a vector from the tail of $\mathbf{u}$ to the tip of $\mathbf{v}$.



Next consider $\mathbf{u}-\mathbf{v}$. This means $\mathbf{u}+(-\mathbf{v})$. From the above geometric description of vector addition, $-\mathbf{v}$ is the vector which has the same length but which points in the opposite direction to $\mathbf{v}$. Here is a picture.


$-\mathbf{v}$
$\mathbf{u}-\mathbf{v}$
$\mathbf{u}$

♠

D.4 Length of a Vector

Outcomes

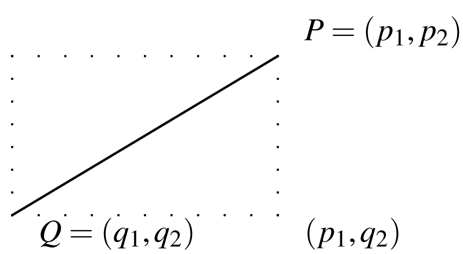
A. *Find the length of a vector and the distance between two points in $\mathbb{R}^n$.*
B. *Find the corresponding unit vector to a vector in $\mathbb{R}^n$.*

In this section, we explore what is meant by the length of a vector in $\mathbb{R}^n$. We develop this concept by first looking at the distance between two points in $\mathbb{R}^n$.

First, we will consider the concept of distance for $\mathbb{R}$, that is, for points in $\mathbb{R}^1$. Here, the distance between two points P and Q is given by the absolute value of their difference. We denote the distance between P and Q by $d(P,Q)$ which is defined as

$$d(P,Q) = \sqrt{(P-Q)^2} \tag{4.1}$$

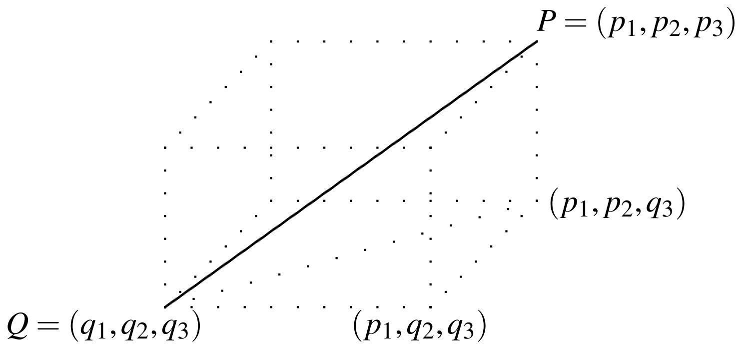
Consider now the case for $n = 2$, demonstrated by the following picture.



There are two points $P = (p_1, p_2)$ and $Q = (q_1, q_2)$ in the plane. The distance between these points is shown in the picture as a solid line. Notice that this line is the hypotenuse of a right triangle which is half of the rectangle shown in dotted lines. We want to find the length of this hypotenuse which will give the distance between the two points. Note the lengths of the sides of this triangle are $|p_1 - q_1|$ and $|p_2 - q_2|$, the absolute value of the difference in these values. Therefore, the Pythagorean Theorem implies the length of the hypotenuse (and thus the distance between P and Q) equals

$$\left(|p_1 - q_1|^2 + |p_2 - q_2|^2\right)^{1/2} = \left((p_1 - q_1)^2 + (p_2 - q_2)^2\right)^{1/2} \tag{4.2}$$

Now suppose $n = 3$ and let $P = (p_1, p_2, p_3)$ and $Q = (q_1, q_2, q_3)$ be two points in $\mathbb{R}^3$. Consider the following picture in which the solid line joins the two points and a dotted line joins the points (q_1, q_2, q_3) and (p_1, p_2, q_3).



Here, we need to use Pythagorean Theorem twice in order to find the length of the solid line. First, by the Pythagorean Theorem, the length of the dotted line joining (q_1, q_2, q_3) and (p_1, p_2, q_3) equals

$$\left((p_1 - q_1)^2 + (p_2 - q_2)^2 \right)^{1/2}$$

while the length of the line joining (p_1, p_2, q_3) to (p_1, p_2, p_3) is just $|p_3 - q_3|$. Therefore, by the Pythagorean Theorem again, the length of the line joining the points $P = (p_1, p_2, p_3)$ and $Q = (q_1, q_2, q_3)$ equals

$$\begin{aligned} &\left(\left(\left((p_1 - q_1)^2 + (p_2 - q_2)^2 \right)^{1/2} \right)^2 + (p_3 - q_3)^2 \right)^{1/2} \\ &= \left((p_1 - q_1)^2 + (p_2 - q_2)^2 + (p_3 - q_3)^2 \right)^{1/2} \end{aligned} \tag{4.3}$$

This discussion motivates the following definition for the distance between points in $\mathbb{R}^n$.

Definition D.10: Distance Between Points

Let $P = (p_1, \cdots, p_n)$ and $Q = (q_1, \cdots, q_n)$ be two points in $\mathbb{R}^n$. Then the distance between these points is defined as

$$\text{distance between } P \text{ and } Q = d(P, Q) = \left(\sum_{k=1}^{n} |p_k - q_k|^2 \right)^{1/2}$$

This is called the ***distance formula****. We may also write $|P - Q|$ as the distance between P and Q.*

From the above discussion, you can see that Definition D.10 holds for the special cases $n = 1, 2, 3$, as in Equations 4.1, 4.2, 4.3. In the following example, we use Definition D.10 to find the distance between two points in $\mathbb{R}^4$.

Example D.11: Distance Between Points

Find the distance between the points P and Q in $\mathbb{R}^4$, where P and Q are given by

$$P = (1, 2, -4, 6)$$

and

$$Q = (2,3,-1,0)$$

Solution. We will use the formula given in Definition D.10 to find the distance between P and Q. Use the distance formula and write

$$d(P,Q) = \left((1-2)^2 + (2-3)^2 + (-4-(-1))^2 + (6-0)^2\right)^{\frac{1}{2}} = 47$$

Therefore, $d(P,Q) = \sqrt{47}$.

♠

There are certain properties of the distance between points which are important in our study. These are outlined in the following theorem.

Theorem D.12: Properties of Distance

Let P and Q be points in $\mathbb{R}^n$, and let the distance between them, $d(P,Q)$, be given as in Definition D.10. Then, the following properties hold .

- $d(P,Q) = d(Q,P)$
- $d(P,Q) \geq 0$, *and equals 0 exactly when $P = Q$.*

There are many applications of the concept of distance. For instance, given two points, we can ask what collection of points are all the same distance between the given points. This is explored in the following example.

Example D.13: The Plane Between Two Points

Describe the points in $\mathbb{R}^3$ which are at the same distance between $(1,2,3)$ and $(0,1,2)$.

Solution. Let $P = (p_1, p_2, p_3)$ be such a point. Therefore, P is the same distance from $(1,2,3)$ and $(0,1,2)$. Then by Definition D.10,

$$\sqrt{(p_1-1)^2 + (p_2-2)^2 + (p_3-3)^2} = \sqrt{(p_1-0)^2 + (p_2-1)^2 + (p_3-2)^2}$$

Squaring both sides we obtain

$$(p_1-1)^2 + (p_2-2)^2 + (p_3-3)^2 = p_1^2 + (p_2-1)^2 + (p_3-2)^2$$

and so

$$p_1^2 - 2p_1 + 14 + p_2^2 - 4p_2 + p_3^2 - 6p_3 = p_1^2 + p_2^2 - 2p_2 + 5 + p_3^2 - 4p_3$$

Simplifying, this becomes

$$-2p_1 + 14 - 4p_2 - 6p_3 = -2p_2 + 5 - 4p_3$$

which can be written as

$$2p_1 + 2p_2 + 2p_3 = -9 \tag{4.4}$$

Therefore, the points $P = (p_1, p_2, p_3)$ which are the same distance from each of the given points form a plane whose equation is given by 4.4. ♠

We can now use our understanding of the distance between two points to define what is meant by the length of a vector. Consider the following definition.

Definition D.14: Length of a Vector

Let $\mathbf{u} = [u_1 \cdots u_n]^T$ be a vector in $\mathbb{R}^n$. Then, the length of $\mathbf{u}$, written $\|\mathbf{u}\|$ is given by

$$\|\mathbf{u}\| = \sqrt{u_1^2 + \cdots + u_n^2}$$

This definition corresponds to Definition D.10, if you consider the vector $\mathbf{u}$ to have its tail at the point $0 = (0, \cdots, 0)$ and its tip at the point $U = (u_1, \cdots, u_n)$. Then the length of $\mathbf{u}$ is equal to the distance between 0 and U, $d(0, U)$. In general, $d(P, Q) = \|\overrightarrow{PQ}\|$.

Consider Example **??**. By Definition D.14, we could also find the distance between P and Q as the length of the vector connecting them. Hence, if we were to draw a vector $\overrightarrow{PQ}$ with its tail at P and its point at Q, this vector would have length equal to $\sqrt{47}$.

We conclude this section with a new definition for the special case of vectors of length 1.

Definition D.15: Unit Vector

Let $\mathbf{u}$ be a vector in $\mathbb{R}^n$. Then, we call $\mathbf{u}$ a ***unit vector*** *if it has length 1, that is if*

$$\|\mathbf{u}\| = 1$$

Let $\mathbf{v}$ be a vector in $\mathbb{R}^n$. Then, the vector $\mathbf{u}$ which has the same direction as $\mathbf{v}$ but length equal to 1 is the corresponding unit vector of $\mathbf{v}$. This vector is given by

$$\mathbf{u} = \frac{1}{\|\mathbf{v}\|}\mathbf{v}$$

We often use the term **normalize** to refer to this process. When we **normalize** a vector, we find the corresponding unit vector of length 1. Consider the following example.

Example D.16: Finding a Unit Vector

Let $\mathbf{v}$ *be given by*

$$\mathbf{v} = \left[\begin{array}{ccc} 1 & -3 & 4 \end{array} \right]^T$$

Find the unit vector $\mathbf{u}$ *which has the same direction as* $\mathbf{v}$ *.*

Solution. We will use Definition D.15 to solve this. Therefore, we need to find the length of $\mathbf{v}$ which, by Definition D.14 is given by

$$\|\mathbf{v}\| = \sqrt{v_1^2 + v_2^2 + v_3^2}$$

Using the corresponding values we find that

$$\begin{aligned} \|\mathbf{v}\| &= \sqrt{1^2 + (-3)^2 + 4^2} \\ &= \sqrt{1+9+16} \\ &= \sqrt{26} \end{aligned}$$

In order to find $\mathbf{u}$, we divide $\mathbf{v}$ by $\sqrt{26}$. The result is

$$\begin{aligned} \mathbf{u} &= \frac{1}{\|\mathbf{v}\|}\mathbf{v} \\ &= \frac{1}{\sqrt{26}} \left[\begin{array}{ccc} 1 & -3 & 4 \end{array} \right]^T \\ &= \left[\begin{array}{ccc} \frac{1}{\sqrt{26}} & -\frac{3}{\sqrt{26}} & \frac{4}{\sqrt{26}} \end{array} \right]^T \end{aligned}$$

You can verify using the Definition D.14 that $\|\mathbf{u}\| = 1$. ♠

D.5 Geometric Meaning of Scalar Multiplication

Outcomes

A. *Understand scalar multiplication, geometrically.*

Recall that the point $P = (p_1, p_2, p_3)$ determines a vector $\mathbf{p}$ from 0 to P. The length of $\mathbf{p}$, denoted $\|\mathbf{p}\|$, is equal to $\sqrt{p_1^2 + p_2^2 + p_3^2}$ by Definition D.10.

Now suppose we have a vector $\mathbf{u} = \left[\begin{array}{ccc} u_1 & u_2 & u_3 \end{array} \right]^T$ and we multiply $\mathbf{u}$ by a scalar k. By Definition D.5, $k\mathbf{u} = \left[\begin{array}{ccc} ku_1 & ku_2 & ku_3 \end{array} \right]^T$. Then, by using Definition D.10, the length of this vector is given by

$$\sqrt{\left((ku_1)^2 + (ku_2)^2 + (ku_3)^2 \right)} = |k| \sqrt{u_1^2 + u_2^2 + u_3^2}$$

Thus the following holds.

$$\|k\mathbf{u}\| = |k| \, \|\mathbf{u}\|$$

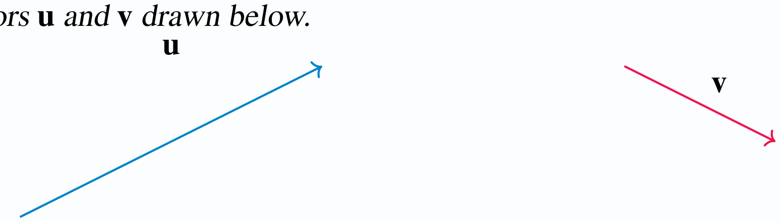
In other words, multiplication by a scalar magnifies or shrinks the length of the vector by a factor of $|k|$. If $|k| > 1$, the length of the resulting vector will be magnified. If $|k| < 1$, the length of the resulting vector will shrink. Remember that by the definition of the absolute value, $|k| > 0$.

What about the direction? Draw a picture of $\mathbf{u}$ and $k\mathbf{u}$ where k is negative. Notice that this causes the resulting vector to point in the opposite direction while if $k > 0$ it preserves the direction the vector points. Therefore the direction can either reverse, if $k < 0$, or remain preserved, if $k > 0$.

Consider the following example.

Example D.17: Graphing Scalar Multiplication

Consider the vectors $\mathbf{u}$ *and* $\mathbf{v}$ *drawn below.*



Draw $-\mathbf{u}$, $2\mathbf{v}$, *and* $-\frac{1}{2}\mathbf{v}$.

Solution.

In order to find $-\mathbf{u}$, we preserve the length of $\mathbf{u}$ and simply reverse the direction. For $2\mathbf{v}$, we double the length of $\mathbf{v}$, while preserving the direction. Finally $-\frac{1}{2}\mathbf{v}$ is found by taking half the length of $\mathbf{v}$ and reversing the direction. These vectors are shown in the following diagram.

♠

Now that we have studied both vector addition and scalar multiplication, we can combine the two actions. Recall Definition D.7 of linear combinations of column matrices. We can apply this definition to vectors in $\mathbb{R}^n$. A linear combination of vectors in $\mathbb{R}^n$ is a sum of vectors multiplied by scalars.

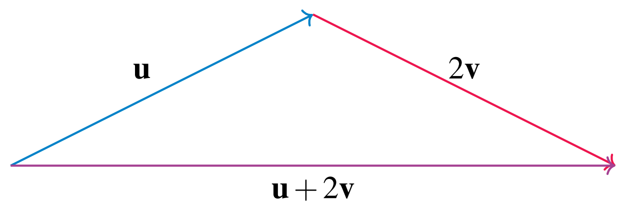
In the following example, we examine the geometric meaning of this concept.

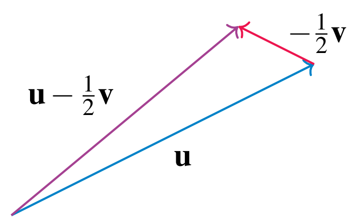
Example D.18: Graphing a Linear Combination of Vectors

Consider the following picture of the vectors $\mathbf{u}$ *and* $\mathbf{v}$

Sketch a picture of $\mathbf{u}+2\mathbf{v}, \mathbf{u}-\frac{1}{2}\mathbf{v}$.

Solution. The two vectors are shown below.





♠

D.6 The Dot Product

Outcomes

A. *Compute the dot product of vectors, and use this to compute vector projections.*

D.6.1. The Dot Product

There are two ways of multiplying vectors which are of great importance in applications. The first of these is called the **dot product**. When we take the dot product of vectors, the result is a scalar. For this reason, the dot product is also called the **scalar product** and sometimes the **inner product**. The definition is as follows.

Definition D.19: Dot Product

Let $\mathbf{u}, \mathbf{v}$ *be two vectors in* $\mathbb{R}^n$*. Then we define the* ***dot product*** $\mathbf{u} \bullet \mathbf{v}$ *as*

$$\mathbf{u} \bullet \mathbf{v} = \sum_{k=1}^{n} u_k v_k$$

The dot product $\mathbf{u} \bullet \mathbf{v}$ is sometimes denoted as $(\mathbf{u}, \mathbf{v})$ where a comma replaces $\bullet$. It can also be written as $\langle \mathbf{u}, \mathbf{v} \rangle$. If we write the vectors as column or row matrices, it is equal to the matrix product $\mathbf{v}\mathbf{w}^T$.

Consider the following example.

Example D.20: Compute a Dot Product

Find $\mathbf{u} \bullet \mathbf{v}$ *for*

$$\mathbf{u} = \left[\begin{array}{r} 1 \\ 2 \\ 0 \\ -1 \end{array} \right], \mathbf{v} = \left[\begin{array}{r} 0 \\ 1 \\ 2 \\ 3 \end{array} \right]$$

Solution. By Definition D.19, we must compute

$$\mathbf{u} \bullet \mathbf{v} = \sum_{k=1}^{4} u_k v_k$$

This is given by

$$\begin{array}{rcl} \mathbf{u} \bullet \mathbf{v} & = & (1)(0) + (2)(1) + (0)(2) + (-1)(3) \\ & = & 0 + 2 + 0 + -3 \\ & = & -1 \end{array}$$

♠

With this definition, there are several important properties satisfied by the dot product.

Proposition D.21: Properties of the Dot Product

Let k *and* p *denote scalars and* $\mathbf{u}, \mathbf{v}, \mathbf{w}$ *denote vectors. Then the dot product* $\mathbf{u} \bullet \mathbf{v}$ *satisfies the following properties.*

- $\mathbf{u} \bullet \mathbf{v} = \mathbf{v} \bullet \mathbf{u}$
- $\mathbf{u} \bullet \mathbf{u} \geq 0$ *and equals zero if and only if* $\mathbf{u} = 0$
- $(k\mathbf{u} + p\mathbf{v}) \bullet \mathbf{w} = k(\mathbf{u} \bullet \mathbf{w}) + p(\mathbf{v} \bullet \mathbf{w})$
- $\mathbf{u} \bullet (k\mathbf{v} + p\mathbf{w}) = k(\mathbf{u} \bullet \mathbf{v}) + p(\mathbf{u} \bullet \mathbf{w})$
- $\|\mathbf{u}\|^2 = \mathbf{u} \bullet \mathbf{u}$

The proof is left as an exercise. This proposition tells us that we can also use the dot product to find the length of a vector.

Example D.22: Length of a Vector

Find the length of

$$\mathbf{u} = \begin{bmatrix} 2 \\ 1 \\ 4 \\ 2 \end{bmatrix}$$

That is, find $\|\mathbf{u}\|$.

Solution. By Proposition D.21, $\|\mathbf{u}\|^2 = \mathbf{u} \bullet \mathbf{u}$. Therefore, $\|\mathbf{u}\| = \sqrt{\mathbf{u} \bullet \mathbf{u}}$. First, compute $\mathbf{u} \bullet \mathbf{u}$.

This is given by

$$\begin{aligned} \mathbf{u} \bullet \mathbf{u} &= (2)(2) + (1)(1) + (4)(4) + (2)(2) \\ &= 4 + 1 + 16 + 4 \\ &= 25 \end{aligned}$$

Then,

$$\begin{aligned} \|\mathbf{u}\| &= \sqrt{\mathbf{u} \bullet \mathbf{u}} \\ &= \sqrt{25} \\ &= 5 \end{aligned}$$

♠

You may wish to compare this to our previous definition of length, given in Definition D.14.

The **Cauchy Schwarz inequality** is a fundamental inequality satisfied by the dot product. It is given in the following theorem.

Theorem D.23: Cauchy Schwarz Inequality

The dot product satisfies the inequality

$$|\mathbf{u} \bullet \mathbf{v}| \leq \|\mathbf{u}\| \|\mathbf{v}\| \tag{4.1}$$

Furthermore equality is obtained if and only if one of $\mathbf{u}$ *or* $\mathbf{v}$ *is a scalar multiple of the other.*

Notice that this proof was based only on the properties of the dot product listed in Proposition D.21. This means that whenever an operation satisfies these properties, the Cauchy Schwarz inequality holds. There are many other instances of these properties besides vectors in $\mathbb{R}^n$.

The Cauchy Schwarz inequality provides another proof of the **triangle inequality** for distances in $\mathbb{R}^n$.

Theorem D.24: Triangle Inequality

For $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$

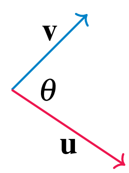
$$\|\mathbf{u}+\mathbf{v}\| \leq \|\mathbf{u}\| + \|\mathbf{v}\| \tag{4.2}$$

and equality holds if and only if one of the vectors is a non-negative scalar multiple of the other. Also

$$|\|\mathbf{u}\| - \|\mathbf{v}\|| \leq \|\mathbf{u}-\mathbf{v}\| \tag{4.3}$$

D.6.2. The Geometric Significance of the Dot Product

Given two vectors, $\mathbf{u}$ and $\mathbf{v}$, the **included angle** is the angle between these two vectors which is given by θ such that $0 \leq \theta \leq \pi$. The dot product can be used to determine the included angle between two vectors. Consider the following picture where θ gives the included angle.



Proposition D.25: The Dot Product and the Included Angle

Let $\mathbf{u}$ *and* $\mathbf{v}$ *be two vectors in* $\mathbb{R}^n$*, and let* θ *be the included angle. Then the following equation holds.*

$$\mathbf{u} \bullet \mathbf{v} = \|\mathbf{u}\|\|\mathbf{v}\| \cos\theta$$

In words, the dot product of two vectors equals the product of the magnitude (or length) of the two vectors multiplied by the cosine of the included angle. Note this gives a geometric description of the dot product which does not depend explicitly on the coordinates of the vectors.

Consider the following example.

Example D.26: Find the Angle Between Two Vectors

Find the angle between the vectors given by

$$\mathbf{u} = \begin{bmatrix} 2 \\ 1 \\ -1 \end{bmatrix}, \mathbf{v} = \begin{bmatrix} 3 \\ 4 \\ 1 \end{bmatrix}$$

Solution. By Proposition D.25,

$$\mathbf{u} \bullet \mathbf{v} = \|\mathbf{u}\|\|\mathbf{v}\| \cos\theta$$

Hence,

$$\cos\theta = \frac{\mathbf{u} \bullet \mathbf{v}}{\|\mathbf{u}\|\|\mathbf{v}\|}$$

First, we can compute $\mathbf{u} \bullet \mathbf{v}$. By Definition D.19, this equals

$$\mathbf{u} \bullet \mathbf{v} = (2)(3) + (1)(4) + (-1)(1) = 9$$

Then,

$$\|\mathbf{u}\| = \sqrt{(2)(2) + (1)(1) + (1)(1)} = \sqrt{6}$$
$$\|\mathbf{v}\| = \sqrt{(3)(3) + (4)(4) + (1)(1)} = \sqrt{26}$$

Therefore, the cosine of the included angle equals

$$\cos\theta = \frac{9}{\sqrt{26}\sqrt{6}} = 0.7205766...$$

With the cosine known, the angle can be determined by computing the inverse cosine of that angle, giving approximately $\theta = 0.76616$ radians. ♠

Another application of the geometric description of the dot product is in finding the angle between two lines. Typically one would assume that the lines intersect. In some situations, however, it may make sense to ask this question when the lines do not intersect, such as the angle between two object trajectories. In any case we understand it to mean the smallest angle between (any of) their direction vectors. The only subtlety here is that if $\mathbf{u}$ is a direction vector for a line, then so is any multiple $k\mathbf{u}$, and thus we will find complementary angles among all angles between direction vectors for two lines, and we simply take the smaller of the two.

Example D.27: Find the Angle Between Two Lines

Find the angle between the two lines

$$L_1: \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix} + t\begin{bmatrix} -1 \\ 1 \\ 2 \end{bmatrix}$$

and

$$L_2: \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 4 \\ -3 \end{bmatrix} + s\begin{bmatrix} 2 \\ 1 \\ -1 \end{bmatrix}$$

Solution. You can verify that these lines do not intersect, but as discussed above this does not matter and we simply find the smallest angle between any directions vectors for these lines.

To do so we first find the angle between the direction vectors given above:

$$\mathbf{u} = \begin{bmatrix} -1 \\ 1 \\ 2 \end{bmatrix}, \mathbf{v} = \begin{bmatrix} 2 \\ 1 \\ -1 \end{bmatrix}$$

In order to find the angle, we solve the following equation for θ

$$\mathbf{u} \bullet \mathbf{v} = \|\mathbf{u}\| \|\mathbf{v}\| \cos\theta$$

to obtain $\cos\theta = -\frac{1}{2}$ and since we choose included angles between 0 and π we obtain $\theta = \frac{2\pi}{3}$.

Now the angles between any two direction vectors for these lines will either be $\frac{2\pi}{3}$ or its complement $\phi = \pi - \frac{2\pi}{3} = \frac{\pi}{3}$. We choose the smaller angle, and therefore conclude that the angle between the two lines is $\frac{\pi}{3}$. ♠

We can also use Proposition D.25 to compute the dot product of two vectors.

Example D.28: Using Geometric Description to Find a Dot Product

Let $\mathbf{u}, \mathbf{v}$ be vectors with $\|\mathbf{u}\| = 3$ and $\|\mathbf{v}\| = 4$. Suppose the angle between $\mathbf{u}$ and $\mathbf{v}$ is $\pi/3$. Find $\mathbf{u} \bullet \mathbf{v}$.

Solution. From the geometric description of the dot product in Proposition D.25

$$\mathbf{u} \bullet \mathbf{v} = (3)(4)\cos(\pi/3) = 3 \times 4 \times 1/2 = 6$$

♠

Two nonzero vectors are said to be **perpendicular**, sometimes also called **orthogonal**, if the included angle is $\pi/2$ radians (90°).

Consider the following proposition.

Proposition D.29: Perpendicular Vectors

*Let $\mathbf{u}$ and $\mathbf{v}$ be nonzero vectors in $\mathbb{R}^n$. Then, $\mathbf{u}$ and $\mathbf{v}$ are said to be **perpendicular** exactly when*

$$\mathbf{u} \bullet \mathbf{v} = 0$$

Consider the following example.

Example D.30: Determine if Two Vectors are Perpendicular

Determine whether the two vectors,

$$\mathbf{u} = \left[\begin{array}{r} 2 \\ 1 \\ -1 \end{array}\right], \mathbf{v} = \left[\begin{array}{r} 1 \\ 3 \\ 5 \end{array}\right]$$

are perpendicular.

Solution. In order to determine if these two vectors are perpendicular, we compute the dot product. This is given by

$$\mathbf{u} \bullet \mathbf{v} = (2)(1) + (1)(3) + (-1)(5) = 0$$

Therefore, by Proposition D.29 these two vectors are perpendicular. ♠

D.7 Contributors

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This project is funded in part by the National Science Foundation, grant no. TUES Phase II DUE-1323785.